

HISTORICAL READING VOLUME FOR REVISION R47

This volume preserves the complete R45 article, supplement, response to R43, and its prior historical volume. They are historical scientific objects, not the current R47 claims. Their source files are retained separately. All original operating models, failed runs, older proofs and counterexamples remain in the inherited repository. The current article, numerical supplement, response to R44 and computation report are separate documents.

We study minimum-cost revision of an installed policy subject to operating guarantees at every state and date, using one common randomized Markov continuation. Backward repair links approximate feasibility to deployment cost. Bellman support envelopes remain valid at operating ties, and a multi-action search theorem separates policy resolution from arithmetic precision. For general Borel dynamics, a repair-density theorem gives an upper approximation without smoothing controlled atoms. Frozen finite experiments compare two complete search constructions with SCIP. Their best independently certified intersection meets an absolute 10^{-3} target in five of sixteen cases; SCIP reports eight numerical target hits, with its lower bounds not independently certified. Earlier root tests met no targets in eight cases; two returned uninformative fallback intervals. Forty-two further root computations expose allowance regimes and numerical floors. The original thirty positive-cost continuous-state randomized intervals remain open. Exact aggregation, observable fibers, and strict randomized savings retain their separate assumptions and economic content.

KEYWORDS: Dynamic programming, Policy revision, Common Markov policies, Global optimization, Verified computation.

1. INTRODUCTION

An organization revising an installed decision rule faces two different questions. How much operating performance can it sacrifice after a particular state and date? How much does implementing the revised rule cost under the states that the rule itself generates? A revision that looks expensive locally can change future occupancy and become expensive globally. A collection of expensive restart-specific plans need not be one implementable Markov rule. We study the minimum implementation cost subject to an operating guarantee after every restart, with a single continuation at any shared state and date.

The choice variable is a randomized Markov policy. Randomization is drawn anew conditional on the current state and date; it does not introduce a hidden initial coin that selects a persistent continuation. Implementation costs attach to realized departures from the installed rule. The objective averages these costs under a specified initial law, while the operating inequalities hold at every state, including states with zero initial mass. Neither a deterministic restriction nor dependent optimization after each restart preserves this economic problem.

The central contribution is an end-to-end certification argument for this common-policy object. Backward repair converts approximate operating feasibility into exact admissibility with an explicit cost modulus. A covering search over shared action probabilities supplies global lower bounds. Bellman support envelopes and simplex-product identities strengthen those bounds without dividing by an operating-action gap. A finite stopping theorem tracks the diameter of the policy box and the error in directed arithmetic separately. Consequently, exact ties do not invalidate certification or finite-model completeness. The theorem does not imply a favorable number of search nodes.

This separation is consequential near indifference. A strictly positive operating-action gap makes nonreference actions rare and yield a quadratic small-allowance bound for a root location. But that is a fixed-model result. A switching surface can drive the minimum gap to zero as a state representation is refined. We retain the strict-gap theorem and its full constants,

precision: the frozen implementation uses denominator 2^{44} for node bounds and 2^{40} for action probabilities, while a theorem covering every positive requested accuracy must have the rounding error to decrease.

In continuous states, two distinctions remain essential. Valid Bellman support functions can approximate action-dependent atomic transitions without treating nearby atoms as close in state space. In addition, a new repair-density argument approximates every feasible Borel measurable policy in implementation value using repaired simple proposals. It applies to controlled continuous-state not only autonomous observable types. For the original affine maintenance model it provides a constructive, atom-preserving upper-approximation route. This result is one-sided: it does not prove that a finite set of support prices supplies convergent lower endpoints. The paired upper and lower envelope results for exact aggregation, reset approximation, and observable fibers therefore rely on their explicit structural hypotheses.

Numerical evaluation now includes actual global trees rather than only root-node ablations. We designed environments from maintenance, inventory, and queueing families and four near-optimal policies were fixed with source and primitive hashes before execution. Both custom methods and SCIP cover the tree; SCIP receives the same common-policy constraints and strengthening cuts. At the registered absolute width 10^{-3} , the Bellman-envelope formulation closes three cases, the aggregate-product formulation closes five, and their valid intersection with independently evaluated candidate policies closes five. SCIP reports eight numerical tests. Its native lower bounds are not passed off as independent rational certificates. The custom formulation has narrower final intervals in eleven cases, but loses on each of the five largest environments. These outcomes measure a computational tradeoff, not uniform optimality.

The original continuous-state problem remains part of the scientific target. Its thirty positive-approximation intervals are still open, even though twenty-seven feasible lotteries cost less than the bound valid for every deterministic feasible rule. The nonlinear two-state experiment provides informative but wide intervals. Those facts remain visible in the article. All proofs, counterexamples, sensitivity exercises, failed calculations, and stopped-control experiments are preserved in the research archive, with the previous complete article and supplement included in a historical volume. The current argument does not depend on reading that history.

1.1. *Relation to the literature*

Constrained Markov decision processes supply the natural framework (Altman, 1999). Occupancy measures are particularly useful for expected constraints under a designated initial state. A separate occupancy measure for every restart would additionally have to agree on the initial action probabilities at shared state-date pairs. Omitting that compatibility would restrict the feasible set. Our direct formulation uses one probability variable at each such pair. It is a formulation requirement, not a claim that occupation measures cannot express the constraints.

Approximation theory for constrained control studies finite representations, policy recovery, and convergence (Dufour and Prieto-Rumeau, 2013, Saldi, 2018, Saldi, Linder, and Saldi, 2018). The particular bridge here is exact backward repair under all-restart inequalities. The repair-density theorem uses a finite family of action kernels to dominate every possible action distribution. Approximation occurs in that measure, while repair restores inequalities almost everywhere, not merely almost everywhere. Its upper approximation must not be confused with the paired global convergence theorem for arbitrary controlled continuous dynamics.

6); the identity that products sum to their common continuation variable follows from the ability simplex. Best-bound spatial search, dynamic programming, and weak duality are new. The additional result is a policy-compatible stopping modulus that charges feasibility violation and arithmetic error. We compare with SCIP rather than excluding a strong general-purpose solver (Hojny et al., 2025). Numerical LP proposals use HiGHS; their associated simplex methods are discussed by Huangfu and Hall (2018). Independent residual corrections show the distinction between computation and verification emphasized by Rump (2010). Bellman support functions are related to information-relaxation and dual bounds (Brown, Smith, and Sun, 2010, Brown and Smith, 2014). No finite price portfolio is asserted to be a complete dual. Whole-region inequalities, as in semi-infinite optimization (Hettich and Kortanek, 1993), are required for a continuous-state certificate. A certificate on sampled restarts is not substituted for the original universal constraint.

The economic motivation is organizational adjustment under a performance requirement, rather than learning an unknown operating model. Safe policy improvement and constrained learning solve distinct information problems (Laroche, Trichelair, and Tachet des Combes, 2019, Wang, Zhang, Basar, and Jovanovic, 2020, Ni and Kamgarpour, 2025). Numerical economics also emphasizes model structure and state-space approximation (Judd, 1998, Brumm and Scheidegger, 2017). In contrast to estimated replacement models such as Rust (1987), our environments are designed known-model exercises. They do not identify empirical implementation costs or net welfare. The institutional meaning of a lottery and its admissible administration charge is stated explicitly below.

2. THE ECONOMIC OBJECT

Let X be a standard Borel space, $A = \{1, \dots, m\}$ a finite action set, and $t = 0, \dots, T - 1$. Known Borel kernels are P_t^a . Borel operating rewards, implementation costs, and terminal payoff satisfy $|r_t| \leq R$, $0 \leq k_t \leq K$, and $|g| \leq G$. Throughout the central results $0 < \beta < 1$ and $S_0 = 0$. Write $S_h = \sum_{s=0}^{h-1} \beta^s$, with $S_0 = 0$.

The operating value and the values of a common policy p obey

$$V_T = g, \quad V_t(x) = \max_a \{r_t(x, a) + \beta P_t^a V_{t+1}(x)\}, \quad (1)$$

$$J_T^p = g, \quad J_t^p = \sum_a p_t(a | x) \{r_t(x, a) + \beta P_t^a J_{t+1}^p(x)\}, \quad (2)$$

$$C_T^p = 0, \quad C_t^p = \sum_a p_t(a | x) \{k_t(x, a) + \beta P_t^a C_{t+1}^p(x)\}. \quad (3)$$

target is

$$K_R(\nu, \varepsilon) = \inf_{p \in \mathcal{F}(\varepsilon)} \int C_0^p d\nu, \quad \mathcal{F}(\varepsilon) = \{p : J_t^p(x) \geq V_t(x) - \varepsilon \text{ for every } (t, x)\}. \quad (4)$$

Policies in this definition are Borel randomized Markov kernels. The finite action set gives well-maximizing selectors by choosing the first maximizer. An operating-optimal policy is implementable, so the set is nonempty. We do not require an optimizer to exist in an infinite-dimensional policy space; infimum arguments use approximate minimizers.

A representative implementation specification is $k_t(x, a) = w_t(x) \mathbf{1}\{a \neq a_t^0(x)\}$ for an installed rule a^0 . The relevant cost is departure from this installed rule, not a switch from the last physical action. A physical switching friction would require that previous action in the state. A fixed cost of administering a randomized rule is also a distinct primitive.

any bounded Borel potential H_t with $H_T = 0$. Define

$$d_t(x, a) = V_t(x) - r_t(x, a) - \beta P_t^a V_{t+1}(x), \quad (5)$$

$$h_t(x, a) = H_t(x) - k_t(x, a) - \beta P_t^a H_{t+1}(x), \quad (6)$$

$$D_t^p = V_t - J_t^p, \quad S_t^p = H_t - C_t^p.$$

≥ 0 , $D_T^p = S_T^p = 0$, and

$$D_t^p = \sum_a p_t(a | x) \{d_t(x, a) + \beta P_t^a D_{t+1}^p(x)\}, \quad (7)$$

$$S_t^p = \sum_a p_t(a | x) \{h_t(x, a) + \beta P_t^a S_{t+1}^p(x)\}. \quad (8)$$

identities for the same policy. In the executed finite formulation H is the imputation value of a first-index operating-optimal reference action a^* . Consequently $h_t(x, a^*) = 0$. Eliminating the reference probability leaves bilinear terms between reference action probabilities and continuation values. The analytic support results below require this particular reference choice, and therefore do not rely on its stability when actions tie.

3. COMMON-POLICY CERTIFICATION WITHOUT A STRICT ACTION GAP

$$D_C = K \sum_{s=0}^{T-1} (s+1) \beta^s, \quad (9)$$

$$D_J = \max_{1 \leq h \leq T} \left\{ 2R \sum_{s=0}^{h-1} (s+1) \beta^s + 2Gh\beta^h \right\}. \quad (10)$$

policies differ by at most θ in action-distribution total variation at every state and date, operating values differ by at most $D_J \theta$, and their costs by at most $D_C \theta$. To see this, subtract Bellman equations. A cost at distance s can be affected by $s+1$ action changes, and a operating payoff by h changes. Using the range, rather than the absolute magnitude, of values gives the stated constants. Total variation is one half of the ℓ_1 distance.

LEM 1—Backward repair: *For $p \in \mathcal{F}(\varepsilon + \delta)$, $\delta \geq 0$, there exists a Borel common $\mathcal{R}_\varepsilon p \in \mathcal{F}(\varepsilon)$ such that $J_t^{\mathcal{R}_\varepsilon p} \geq J_t^p$ at every state and date, and*

$$\sup_{t,x} \text{TV}((\mathcal{R}_\varepsilon p)_t, p_t) \leq \rho_\varepsilon(\delta) := \frac{\delta}{(1-\beta)\varepsilon + \delta}. \quad (11)$$

exceeds that of p by at most $D_C \rho_\varepsilon(\delta)$. In particular,

$$0 \leq K_R(\nu, \varepsilon) - K_R(\nu, \varepsilon + \delta) \leq D_C \rho_\varepsilon(\delta). \quad (12)$$

$\sum_a p_t(a \mid x) q_a$, $q_* = \max_a q_a$, and $b = V_t(x) - \varepsilon$. The induction hypothesis gives $q_p \geq \varepsilon - \delta$ and $q_* \geq V_t - \beta\varepsilon$. If $q_p \geq b$, retain the original action distribution. Otherwise mix it a first maximizing action in proportion

$$\alpha = \frac{b - q_p}{q_* - q_p}.$$

denominator is at least $(1 - \beta)\varepsilon + (b - q_p)$, so $\alpha \leq \rho_\varepsilon(\delta)$. The repaired value is b , weakly above both q_p and the original value. Every operation is Borel, and the division is used only with a positive denominator. This constructs one continuation, not a mixture of persistent plans. The perturbation bound gives the cost inequality; apply it to approximate minimizers to obtain *Q.E.D.*

The same formula defines $\mathcal{R}_\varepsilon q$ for an arbitrary Borel proposal q , because the repaired continuation always supplies $q_* \geq V_t - \beta\varepsilon$. A small uniform cost bound requires a small perturbation, but existence of the repair does not. This distinction will be used for continuous states.

3.1. Tie-safe support envelopes

For a fixed real price λ , define $W_{T,\lambda}^- = W_{T,\lambda}^+ = 0$ and

$$W_{t,\lambda}^- = \min_a \{h_t - \lambda d_t + \beta P_t^a W_{t+1,\lambda}^-\}, \quad W_{t,\lambda}^+ = \max_a \{h_t - \lambda d_t + \beta P_t^a W_{t+1,\lambda}^+\}. \quad (13)$$

THEOREM 2—Bellman supports at and near ties: *For every Borel common policy, without a uniqueness or positive-gap assumption,*

$$W_{t,\lambda}^- \leq S_t^p - \lambda D_t^p \leq W_{t,\lambda}^+. \quad (14)$$

Consequently any feasible policy satisfies all inequalities (14) together with $0 \leq D_t^p \leq \varepsilon$. For $0, \text{ let}$

$$Z_{T,\lambda} = \lambda g, \quad Z_{t,\lambda} = \max_a \{\lambda r_t - k_t + \beta P_t^a Z_{t+1,\lambda}\}. \quad (15)$$

the $\ell_{t,\lambda} := \lambda(V_t - \varepsilon) - Z_{t,\lambda}$ is a lower bound for C_t^p , and it is independent of H . For any price set $\Lambda \subset [0, \infty)$,

$$\int \max\{0, \max_{\lambda \in \Lambda} \ell_{0,\lambda}(x)\} \nu(dx) \leq K_R(\nu, \varepsilon). \quad (16)$$

PROOF: Subtract λ times (7) from (8). Backward induction and the fact that a convex combination lies between its smallest and largest action value give (14). Direct substitution yields $C_t^p = H_t - \lambda V_t + Z_{t,\lambda}$. Thus $C_t^p \geq H_t - W_{t,\lambda}^+ - \lambda\varepsilon = \ell_{t,\lambda}$. Pointwise maxima and nonnegativity preserve the lower inequality before integration. No transition has been approximated in the argument. *Q.E.D.*

For a finite model, each inequality in (14) is linear in D, S . The frozen enhanced formulation with $\lambda = 0$ and $\lambda = \pm 2^j$, $j \in \{-2, 0, 2, 4, 6, 8, 10, 12\}$. The prices are fixed before optimization. The finiteness makes them inexpensive cuts, not a claim of exact global duality.

ative numbers e_t satisfy

$$\lambda_- d_t(x, a) - e_t \leq h_t(x, a) \leq \lambda_+ d_t(x, a) + e_t \quad \text{for every } (x, a).$$

$$= 0 \text{ and } B_t = e_t + \beta B_{t+1},$$

$$\lambda_- D_t^p - B_t \leq S_t^p \leq \lambda_+ D_t^p + B_t \quad (17)$$

y policy. In particular the result permits $d_t = 0$ and nonzero h_t . If, instead, only actions $\geq \tau > 0$ are classified as separated, every feasible policy assigns their total probability ε/τ .

F: Subtract $\lambda_+ D$ from S in the joint Bellman recursion and bound the stage term by action gives the upper inequality. The lower inequality is identical with $-e_t$. For the last nonnegativity in (7) gives $\sum_a p_t(a | x) d_t(x, a) \leq \varepsilon$. *Q.E.D.*

ffine intercept records the cost of a near-tie group instead of hiding it inside an in-imum gap. It need not be small. In particular, an exact operating tie may allow a ial implementation-cost difference. The frozen solver uses the Bellman supports, not this ated affine-cone variant; the latter is an analytic strengthening available for subsequent entations.

3.2. The strict-gap specialization

earlier regret-scaled rate remains a useful specialization. With an operating-optimal e and strictly positive nonreference disadvantages, put

$$\underline{d} = \min_{t, i, a \neq a^*} d_{tia}, \quad \chi = \min\{1, \varepsilon/\underline{d}\},$$

$$\lambda_- = \min\{0, \min h_{tia}/d_{tia}\}, \quad \lambda_+ = \max\{0, \max h_{tia}/d_{tia}\}.$$

nonreference mass is at most χ , and $\lambda_- D \leq S \leq \lambda_+ D$. With $\Delta P_{tia} = P_i^a - P_i^{a^*}$, $P_{D, T-1} = 0$ and, for $t < T - 1$,

$$e_{D,t} = \frac{\beta \varepsilon^2}{4} \max_i \sum_{a \neq a^*} \frac{\|\Delta P_{tia}\|_1}{d_{tia}}, \quad E_{D,t} = e_{D,t} + \beta E_{D,t+1}, \quad (18)$$

$$E_{S,t} = (\lambda_+ - \lambda_-) e_{D,t} + \beta E_{S,t+1}, \quad \overline{E}_D = \max_t E_{D,t},$$

o terminal errors. An exact optimizing root-LP solution, repaired toward the operating e, has width bounded by

$$E_{S,0} + D_C \chi \frac{\overline{E}_D}{(1 - \beta)\varepsilon + \overline{E}_D}. \quad (19)$$

and follows from the McCormick product error, at most one quarter of the product of interval widths, followed by the rare-departure repair. The full derivation is retained in orical technical material. At fixed primitives, fixed $\beta < 1$, and $\underline{d} > 0$, (19) is $O(\varepsilon^2)$. A al LP optimality loss and directed endpoint errors must be added when an implemented

ero. The rate neither covers ties nor remains uniform when $\underline{d}$ vanishes along state refinements. Formula (17) and Theorem 3 below give valid alternatives without declaring the strictly separated from uniform. The tables report the actual constants and exclude tied cases from the inverse-formula.

3.3. A complete multi-action search and its numerical floor

For finite X , use one coordinate for every nonreference probability at every state and date. Node is a box in these $d = nT(m - 1)$ coordinates, intersected with the local probability simplexes. The reference probability is the remaining mass. The lower corner belongs to every empty such row because the sum of its nonreference lower bounds cannot exceed one.

Restricted Bellman minimization of regret and implementation cost over these local simplexes is, respectively, a necessary operating test and a cost lower bound. These restricted dynamic programs are not the final constrained optimization. They are admissible relaxations of all the common policies in the box. The linear formulation additionally imposes (7)–(8), the support and product envelopes. In the disaggregated formulation the products also obey

$$\sum_{a \in A} w_{tia,j} = z_{t+1,j}, \quad w_{tia,j} = p_{tia} z_{t+1,j} \quad \text{before relaxation,} \quad (20)$$

each retained continuation coordinate $z = D, S$. The aggregate formulation instead relaxes $(\Delta P_{tia} z_{t+1})$. These are different relaxations of the same Bellman equations. Disaggregation is more variables and arithmetic per node.

Floating-point LP supplies only a candidate policy and multipliers. For a minimization LP $c'x$ subject to $Ax = b$, $Gx \leq h$, and $l \leq x \leq u$, any equality multipliers y and nonnegative inequality multipliers μ give

$$b'y - h'\mu + \sum_j \min\{(c - A'y + G'\mu)_j l_j, (c - A'y + G'\mu)_j u_j\} \quad (21)$$

lower bound. Directed arithmetic encloses each term outward. Zero multipliers remain valid if the proposal is unavailable. A floating-point infeasibility flag alone never removes a box.

The incumbent is the smallest exactly evaluated repaired-policy cost. Every split retains both children. Infeasible leaves require a rational necessary inequality; all other terminal leaves retain the lower endpoint. The global lower endpoint is the minimum of those leaf endpoints and the incumbent. This full cover is part of the proof object.

THEOREM 3—Gap-free finite stopping with explicit precision: *Consider a surviving probability box of maximum nonreference coordinate width w . Suppose each restricted Bellman scalar is rounded outward with error at most $q = 2^{-b}$ per date, and all other node removals and additional lower bounds are valid. Put*

$$\begin{aligned} \theta(w) &= \min\{1, (m - 1)w\}, & \gamma_b &= 2^{-b} S_T, \\ \mathcal{H}(w, b) &= D_C \theta(w) + \gamma_b + D_C \rho_\varepsilon(D_J \theta(w) + \gamma_b). \end{aligned} \quad (22)$$

the cost of exact backward repair of the lower-corner policy minus the rounded restricted cost per bound is at most $\mathcal{H}(w, b)$. For every $\eta > 0$, choose b and $w_ > 0$ with $\mathcal{H}(w_*, b) \leq \eta$.*

minimates at gap η without node or time caps. One sufficient full-tree depth bound is

$$8d \lceil \log_2(1/w_*) \rceil, \quad (23)$$

the ceiling is replaced by zero if $w_* \geq 1$. No minimum positive action disadvantage enters statements.

F: The detailed argument is in Appendix A. Restricted Bellman minimization attains ϵ regret simultaneously at every state with one Markov policy in the box. Its exact ϵ regret is at most $\epsilon + \gamma_b$ in a surviving node. All policies in that box differ by at most ϵ action-distribution total variation. The corner therefore violates operating feasibility by $D_J\theta(w) + \gamma_b$. Its cost is at most $D_C\theta(w) + \gamma_b$ above the rounded restricted minimum. This proves (22). A small enough box cannot require further splitting when its repaired corner has been considered. Widest-coordinate splits force all widths below w_* by the stated bound independently of heuristic intervening splits. There are finitely many nodes up to that bound. *Q.E.D.*

Theorem 3 is uniform over finite models with common bounds $R, K, G, T, m, \beta, \epsilon$ for the cost modulus, even when action gaps vanish. It is not uniform in computational dimension: d contains d , and the number of tree nodes is exponential in depth. Nor does it give a ϵ -approximation theorem merely by substituting a finer state grid for the original kernel. The frozen code uses two grids: 2^{-44} for node-bound arithmetic and 2^{-40} for deployed reference probabilities. It also uses an operating-reference repair with a rounding guard, rather than the exact first-maximizer repair in Theorem 1. The following additional bound quantifies this implementation difference.

COLLARY 1—Quantized operating-reference repair: *Let b_v denote the bound precision and b_p the deployed-probability precision. Define*

$$s = (1 - \beta)\epsilon, \quad B = \max_{t,i,a} d_{tia} + \beta\epsilon, \quad \kappa = (m - 1)2^{-b_p}, \quad \mu = \kappa B, \quad (24)$$

$$\delta = D_J\theta(w) + \gamma_{b_v}, \quad \Delta = \delta + \mu S_T.$$

At each row, mix the proposal toward the operating reference until its regret is at most $\kappa \max_a |q_a - q_{a^}|$, where $q_a = d_{tia} + \beta P_i^a \widehat{D}_{t+1}$ uses the already repaired continuation. Then decrease each nonreference probability downward to a multiple of 2^{-b_p} and assign the remaining probability to the reference. This is the frozen repair rule. Its repaired lower-corner cost minus the node cost lower bound is at most*

$$\mathcal{H}_Q(w, b_v, b_p) = D_C\theta(w) + \gamma_{b_v} + D_C \min \left\{ 1, \kappa + \frac{\Delta}{s - \mu + \Delta} \right\}. \quad (25)$$

The ϵ -depth conclusion of Theorem 3 holds with $\mathcal{H}_Q$ in place of $\mathcal{H}$. Arbitrary positive target ϵ requires choosing both precisions sufficiently large; fixing either grid need not give a ϵ -gap sufficient floor.

F: Appendix A bounds the accumulated regret increase due to rounding by μS_T , the rounding fraction by $\Delta/(s - \mu + \Delta)$, and the extra row total variation by κ . Apply the cost-modulus to the repaired lower corner. The reference slack s is positive if there is any strict action gap. *Q.E.D.*

els; the largest is about 3.715×10^{-4} for Q3. The corresponding exact-repair floor is smaller and is not substituted for this implemented-repair bound. The supplementary tables give both floors and their sufficient widths. Actual certificate endpoints remain valid independently of these conservative bounds. The pre-existing fixed-precision sources and measurements are unchanged; no adaptive-precision experiment is claimed.

At $\varepsilon = 0$, a different exact argument applies. Because all terms in (7) are nonnegative, every admissible policy uses only operating-maximizing actions at every state and date. Conversely those policies give zero regret by backward induction. Minimizing implementation cost by Bellman iteration within that action set therefore solves the zero-allowance problem, including ties. This statement does not require taking the singular limit of (11). The retained finite-horizon allowance-ladder extension likewise has its own assumptions when $\beta = 1$.

4. CONTROLLED ATOMIC STATES AND APPROXIMATION

A continuously indexed state is economically relevant when it affects operating rewards, implementation costs, or action-dependent transitions. It cannot be dismissed as a nuisance coordinate merely because a separate aggregation example has that property. The following results retain Borel common policies and allow controlled transitions. They distinguish an approximation of an upper policy from a certificate covering the unrestricted optimum.

4.1. Repair density for unrestricted Borel policies

Define finite, not necessarily probability, measures

$$\mu_0 = \nu, \quad \mu_{t+1}(B) = \sum_{a \in A} \int P_t^a(x, B) \mu_t(dx). \quad (26)$$

For masses are m^t . Every state distribution induced from ν by any admissible or inadmissible Markov policy is dominated by μ_t . Let Π_N be increasing finite Borel partitions generating the Borel sigma field of X . Such partitions exist on a standard Borel space. Let $\mathcal{Q}$ be the countable set of all rational-probability policies that are constant on some Π_N at each date.

THEOREM 4—Repair-complete upper approximation: *Under the bounded Borel assumptions of Section 2,*

$$\inf_{q \in \mathcal{Q}} \int C_0^{\mathcal{R}_{\varepsilon} q} d\nu = K_R(\nu, \varepsilon). \quad (27)$$

where $(q_j)_{j \geq 1}$ enumerates $\mathcal{Q}$, the exact feasible upper values $U_N = \min_{j \leq N} \int C_0^{\mathcal{R}_{\varepsilon} q_j} d\nu$ decrease to the unrestricted Borel-policy infimum. No exogeneity, recoverable type label, positive action continuity of the policy, or smoothing of an atomic kernel is assumed. This is an upper-approximation theorem; it supplies neither a finite N error rate nor a matching global lower bound.

PROOF: For any feasible p , conditional expectations of its probabilities on Π_N converge in $L_1(\mu_t)$ to p_t . Rational simplex approximations preserve that convergence. Write the resulting proposal as q^N . A backward projection argument shows that the operating value of q^N approaches J^p in $L_1(\mu_t)$: its scalar value is the maximum of the proposal value against repaired continuation and $V_t - \varepsilon$, and J_t^p is above that threshold. The repair probability is

in p in the same measures. A performance-difference identity and occupancy domination imply convergence of initial implementation costs. Appendix B supplies the complete inductive proof, including measure-zero states. Every repaired policy is globally feasible, and the reverse inequality in (27) is immediate. Approximate feasible minimizers finish the proof. *Q.E.D.*

The result does not restrict the target policy class to a sieve. Only proposals are simple; repairs need not be. All-state feasibility is restored by pointwise Bellman construction, excluding states excluded from every μ_t -integral. Dominating measures can be expensive: their costs grow as m^t . The theorem therefore removes an expressivity obstruction without incurring a practical dimension-free rate.

The original maintenance primitives in Section 6, rational interval partitions give a concrete computational consequence. With rational piecewise-affine rewards and finitely many affine repair maps, V is piecewise affine on a finite partition at every finite horizon. If q is rational and ε is constant, the repaired operating recursion is

$$J_t^{\mathcal{R}_{\varepsilon}q} = \max \left\{ \sum_a q_t(a | x) (r_t + \beta P_t^a J_{t+1}^{\mathcal{R}_{\varepsilon}q}), V_t - \varepsilon \right\}. \quad (28)$$

$J_t^{\mathcal{R}_{\varepsilon}q}$ is piecewise affine. First-index maximizing action selection and the repair fraction are rational, and the fraction's denominator on its active region is at least $(1 - \beta)\varepsilon$. For rational rules with finitely certified regions and piecewise-affine costs, implementation values are piecewise rational. This representation condition must be checked for a compiled incumbent; it is not inferred for an arbitrary floating-point neural network. Exact breakpoint refinement by divergent outward integration under the uniform initial law can consequently certify each value. Jumps and isolated breakpoints are assigned explicitly; an integral ignoring a point cannot excuse to omit its operating constraint. This establishes a constructive upper route for rational controlled atoms. No new optimization of the thirty original positive-cost rows is required in this revision.

4.2. Support witnesses that preserve moving atoms

Lemma 2 gives global lower bounds even when transitions interact across continuous labels. If certified functions $\underline{V}_t \leq V_t$ and $\bar{Z}_{t,\lambda} \geq Z_{t,\lambda}$ are available. Then

$$\int \max\{0, \max_{\lambda \in \Lambda} [\lambda(\underline{V}_0 - \varepsilon) - \bar{Z}_{0,\lambda}]\} d\nu \leq K_R(\nu, \varepsilon) \quad (29)$$

for nonnegative prices. A terminal inequality and a Bellman supersolution for Z establish the witness by induction. In a finitely atomic model the operator is evaluated as the original weighted sum at its exact successor maps. No weak-to-total-variation substitution is used. The bounds also have a gap-free primitive stability statement. Suppose corresponding functions satisfy $\|V_t - \tilde{V}_t\|_{\infty} \leq a_t$ and $\|Z_{t,\lambda} - \tilde{Z}_{t,\lambda}\|_{\infty} \leq z_{t,\lambda}$. Then the support lower functions are at most $\lambda a_t + z_{t,\lambda}$ at the same allowance. The bound does not involve a reference function. For example, with uniform reward error e_r , cost error e_k , terminal error e_g , and total variation at most α_t , valid backward errors are

$$\begin{aligned} a_T &= e_g, & a_t &= e_r + \beta a_{t+1} + 2\beta \alpha_t M_{t+1}^V, \\ z_{T,\lambda} &= \lambda e_g, & z_{t,\lambda} &= \lambda e_r + e_k + \beta z_{t+1,\lambda} + 2\beta \alpha_t M_{t+1,\lambda}^Z, \end{aligned} \quad (30)$$

the total-variation convention. More generally the last terms may be replaced by verified expectation errors on the particular comparison functions. For controlled affine atoms and Lipschitz comparison functions, those errors can be bounded by weighted displacements of atoms times the Lipschitz constant. That is a test-function error estimate, not a bound on variation between distinct Dirac masses. Discontinuities require one-sided cell bounds or implicit breakpoint treatment.

An approximate operating solution must also be charged on the upper side. The retained witness-driven repair theorem uses globally valid operating enclosures and their Bellman duals, rather than treating an approximate V as exact. The frozen finite experiments use exact rational operating dynamic programming, so they do not test a difficult operating-oracle construction. The new witness formulas state what an application must verify; they are not certified as a completed high-dimensional oracle experiment.

4.3. Where paired continuum convergence is available

The earlier exact aggregation and reset-transfer results are preserved in full. Exact exogenous aggregation requires both pointwise aggregate transition identities and a compatible family of within-cell measures. These hypotheses can hold for atomic transitions. The resulting cost problem is an ordinary finite linear program, and the continuous nuisance coordinates of its witness examples do not affect rewards, implementation costs, or regime transitions. This is an exactness test, not evidence of solving an economically coupled high-dimensional continuous problem.

Observable fibers give a different paired route. Suppose the state is (i, z) and the next state is $(j, \Phi_t(z))$ with action-dependent regime probability $P_t^a(i, j; z)$. Each Φ_t is an autonomous Borel function with Borel inverse. With $F_t = \Phi_{t-1} \circ \dots \circ \Phi_0$, the label $\zeta = F_t^{-1}(z)$ is observable. If the initial law is $\nu(di, d\zeta) = q_i(\zeta)\xi(d\zeta)$ for any Borel probability vector $q(\zeta)$, then

$$K_R(\nu, \varepsilon) = \int v(\zeta, q(\zeta), \varepsilon) \xi(d\zeta), \quad (31)$$

where v is the finite common-policy optimum in that fiber. Independence of initial regime and initial law is unnecessary. Appendix C gives the full Borel finite-selection construction and the error in approximating the conditional initial law.

Refining a fiber partition alone cannot eliminate residual finite optimization gaps. Both the continuous variation and the finite global intervals must be refined. Nor can (31) apply to the initial condition process merely because its shocks are discrete: actions move its continuous condition and do not preserve an autonomous observable label. The repair-density result and the witness support witnesses address that class directly, but a convergent pairing of their two sides is not established here.

5. FROZEN GLOBAL-SEARCH EVIDENCE

5.1. Design, information, and computational accounting

The global experiment consists of twelve designed cases from three families and four variants of separate near-tie construction. Within each primary family, (n, T, m) takes values $(4, 4, 3)$, $(8, 16, 4)$, and $(16, 32, 4)$. The corresponding (β, ε) pairs are $(0.9, 0.01)$, $(0.95, 0.01)$, $(0.95, 0.005)$, and $(0.95, 0.001)$. Every method receives identical rational rewards, costs, controlled transition probabilities, initial law, and restart inequalities. The inventory and queueing

transitions. The exact generators are specified in the supplement. The external source freeze precedes these executions and includes eleven scientific source hashes between primitive hashes. The registered allowance is 120 seconds and 2,047 nodes per second with 15 seconds per node LP and at most five seconds for a local root proposal. A check occurs between proof-producing node operations: an iteration can exceed the wall cap while completing a valid node or its sibling. Measured time, rather than the wall cap, is reported. Exact checking time is charged separately. Every method was run with a single thread; the environment records Python 3.13.15, NumPy 2.3.5, SciPy 1.17.0, and CPython 6.2.1 on a four-CPU Linux runner. Paired methods in a case use the same runner. No equivalence is not inferred across separately scheduled cases.

Method B uses Bellman supports, disaggregated products, and simplex-product consistency. Method A uses the aggregate-product relaxation. Both use the same common-policy repair and a g-tree contract. A local nonlinear optimizer can propose a root candidate, but its output is checked and repaired before it affects an upper endpoint. Every node also evaluates its lower policy. Thus soundness and theoretical completeness do not depend on success of the local optimizer or on a favorable LP proposal.

Method B uses the common-policy bilinear equations, the same root domains, Bellman supports, and gap cones when applicable. Its objective, feasibility tolerance, random seed, one-thread absolute-gap target, and budgets are recorded. This grants the external solver the same timing information rather than withholding the method's cuts. Its native lower endpoint is a cutting-point solver assertion; its proposed policy is independently evaluated after rationalization. No claim is made that SCIP cannot provide other verification modes. The comparison is of this executed nonlinear adapter.

5.2. Every global interval, including unsuccessful stops

Method A keeps all sixteen cases. B has narrower final intervals in eleven cases, A in four, and the rest is equal. Nevertheless B meets the target in four cases and A in five. Their certified solution, including the independently evaluated SCIP policy, meets it in five. A narrower interval and or a larger formulation is not enough to predict final target attainment under a fixed

budget. B is now successful closure after branching. On Q1, B reduces its gap below 10^{-3} in 49 nodes, whereas A requires 153 nodes. On M0, A closes in 13 nodes, whereas B already closes at its root. On the zero-cost case, and Q0 is nearly exact at the root; they are not counted as demonstrations of global search. Q3, the largest queueing case, closes under A at the root. B explores only 10 nodes and leaves a width about 0.0322 there. All three largest cases favor A's final width. The custom construction and stronger per-node relaxations consume resources that otherwise support branching.

Method B reports eight native target hits. It often returns smaller numerical gaps substantially than the custom proof-producing search. Its failures, including node limits on inventory cases and time limits on the largest maintenance and inventory cases, are also retained. The evidence does not establish computational superiority of the new cuts or implementation of a modern external global solver. What the custom method adds in the reported comparison is a verifiable rational covering certificate. Both the cost of that certificate and the stronger numerical performance are visible.

Case	<i>n</i>	<i>T</i>	<i>m</i>	B width	A width	Intersection	Target met
M0	4	4	3	0.000497	0.000567	0.000497	Yes
M1	8	8	3	0.007157	0.0222	0.007157	No
M2	8	16	4	0.008723	0.0255	0.008723	No
M3	16	32	4	0.7286	0.1150	0.1139	No
I0	4	4	3	0	0	0	Yes
I1	8	8	3	0.0179	0.0268	0.0179	No
I2	8	16	4	0.0142	0.0349	0.0142	No
I3	16	32	4	0.0640	0.004768	0.004630	No
Q0	4	4	3	7.607×10^{-12}	6.129×10^{-12}	6.129×10^{-12}	Yes
Q1	8	8	3	0.000980	0.000997	0.000980	Yes
Q2	8	16	4	0.001151	0.004986	0.001151	No
Q3	16	32	4	0.0322	0.000694	0.000660	Yes
T0	4	8	3	0.004995	0.1606	0.004995	No
T1	4	8	3	0.005256	0.2000	0.005256	No
T2	4	8	3	0.003211	0.1678	0.003211	No
T3	4	8	3	0.1345	1.3603	0.1345	No

aintenance; I: inventory; Q: queueing; T: near-tie variants. B and A are independently checked whole-tree intervals, not root LP statuses. The intersection uses the largest checked lower endpoint and the smallest checked feasible upper cost, including SCIP's repaired candidate. It combines separate computations and is not a single second algorithm. Target tests use exact fractions, not rounded displays. Full endpoints, relative widths, work, and stop reasons are in the supplement.

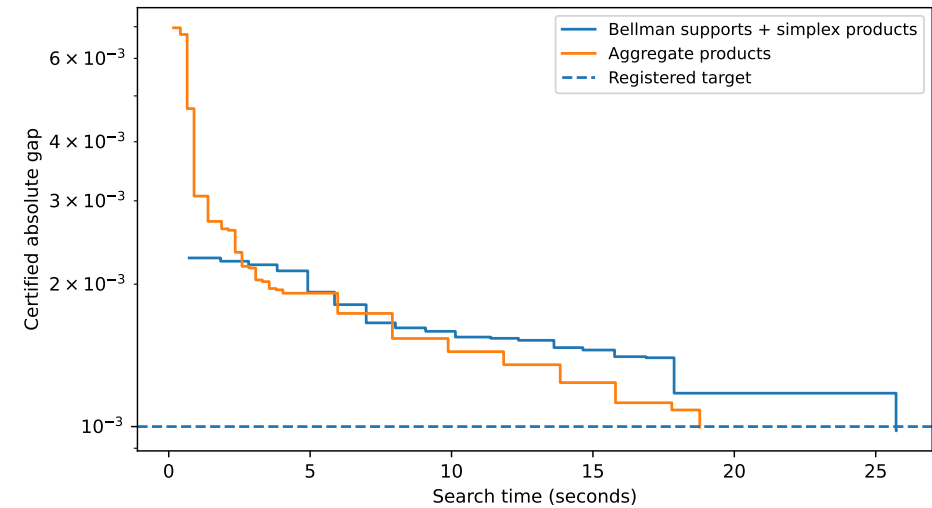


FIGURE 1.—Certified gap along the recorded Q1 searches. Every plotted point is an archived full-cover lower point paired with the current feasible incumbent. Time excludes the final independent checker, whose cost is tested separately. The horizontal line is the registered absolute target.

Case	SCIP stop	Native width	Nodes	Seconds	Checked intersection
M0	gaplimit	0.000497	1	0.11	0.000497
M1	gaplimit	0.000832	11	3.86	0.007157
M2	gaplimit	0.000835	1	12.96	0.008723
M3	timelimit	0.0520	1	120.52	0.1139
I0	optimal	0	0	0.05	0
I1	odelimit	0.001326	2047	14.26	0.0179
I2	odelimit	0.003895	2047	72.75	0.0142
I3	timelimit	0.001470	1	120.43	0.004630
Q0	gaplimit	7.000×10^{-9}	1	0.06	6.129×10^{-12}
Q1	gaplimit	0.000243	1	1.35	0.000980
Q2	gaplimit	2.670×10^{-5}	1	4.98	0.001151
Q3	gaplimit	0.000149	1	13.14	0.000660
T0	odelimit	0.002574	2047	6.55	0.004995
T1	timelimit	0.002388	1359	120.01	0.005256
T2	odelimit	0.002624	2047	6.17	0.003211
T3	odelimit	0.1092	2047	10.85	0.1345

Methods use SCIP’s floating-point primal and dual bounds; they are not independent lower certificates. The final has the same certified construction as Table I. Native run time includes model construction and candidate dependent candidate checking and the two custom lower-bound computations are not hidden inside that time.

5.3. Exact and near operating ties

The model has four states, eight periods, three actions, $\beta = 0.95$, and $\varepsilon = 0.01$. The first operating rewards are 0 and $-\tau$, while the third is $-1/5 - i/40$; terminal operating reward is $i/8$, and zero. The four values of τ are 0, 10^{-8} , 10^{-4} , and 10^{-2} . Thus $\tau = 0$ is an exact tie with distinct implementation costs and transitions, not two duplicated actions. Native widths in these four cases are approximately 0.004995, 0.005256, 0.003211, and 0.1345. A’s corresponding widths are 0.160574, 0.200032, 0.167849, and 1.360331. All four are above target, even though the support formulation substantially improves the intervals. Primal validity tests succeed at $\tau = 0$. This demonstrates validity and useful tightening difference, not target-level tie closure. The gap-free theorem explains why a positive gap is unnecessary for eventual finite-model certification; the observed budgets remain within the constraint.

5.4. Allowance scaling and the exact-LP distinction

Each frozen calculation uses one $n = 3, T = 8, m = 3$ model from each primary family, with allowances from 10^{-1} to 10^{-7} , and both earlier root relaxations: 42 root outputs, not 42 independent environments. Each output has its own rational certificate. The complete tables of measured widths, width divided by ε^2 , $\bar{E}_D$, $E_{S,0}$, D_C , χ , the repair fraction, and the tight exact-LP bound (19).

Maintenance scaled widths at allowances $10^{-3}, 10^{-4}, 10^{-5}$ are approximately 0.0140484, 1.44446×10^{-4} , and 1.44446×10^{-6} . Their ratios to ε^2 are close to 1.4×10^4 over that range. At 10^{-6} and 10^{-7} the ratios increase, so the finite-precision output is not assigned a rational quadratic exponent. The inventory model contains ties; the strict-gap bound does not and its scaled and unscaled widths coincide. Queueing widths reach roughly 10^{-10} over

T	Positive	Median width (%)	Max. (%)	Strict gains	Min. gain	Max. gain
4	10	5.24	14.61	10	0.021898	1.041948
8	10	14.33	32.29	8	0.058701	0.896320
12	10	15.75	34.23	9	0.044193	0.747177

width is $(U_R - L_R)/U_R$ among positive-cost randomized intervals. A strict gain is $L_D - U_R > 0$, with L_D a deterministic lower bound and U_R a feasible randomized cost. Gain extrema condition on positive gains. The twelve positive-cost configurations are not included in the width statistics. These are retained certificates, not newly optimized rows.

practical allowances, making a log-log slope a description of numerical residuals rather than the theoretical rate. The unscaled queueing run at 10^{-6} returns a width about 5.3153 and is retained. The theorem’s clean bound and the realized interval measure different quantities. The available artifacts do not certify an optimizing primal–dual pair for every numerical root relaxation separately from the deployment repair. Therefore an exact numerical LP-optimality-loss decomposition is marked unavailable, not filled with zeros from an “optimal” status. The full measured interval is nevertheless valid. The former four-point scaling calculation and all eight earlier controlled root pairs are retained; none of those eight earlier environments met the 10^{-3} absolute target, and two returned uninformative $[0, U]$ intervals under both relaxations.

6. ECONOMIC INTERPRETATION AND THE RETAINED CONTINUOUS-STATE EVIDENCE

The original maintenance state is $x \in [0, 1]$, with three actions and shock probabilities $3/5, 2/5$. Discounting is $19/20$, the stage operating reward is $x - c_a$ for $c = (0, 3/20, 9/20)$, terminal payoff is $x/2$, and revision weight is $1 + x$. The two successors under each action are

$$(x/4, 3x/4 + 1/20), \quad (17x/20 + 1/10, 17x/20 + 7/50), \quad (x/10 + 4/5, x/10 + 17/20).$$

These are controlled continuous-state atoms. The primary cohort uses seven installed rules, horizons 4, 8, 12, allowances 0.01, 0.05, and the uniform initial law. The forty-two configurations are a development cohort, not independent observations. All their endpoints and alternative initial-law calculations are preserved.

All thirty positive-cost randomized intervals remain open. Their median relative widths decrease from 5.24% at horizon four to 14.33% at horizon eight and 15.75% at horizon twelve. Forty-seven randomized upper policies nevertheless lie strictly below deterministic lower bounds. These two statements are compatible: a policy-class separation does not identify the randomized optimum. The finite frozen cohort is not a replacement for closing these controlled continuous-state intervals.

6.1. Randomization as an institutional choice

The admissible lottery is a fresh state-dependent action draw at each date, with realized outcome charged under k_t . Suppose an organization additionally charges $f \geq 0$ per use of a degenerate lottery. For a certified policy p , let

$$B(p) = \mathbb{E}_\nu^p \sum_{t=0}^{T-1} \beta^t \mathbf{1}\{p_t(\cdot | X_t) \text{ is nondegenerate}\}.$$

stochastic rule whenever $fB(p) < s$. A conservative sufficient condition is $fS_T < s$. A fixed governance cost F instead requires $F < s$. This is a break-even implication in the implementation units, not an estimate of actual administrative costs. Such overhead is the economic objective and must be specified before treating the new optimum as a final result. Deterministic lower bounds remain a valid comparator because deterministic pay no lottery overhead.

Reporting the scale of each separation, the relevant normalized gains are s/L_D and s/U_R , operating-welfare percentage. The historical numerical sources retain the signed gains, deterministic bounds, and upper-policy costs. Any percentages computed only from rounded tables must be labeled approximations rather than exact certificates. No empirical evidence of feasible organizational randomization is inferred from the designed cohort.

6.2. Other continuous-state classes and the nonlinear experiment

Twelve exact-aggregation service examples retain their tightly closed finite-LP intervals, economic but economically ancillary continuous coordinates. The density-transfer exercise failed sufficient tightening at its largest perturbation. The observable-fiber example re-sixteen-cell interval $[10.074215, 10.184476]$, with width contributions about 0.074982 at endpoints and 0.035279 from partition variation. Seventeen endpoint solves represent a metric family, not seventeen model tests. No newly refined or higher-dimensional fiber partition is claimed.

Nonlinear two-state model has four actions, bilinear successors, and two shocks. At $s = 1/2$, retained relative widths at $(T, N) = (8, 64), (16, 128), (32, 64), (32, 128)$, and $(64, 128)$ are 76.01%, 45.32%, 61.05%, 35.11%, 25.13%. Full independent arithmetic coverage is explicitly marked in the retained tables; in particular the finest $(32, 256)$ object has been independently rebuilt. That audit does not turn a 25.13% interval into a near-exact result. At $s = 0.05$, the retained calculations remain lower-only because no matching upper policy is available. A failed sufficient enclosure is not proof of policy infeasibility. This revision does not provide a per-candidate failure classification when the retained records do not provide one.

Inherited binary finite-state comparison also remains unfavorable to interval Bellman on every difficult pair: the verified McCormick formulation gives stronger intervals than the stopped-control derivations, economic sensitivities, deterministic frontiers, and earlier examples remain in the historical volume without being counted as new completed verification targets.

7. VERIFICATION AND IMPLEMENTATION BOUNDARIES

Distinct claims require separate evidence. The economic and mathematical reduction reserve (4). The arithmetic must prove each serialized inequality. The publication must be checked source and outputs. Hashes address identity, not theorem validity. Finite model tests probe the checker, not the full space of possible bugs.

A standalone tree reader imports neither optimizer nor constructor. It independently rebuilds the model, operating value, reference, supports, product domains, dual residual bounds, policy set, and every split of the global cover. Every feasible upper is checked after every state and SCIP candidate checks verify only the feasible policy and its cost, not SCIP's native global bound. The local replication of this revision checks all 32 complete custom trees and all policies against their frozen model files; the exact reports accompany the manuscript. Root-scaling certificates are separately identified and checked by the earlier standalone reader.

port inequalities, including an exact operating tie. Seventeen archived adversarial mutations are rejected. They include forged endpoints, invalid probabilities, dual signs, nonstochastic positions, altered discount factors, incorrect primitive disadvantages, exchanged action labels, formed probability/product boxes, forged repair rows, missing leaves or siblings, a coverage and a cyclic tree. These are finite tests, not machine-checked proofs of the general support stopping theorems. The theorem–code matrix records which theorem is implemented, which assumptions are analytic, and which calculations were actually executed.

The final manuscript and response introduce analytic results beyond the frozen code: the explicit multi-action numerical-floor theorem, affine approximate-tie cones, and repair-density theorem. They are accompanied by proofs and separated from the predeclared numerical results. The frozen implementations and unfavorable outputs remain unchanged. In particular, adaptive-precision extension and the atom-preserving upper enumeration are not described as completed numerical experiments.

The new precision contract was also checked with a separate exact-rational implementation of 56 seeded two-state, three-date, three-action boxes, including an exact operating tie. Every surviving box satisfied the ideal-repair stopping inequality and all repaired restart constraints. A separate exact implementation checked the quantized operating-reference repair on those same boxes, including its rounding guard; all also satisfied the two-grid bound. These are two exact checks on the same boxes, not 512 independent test environments. Sixteen additional test calculations report sufficient precision floors and box widths for the frozen models. These checks are instances of the mathematical contract; the proofs, not the test count, establish the theorems. All forty-two frozen scaling proof objects were independently replayed as well.

8. CONCLUSION

Minimum-cost revision under all-restart operating protection is a common-continuation problem. Backward feasibility repair provides an explicit connection between numerical approximation and admissible deployment. Bellman supports and a multi-action covering search extend this connection to operating ties, while a precision-dependent stopping modulus makes the role of finite arithmetic explicit. The strictly separated small-allowance rate is retained as a separate conditional result, not a universal numerical law.

For controlled continuous states, the repair-density theorem keeps the unrestricted Borel target and permits action-dependent atoms. It supplies a convergent upper approximation; dual support witnesses supply lower bounds without atom smoothing. A matched two-sided convergence rate for the original controlled continuum remains a distinct task. Exact aggregation of observable-fiber identities continue to give paired results under their stated structure.

The complete-search evaluation now records real branching, a modern external baseline, all end stops, and complete arithmetic covers. Five of sixteen combined rational intervals meet the registered target. The stronger custom relaxation does not dominate the cheaper one at the test scale, and SCIP often delivers better native numerical performance. Those results identify value and cost of independently verifiable certification. They do not erase the original policy open positive-cost randomized intervals or the economic assumptions behind admissible optimization. The policy-class savings and the certification methods are substantive results toward their stated objects, not substitutes for unperformed calibration or unsupported claims of universal tractability.

node Q , the admissible distributions in each row form a compact simplex intersected by coordinate intervals. If the sum of lower nonreference bounds exceeds one, the row is empty and the interval is exact. Otherwise its lower corner is admissible. Given two row distributions p, q on $\mathcal{A}$,

$$V(p, q) = \frac{1}{2} \left(\sum_{a \neq a^*} |p_a - q_a| + \left| \sum_{a \neq a^*} (p_a - q_a) \right| \right) \leq \sum_{a \neq a^*} |p_a - q_a| \leq (m-1)w.$$

and is also at most one. It does not depend on which action was used as the reference. The $D_{t,Q}^{\min}$ by restricted Bellman minimization of nonnegative disadvantages d on those nodes, with zero terminal value. A minimizing action distribution can be selected at each node, and the collection of these distributions is one common Markov policy. Backward induction proves that its regret equals $D_{t,Q}^{\min}$ simultaneously at all state–date pairs. Likewise minimization of k produces $C_{t,Q}^{\min}$. The two minimizing policies can differ: no equality of choices is asserted or needed.

The minimum is rounded downward by at most q , the error relative to the exact restricted minimum at a date with h decisions remaining is at most qS_h . This follows from the stochastic bounds and the recursion $e_t \leq q + \beta e_{t+1}$. A surviving node has rounded minimum regret at every pair, hence $D_{t,Q}^{\min} \leq \varepsilon + \gamma_b$. Let p^c denote its lower corner. The policy perturbation gives

$$D_t^{p^c} \leq D_{t,Q}^{\min} + D_J \theta(w) \leq \varepsilon + \gamma_b + D_J \theta(w).$$

the initial-law average of the rounded minimum cost, then

$$\int C_0^{p^c} d\nu - \underline{c}_Q \leq D_C \theta(w) + \gamma_b.$$

Using p^c and using Theorem 1 gives $U_Q - \underline{c}_Q \leq \mathcal{H}(w, b)$. Raising the lower endpoint by the LP residual or inherited parent bound preserves this inequality. Additional valid nodes cannot discard an actually feasible point.

Since the incumbent is U and a node’s valid lower endpoint is L_Q . Once its corner has been evaluated, $U \leq U_Q$. If $\mathcal{H}(w, b) \leq \eta$, then $U - L_Q \leq \eta$ and the node need not be refined. Refinement and processing therefore cannot continue splitting such nodes while a positive gap larger than η remains. At least one widest-coordinate split in each block of eight levels reduces all widths below w_* after at most $8d \lceil \log_2(1/w_*) \rceil$ levels: after d widest splits the largest width is at most half its previous worst-case bound, and intervening splits only help. A full binary tree at finite depth bounds the number of needed evaluations.

For any fixed positive ε and $\beta < 1$, first choose b so that $\mathcal{H}(0, b) < \eta$, then choose $w_* > 0$ sufficiently small. Fixed b alone yields a nonzero sufficient numerical floor. The implemented 2^{-44} precision arithmetic is rational and valid, but the statement “every prescribed positive accuracy” is achieved only by this precision choice or a precision-refinement schedule. Completeness does not depend on obtaining exact LP optima, because the restricted Bellman lower bound and the repair suffice.

A.1. The deployed-probability grid and its repair guard

Prove Corollary 1. Use the original lower-corner policy p^c as the comparison, not its repair. The implemented repair works backwards. If future rows satisfy $0 \leq \widehat{D}_{t+1} \leq \varepsilon$, the

and $g_{ti} = \kappa \max_a |q_a - q_{a^*}|$ is at most $\mu < s$. Its target $\varepsilon - g_{ti}$ is strictly above q_{a^*} , so the back to the reference due to an inadequate target is not needed under this sufficient precision condition.

Mixing changes expected regret to its minimum with that target. Downward rounding of the 1 free probabilities moves total variation by at most κ and changes expected regret upward at most g_{ti} . The final regret therefore remains at most ε . It is also at most the proposal regret against the implemented future plus μ . Let $e_t = \sup_i (\widehat{D}_t(i) - D_t^{p^c}(i))_+$. Then $e_T = 0$ and

$$e_t \leq \mu + \beta e_{t+1} \leq \mu S_{T-t}.$$

An unmodified corner is feasible at allowance $\varepsilon + \delta$ by the preceding box argument. Consequently the excess of the current proposal regret over its guarded target is at most $\delta + \mu S_T = \Delta$. That excess is $u > 0$, the mixing fraction is

$$\frac{u}{u + \varepsilon - g_{ti} - q_{a^*}} \leq \frac{u}{u + s - \mu} \leq \frac{\Delta}{\Delta + s - \mu}.$$

The total row variation from p^c to the deployed policy is at most this fraction plus κ , and at most one. Its implementation-cost difference is bounded by D_C times that quantity. Adding the corner's $D_C \theta(w) + \gamma_{b_v}$ distance from the rounded restricted cost minimum proves (25). The proof explicitly accommodates regret increases caused by rounding; it does not assume the implemented rounding step is operating-value monotone.

For any fixed model and positive target, increasing b_p makes $\mu < s$ and sends both μ and κ to zero. Increasing b_v sends γ_{b_v} to zero. Hence $\mathcal{H}_Q(0, b_v, b_p)$ tends to zero, after which the same all-box and widest-split argument supplies a finite stopping depth. The fixed-grid sufficient bounds in the supplement are exact calculations on the serialized models, not claims of an efficient search complexity.

APPENDIX B: QUANTITATIVE REPAIR-DENSITY ARGUMENT

Let $p \in \mathcal{F}(\varepsilon)$, let q be any Borel proposal, and write $\widehat{q} = \mathcal{R}_\varepsilon q$. Define

$$E_t = \int \text{TV}(q_t, p_t) d\mu_t, \quad E_t = \int |J_t^{\widehat{q}} - J_t^p| d\mu_t, \quad M_t = RS_{T-t} + G\beta^{T-t}, \quad s = (1 - \beta)\varepsilon.$$

Since $E_T = 0$. The action values against any valid continuation are in $[-M_t, M_t]$. Subtracting the proposal Bellman value against the repaired continuation from J_t^p gives a term bounded by $M_t \text{TV}(q_t, p_t)$ and the continuation term $\beta P_t^{p_t} |J_{t+1}^{\widehat{q}} - J_{t+1}^p|$. The scalar repair replaces the prior proposal value by its maximum with $V_t - \varepsilon$. Since $J_t^p \geq V_t - \varepsilon$, this projection cannot increase its distance from J_t^p . Moreover, for any nonnegative f ,

$$\int P_t^{p_t} f d\mu_t \leq \sum_a \int P_t^a f d\mu_t = \int f d\mu_{t+1}.$$

Consequently

$$E_t \leq 2M_t a_t + \beta E_{t+1}. \quad (32)$$

The repair fraction at each point is at most the positive deficit divided by s , and that deficit is at most the absolute difference of the proposal value from J_t^p . The triangle inequality therefore

$$\int \text{TV}(\hat{q}_t, p_t) d\mu_t \leq a_t + \frac{2M_t a_t + \beta E_{t+1}}{s}. \quad (33)$$

equalities hold without a uniform approximation of p .

any policy u , its state laws η_t^u from ν satisfy $\eta_t^u \leq \mu_t$ by induction. The implementation difference-identity between $\hat{q}$ and p uses the state law of $\hat{q}$ and the implementation values of p . Those action values belong to $[0, KS_{T-t}]$. Hence

$$\left| \int (C_0^{\hat{q}} - C_0^p) d\nu \right| \leq \sum_{t=0}^{T-1} \beta^t KS_{T-t} \left(a_t + \frac{2M_t a_t + \beta E_{t+1}}{s} \right). \quad (34)$$

identity follows directly by telescoping the difference of the two cost Bellman equations; it impose an almost-sure operating constraint in place of the pointwise one.

ounded conditional expectations $\mathbb{E}_{\mu_t}[p_t(a \mid \cdot) \mid \Pi_N]$ converge in $L_1(\mu_t)$ on increasing partitions. If a cell has zero measure, choose an arbitrary probability vector there. At the partition round all but one probabilities down to multiples of an increasing denominator, adding the remainder to the last action. The resulting vectors remain probabilities, and the total-variation error tends uniformly to zero. Because each μ_t is finite, this supplies simple proposals q^N with every $a_t \rightarrow 0$.

ions (32)–(34) imply initial implementation-cost convergence. Meanwhile backward enforces $J_t^{\hat{q}^N}(x) \geq V_t(x) - \varepsilon$ for every x , including cells and singletons that have zero measure. For any $\zeta > 0$, choose a feasible p with cost less than $K_R + \zeta$ and then a simple proposal within ζ in cost. Since all members of the countable repaired class are its infimum equals (4). Enumeration gives the monotone exact upper sequence.

numerical use, the Bellman repair and integration must themselves be computable with errors tending to zero. This is an additional evaluation requirement, not implied merely by measurability. The finite-horizon rational piecewise-affine maintenance specification is a finite piecewise-rational construction as described in the main text. Its number of pieces grows rapidly. Neither the present proof nor the frozen finite experiments supply a uniform error bound for that construction or a two-sided convergence rate for its global constrained minimum.

APPENDIX C: CONDITIONAL INITIAL LAWS IN OBSERVABLE FIBERS

each label ζ , let the finite fiber use the primitives evaluated at $F_t(\zeta)$ and initial law $q(\zeta)$. The global policy restricts to a feasible common policy in every fiber. Its cost is therefore at most $v(\zeta, q(\zeta), \varepsilon)$ before integration.

For the reverse inequality, fix a denominator M . There are finitely many simplex-lattice policies for an n -state, T -date fiber. Their Bellman values are Borel in ζ , as finite sums and compositions of the Borel primitives. Their initial costs, weighted by the Borel vector $q(\zeta)$, are also Borel. Let $\theta_M = \min\{1, (m-1)/M\}$. Rounding an arbitrary feasible policy shows that the feasible set at allowance $\varepsilon + D_J \theta_M$ is nonempty. Its first cost minimizer under a Borel ordering is Borel: every selection event is a finite intersection of Borel comparison sets, and the feasibility comparisons at all finite state–date pairs.

Let the selected array by Theorem 1. Its dependence on ζ remains Borel, including at ties, since the first-index finite maximizers and division on a strictly positive denominator are Borel. The total cost is at most

$$v(\zeta, q(\zeta), \varepsilon) + D_C \{\theta_M + \rho_\varepsilon(D_J \theta_M)\},$$

er and upper endpoints. Integrating the selected array at date t using $F_t^{-1}(z)$ yields one Borel
kov policy, with no remembered latent label. Integrating and letting $M \rightarrow \infty$ proves (31).
or cell approximation, let a representative use q_c while the true conditional initial law obeys
 $\zeta \in B_c$, $\text{TV}(q(\zeta), q_c) \leq \delta_{q,c}$. Every feasible policy has statewise initial implementation values
 $[\delta_{q,c}, KS_T]$. Changing its initial regime law therefore changes cost by at most $KS_T \delta_{q,c}$. The
bound applies to the infimum because its all-restart feasibility set is independent of the
al law. It is added to each cell’s existing primitive/solver cost error. Alternatively one
integrate the representative policy’s statewise costs against the exact varying conditional
Independence between regime and label is a convenience of the earlier example, not an
mption needed for disintegration.

APPENDIX: REFERENCES

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1. READING CONVENTIONS AND PROVENANCE

This supplement accompanies revision R45. All numerical optimization outcomes below are from the preserved R43 development experiments or the R44 frozen global experiment. R45 contributes their integrated analysis, independent rechecking, new analytic results, and exact values of the stopping contract. It does not rebrand an already observed experiment as a new experimental sample.

SCIP denotes Bellman supports with disaggregated simplex-consistent products. A denotes the aggregate-product relaxation. A certified intersection takes the larger of the two independently checked lower bounds and the smallest independently checked feasible upper cost, also allowing for rationally repaired SCIP proposal. SCIP's native lower bound is not used in that intersection. Absolute widths are compared with the registered target 10^{-3} . Rounded table endpoints are non-intrusive; the exact fractions in the machine-readable results and proof objects determine validity and target status.

The global cases are twelve designed environments (four each in maintenance, inventory, and queueing) plus four perturbations of one near-tie construction. Forty-two additional scaling experiments are three model families, seven allowances, and two root formulations. They are not two independent environments. The historical controlled suite consists of eight cases and two root formulations. Its fiber calculation is one parametric family with seventeen endpoints.

1.1. *Designed family primitives*

Inventory has state $i \in \{0, \dots, n-1\}$, action $a \in \{0, \dots, m-1\}$, and demand shock $z \in \{0, 1, 2\}$. Set $o = \min\{a, n-1-i\}$, $b = i + o$, $s = \min\{b, z\}$, and $i' = b - s$. The stage operating payoff is $2s/n - 3o/(4n) - i'/(5n) - (z-b)_+/(2n)$. The installed action is $\min\{m - \max\{0, n/2 - i\}\}$ for the executed even state counts, the departure charge is $(1 + i/n)\mathbf{1}\{a \neq 0\}$, and terminal reward is $i/(4n)$.

Queueing uses $i' = \min\{n-1, (i-a)_+ + z\}$ and stage operating payoff $-i/n - a^2/(3m^2) - (a+z - (n-1))_+/n$. Its installed action is one below $n/2$ and $\min\{2, m-1\}$ otherwise; departure costs are again $1 + i/n$, and terminal reward is $-i/(2n)$. In both families the initial distribution is uniform. Shock probabilities are rational $(w_0, w_1, 8 - w_0 - w_1)/8$ with $w_0 \in \{1, 2\}$ and $w_1 \in \{3, 4\}$ chosen by the frozen seed. Maintenance uses the preserved controlled rational decision constructor and departure charges from R43; every resulting primitive is serialized in the frozen model files. All three families are known-model designs, not estimated data.

2. FULL GLOBAL RESULTS

Case	Lower	Upper	Width	Width/upper (%)	Target
M0	1.2487	1.2491	0.000497	0.0398	Yes
M1	4.8670	4.8741	0.007157	0.1468	No
M2	10.6217	10.6304	0.008723	0.0821	No
M3	16.2520	16.3659	0.1139	0.6957	No
I0	0	0	0	0	Yes
I1	4.5308	4.5486	0.0179	0.3925	No
I2	11.2217	11.2359	0.0142	0.1263	No
I3	15.5942	15.5988	0.004630	0.0297	No
Q0	0.6363	0.6363	6.129×10^{-12}	0.0000	Yes
Q1	3.7682	3.7692	0.000980	0.0260	Yes
Q2	7.5909	7.5920	0.001151	0.0152	No
Q3	11.6764	11.6771	0.000660	0.0057	Yes
T0	5.2227	5.2277	0.004995	0.0956	No
T1	5.2224	5.2277	0.005256	0.1005	No
T2	5.2272	5.2305	0.003211	0.0614	No
T3	8.3022	8.4367	0.1345	1.5941	No

TABLE II: Covering-search progress and stopping events

Case	Method	Nodes	Stop	Root width	Final width
M0	B	1	target	0.000497	0.000497
M0	A	13	target	0.005303	0.000567
M1	B	363	time_cap	0.015462	0.007157
M1	A	1125	time_cap	0.055623	0.022168
M2	B	45	time_cap	0.010616	0.008723
M2	A	209	time_cap	0.033659	0.025514
M3	B	7	time_cap	0.728567	0.728567
M3	A	13	time_cap	0.115013	0.115011
I0	B	1	target	0	0
I0	A	1	target	0	0
I1	B	239	time_cap	0.030765	0.017855
I1	A	911	time_cap	0.058425	0.026761
I2	B	25	time_cap	0.014884	0.014188
I2	A	209	time_cap	0.047759	0.034881
I3	B	7	time_cap	0.064000	0.064000
I3	A	13	time_cap	0.005123	0.004768
Q0	B	1	target	7.607×10^{-12}	7.607×10^{-12}
Q0	A	1	target	6.129×10^{-12}	6.129×10^{-12}
Q1	B	49	target	0.002272	0.000980
Q1	A	153	target	0.006964	0.000997
Q2	B	81	time_cap	0.002249	0.001151
Q2	A	331	time_cap	0.007350	0.004986
Q3	B	7	time_cap	0.032158	0.032158
Q3	A	1	target	0.000694	0.000694
T0	B	879	time_cap	0.029761	0.004995

T0	A	2047	node_cap	0.210087	0.160574
T1	B	789	time_cap	0.029761	0.005256
T1	A	2047	node_cap	0.210088	0.200032
T2	B	879	time_cap	0.029263	0.003211
T2	A	2047	node_cap	0.212848	0.167849
T3	B	1053	time_cap	0.176091	0.134494
T3	A	2047	node_cap	3.004802	1.360331

target stop certifies the stated numerical tolerance, not exact equality of the two endpoints. A or node stop preserves the surviving global cover. In particular, no numerical LP infeasibility alone is allowed to discard a policy box. Q1 closes after 49 B nodes and 153 A nodes. M0 es at B's root and after 13 A nodes. Several target hits, including the zero-cost I0 example, ur at the root. The largest cases in all three primary families favor A under the common get.

TABLE III: Search, LP, and independently checking work

Case	Method	Search (s)	LP (s)	Check (s)	Proof (KiB)
M0	B	0.100	0.016	0.077	2.7
M0	A	0.367	0.080	0.237	6.4
M1	B	120.396	40.424	80.470	4005.7
M1	A	120.223	23.048	77.318	2592.6
M2	B	120.178	83.605	34.056	2189.0
M2	A	120.679	46.165	58.524	7093.8
M3	B	128.174	87.564	27.164	2105.0
M3	A	124.350	90.002	18.922	4182.9
I0	B	0.082	0.012	0.065	0.8
I0	A	0.028	0.005	0.018	0.7
I1	B	120.582	53.186	69.526	1217.5
I1	A	120.108	27.079	74.702	1821.5
I2	B	120.592	96.763	20.994	352.6
I2	A	120.132	55.471	50.890	1689.6
I3	B	126.244	92.112	25.605	187.4
I3	A	120.886	94.694	13.401	568.2
Q0	B	0.067	0.011	0.054	1.4
Q0	A	0.021	0.004	0.012	1.3
Q1	B	25.720	11.935	13.877	272.0
Q1	A	18.778	3.902	11.701	206.2
Q2	B	121.181	69.953	51.340	519.5
Q2	A	120.178	51.433	53.891	673.5
Q3	B	127.730	85.776	30.474	51.5
Q3	A	16.055	7.098	1.456	18.3
T0	B	120.073	39.233	79.651	1309.3
T0	A	105.073	25.732	62.810	2957.8
T1	B	120.186	43.859	74.915	1263.4
T1	A	100.177	21.173	61.847	833.4
T2	B	120.067	43.136	75.872	1326.2
T2	A	100.465	23.915	59.707	2722.9

T3	B	120.118	53.741	64.360	1582.6
T3	A	70.653	18.852	40.171	3322.7

times above are the original per-case execution records, not a new benchmark on the machine. Search includes non-LP preparation and exact node operations. Check time is al. The nominal time budget is tested between proof-producing operations, so completion operation or sibling pair can exceed it. The JSON records retain the complete trace and d totals.

TABLE IV: Global root proposal status

Case	Method	LP status	Primal	Dual	Variables
M0	B	0	Yes	Yes	296
M0	A	0	Yes	Yes	160
M1	B	0	Yes	Yes	1496
M1	A	0	Yes	Yes	704
M2	B	0	Yes	Yes	4128
M2	A	0	Yes	Yes	2080
M3	B	1	No	No	17952
M3	A	0	Yes	Yes	8512
I0	B	0	Yes	Yes	278
I0	A	0	Yes	Yes	136
I1	B	0	Yes	Yes	1580
I1	A	0	Yes	Yes	648
I2	B	0	Yes	Yes	5088
I2	A	0	Yes	Yes	1900
I3	B	1	No	No	23656
I3	A	0	Yes	Yes	8140
Q0	B	0	Yes	Yes	278
Q0	A	0	Yes	Yes	124
Q1	B	0	Yes	Yes	1580
Q1	A	0	Yes	Yes	620
Q2	B	0	Yes	Yes	5088
Q2	A	0	Yes	Yes	1720
Q3	B	1	No	No	23904
Q3	A	0	Yes	Yes	7768
T0	B	0	Yes	Yes	538
T0	A	0	Yes	Yes	324
T1	B	0	Yes	Yes	538
T1	A	0	Yes	Yes	324
T2	B	0	Yes	Yes	538
T2	A	0	Yes	Yes	324
T3	B	0	Yes	Yes	538
T3	A	0	Yes	Yes	324

s zero denotes a returned successful numerical LP proposal; it does not certify the r problem. The full tree contains the rounded multipliers and their reconstructed outward

nd. No missing optimizer-specific scalar is invented to replace that calculation.

TABLE V: External SCIP baseline and the separate certified intersection

Case	SCIP status	Native width	Nodes	Total (s)	Certified width
M0	gaplimit	0.000497	1	0.11	0.000497
M1	gaplimit	0.000832	11	3.86	0.007157
M2	gaplimit	0.000835	1	12.96	0.008723
M3	timelimit	0.0520	1	120.52	0.1139
I0	optimal	0	0	0.05	0
I1	nodelimit	0.001326	2047	14.26	0.0179
I2	nodelimit	0.003895	2047	72.75	0.0142
I3	timelimit	0.001470	1	120.43	0.004630
Q0	gaplimit	7.000×10^{-9}	1	0.06	6.129×10^{-12}
Q1	gaplimit	0.000243	1	1.35	0.000980
Q2	gaplimit	2.670×10^{-5}	1	4.98	0.001151
Q3	gaplimit	0.000149	1	13.14	0.000660
T0	nodelimit	0.002574	2047	6.55	0.004995
T1	timelimit	0.002388	1359	120.01	0.005256
T2	nodelimit	0.002624	2047	6.17	0.003211
T3	nodelimit	0.1092	2047	10.85	0.1345

SCIP’s native lower endpoints are solver-reported floating-point quantities. The native target count is eight; the independently certified intersection count is five. Every SCIP proposed root policy is separately rationalized, repaired, and checked at all restarts. These distinct checks are not a certification of SCIP’s native lower endpoint. The native solver often supplies tighter bounds: the result is retained rather than filtered out. The adapter gives SCIP the same mon-policy equations, admissible root domains, Bellman supports, and valid strict-gap cuts.

3. ACTION GAPS, SUPPORT CONSTANTS, AND ARITHMETIC PRECISION

TABLE VI: Constants across all global cases

Case	Strict	$\underline{d}$	λ_-	λ_+	Exact-LP bound
M0	Yes	0.002320	-190.5950	767.8502	16.1024
M1	Yes	0.003793	-166.1942	183.0214	54.5307
M2	Yes	0.001033	-111.7540	1660.1008	162.9533
M3	Yes	5.146×10^{-5}	-25440.8333	1774.2602	420.1714
I0	No	0	-19.3268	0	–
I1	No	0	-37.9259	24.4222	–
I2	No	0	-19.6491	25.1542	–
I3	No	0	-42.1053	47.3684	–
Q0	Yes	0.001389	-1080.0000	27.0000	4.7987
Q1	Yes	0.0223	-61.5544	69.1200	18.1096
Q2	Yes	0.003125	-440.0000	560.0000	134.2563

Q3	Yes	0.008854	-162.3529	94.4762	35.9911
T0	No	0	0	8.0902	—
T1	Yes	1.000×10^{-8}	0	9.953×10^7	1.426×10^{12}
T2	Yes	0.000100	0	9952.9688	14318.5489
T3	Yes	0.0100	0	99.5297	49.7421

h is not a zero bound. If an operating tie exists, the inverse-gap theorem is inapplicable. The extrema shown in those rows describe the nonzero disadvantages only and do not form a cone for the zero-disadvantage actions. Bellman support envelopes and the policy-box theorem do not require strict separation. The first-index reference is computed in exact arithmetic for the stored primitives. Its identity can change at a tie; the reference-free expression remains valid on either side.

TABLE VII: Exact-repair precision contract evaluated on the frozen models

Case	D_C	D_J	$H(0, 44)$	Width level k	Depth bound
I0	14.255500	11.675775	2.787×10^{-9}	29	7424
I1	53.408547	17.098336	4.087×10^{-8}	32	32768
I2	155.828997	40.479846	3.967×10^{-7}	37	113664
I3	384.671359	71.774254	7.052×10^{-6}	41	503808
M0	16.292000	18.916400	3.185×10^{-9}	30	7680
M1	56.969117	62.276480	4.360×10^{-8}	34	34816
M2	166.217597	173.259624	4.232×10^{-7}	39	119808
M3	397.080112	403.278880	7.280×10^{-6}	44	540672
Q0	14.255500	18.260300	2.787×10^{-9}	29	7424
Q1	53.408547	62.931790	4.087×10^{-8}	34	34816
Q2	155.828997	182.767970	3.967×10^{-7}	39	119808
Q3	384.671359	452.526471	7.052×10^{-6}	44	540672
T0	49.847977	15.666507	3.815×10^{-8}	32	16384
T1	49.847977	15.666507	3.815×10^{-8}	32	16384
T2	49.847977	15.666507	3.815×10^{-8}	32	16384
T3	49.847977	15.666507	3.815×10^{-8}	32	16384

$q = 2^{-44}$, the arithmetic accumulation is $\gamma = q \sum_{s=0}^{T-1} \beta^s$, and H is the stopping bound in the main article. The width level means $w = 2^{-k}$, sufficient for $H(w, 44) \leq 10^{-3}$ using the model bounds. The depth bound is $8nT(m-1)k$, not an observed tree size or an complexity guarantee. These are exact-repair floors. They do not by themselves charge on probability grid or its repair guard; the separate table below does. A smaller arbitrary ϵ can require increasing both precisions. The exact fractions and inequalities are in `precision_contract.json`.

The independent modulus test constructs 256 rational boxes in a two-state, three-date, three-model with an exact operating tie. It separately evaluates restricted Bellman minima, the lower-corner policy, and verifies the stopping inequality with exact fractions. All 256 of the necessary tests and passed. This is finite property testing, not a machine-checked proof of the theorem. The mathematical argument is given in the main article’s Appendix A.

Case	$\max d$	$\mathcal{H}_Q(0, 44, 40)$	Prob. bits	Width level	Depth bound
I0	0.558594	5.343×10^{-8}	40	29	7424
I1	0.265971	4.013×10^{-7}	40	32	32768
I2	0.394211	7.995×10^{-6}	40	37	113664
I3	0.385802	0.000138	40	41	503808
M0	0.285941	3.327×10^{-8}	40	30	7680
M1	0.400005	6.150×10^{-7}	40	34	34816
M2	0.527327	1.123×10^{-5}	40	39	119808
M3	0.542239	0.000197	40	44	540672
Q0	0.629561	5.976×10^{-8}	40	29	7424
Q1	1.064878	1.446×10^{-6}	40	34	34816
Q2	0.932956	1.825×10^{-5}	40	39	119808
Q3	1.075412	0.000371	40	44	540672
T0	0.275000	3.855×10^{-7}	40	32	16384
T1	0.275000	3.855×10^{-7}	40	32	16384
T2	0.275000	3.855×10^{-7}	40	32	16384
T3	0.275000	3.855×10^{-7}	40	32	16384

the frozen node-bound precision is 44 bits, while the inherited repair routine floors nonreference action probabilities at 40 bits and assigns the residual mass to the operating reference. Exact guard is charged in Corollary 1 of the main article. All sixteen models satisfy the allary’s positive-slack condition and have sufficient two-grid floors below 10^{-3} ; the largest approximately 0.0003714041. The displayed width levels refer to this quantized repair, not to ideal repair. The worst-case depth is an analytic bound, not executed work. A second exact implementation verified this contract on the same 256 boxes used for the ideal-repair test, at aral value and probability precisions. No adaptive-precision search was executed.

4. COMPLETE ALLOWANCE DIAGNOSTICS

TABLE IX: Seven allowances in three families, with both root formulations

Family	ε	Unscaled width	Scaled width	Scaled/ ε^2	Exact-LP bound
M	10^{-1}	1.8633	1.8633	186.3299	1149.4310
M	10^{-2}	0.9721	0.9567	9567.2260	64.9860
M	10^{-3}	0.1961	0.0140	14048.4194	34.2205
M	10^{-4}	0.0187	0.000144	14406.2818	0.8230
M	10^{-5}	0.001858	1.444×10^{-6}	14444.6438	0.009579
M	10^{-6}	0.000186	2.113×10^{-8}	21125.6740	9.739×10^{-5}
M	10^{-7}	1.857×10^{-5}	8.648×10^{-10}	86480.7821	9.755×10^{-7}
I	10^{-1}	2.6080	2.6080	260.8030	—
I	10^{-2}	0.2168	0.2168	2168.1330	—
I	10^{-3}	0.0212	0.0212	21213.1715	—
I	10^{-4}	0.002117	0.002117	2.117×10^5	—
I	10^{-5}	0.000212	0.000212	2.116×10^6	—
I	10^{-6}	2.116×10^{-5}	2.116×10^{-5}	2.116×10^7	—
I	10^{-7}	2.116×10^{-6}	2.116×10^{-6}	2.116×10^8	—

Q	10^{-1}	0.6245	0.1050	10.5035	48.3406
Q	10^{-2}	0.0444	1.259×10^{-10}	1.259×10^{-6}	8.0827
Q	10^{-3}	0.004533	1.156×10^{-10}	0.000116	0.1854
Q	10^{-4}	0.000453	1.197×10^{-10}	0.0120	0.002130
Q	10^{-5}	4.525×10^{-5}	1.217×10^{-10}	1.2172	2.162×10^{-5}
Q	10^{-6}	5.3153	7.825×10^{-10}	782.5364	2.165×10^{-7}
Q	10^{-7}	4.527×10^{-7}	1.354×10^{-10}	13540.7430	2.166×10^{-9}

TABLE X: Complete strict-gap bound components for the scaled formulation

y	ε	$\bar{E}_D$	$E_{S,0}$	D_C	χ	ρ	Repair term
10^{-1}		0.9286	1092.7670	56.9691	1.0000	0.9946	56.6640
10^{-2}		0.009286	10.9277	56.9691	1.0000	0.9489	54.0583
10^{-3}	9.286×10^{-5}		0.1093	56.9691	0.9212	0.6500	34.1112
10^{-4}	9.286×10^{-7}		0.001093	56.9691	0.0921	0.1566	0.8220
10^{-5}	9.286×10^{-9}	1.093×10^{-5}		56.9691	0.009212	0.0182	0.009568
10^{-6}	9.286×10^{-11}	1.093×10^{-7}		56.9691	0.000921	0.001854	9.728×10^{-5}
10^{-7}	9.286×10^{-13}	1.093×10^{-9}		56.9691	9.212×10^{-5}	0.000186	9.744×10^{-7}
10^{-1}		—	—	—	—	—	—
10^{-2}		—	—	—	—	—	—
10^{-3}		—	—	—	—	—	—
10^{-4}		—	—	—	—	—	—
10^{-5}		—	—	—	—	—	—
10^{-6}		—	—	—	—	—	—
10^{-7}		—	—	—	—	—	—
10^{-1}		0.0843	3.5231	47.4743	1.0000	0.9440	44.8175
10^{-2}		0.000843	0.0352	47.4743	0.2700	0.6278	8.0475
10^{-3}	8.435×10^{-6}		0.000352	47.4743	0.0270	0.1443	0.1850
10^{-4}	8.435×10^{-8}	3.523×10^{-6}		47.4743	0.002700	0.0166	0.002126
10^{-5}	8.435×10^{-10}	3.523×10^{-8}		47.4743	0.000270	0.001684	2.159×10^{-5}
10^{-6}	8.435×10^{-12}	3.523×10^{-10}		47.4743	2.700×10^{-5}	0.000169	2.162×10^{-7}
10^{-7}	8.435×10^{-14}	3.523×10^{-12}		47.4743	2.700×10^{-6}	1.687×10^{-5}	2.162×10^{-9}

mean bound equals $E_{S,0} + D_C \chi \rho$ with $\rho = \bar{E}_D / ((1 - \beta)\varepsilon + \bar{E}_D)$. It describes an exactly relaxed solution and exact repair. The published numerical interval additionally includes proposal suboptimality, directed rounding, dual-residual corrections, and candidate selection. These are not generally identified by the numerical LP status. An independently computed numerical LP optimality-loss scalar is not available for these records; it is left unquantified and set to zero. The exact residual construction still establishes each reported endpoint.

Inventory models contain ties, so the strict-gap theorem does not apply and both root nodes coincide in the small-allowance sequence. Maintenance exhibits a roughly quadratic intermediate range followed by arithmetic effects. Queueing exhibits a small-width floor and an unscaled endpoint at allowance 10^{-6} . All points are retained. Descriptive secants are included in the JSON, but no common asymptotic exponent is estimated from these different

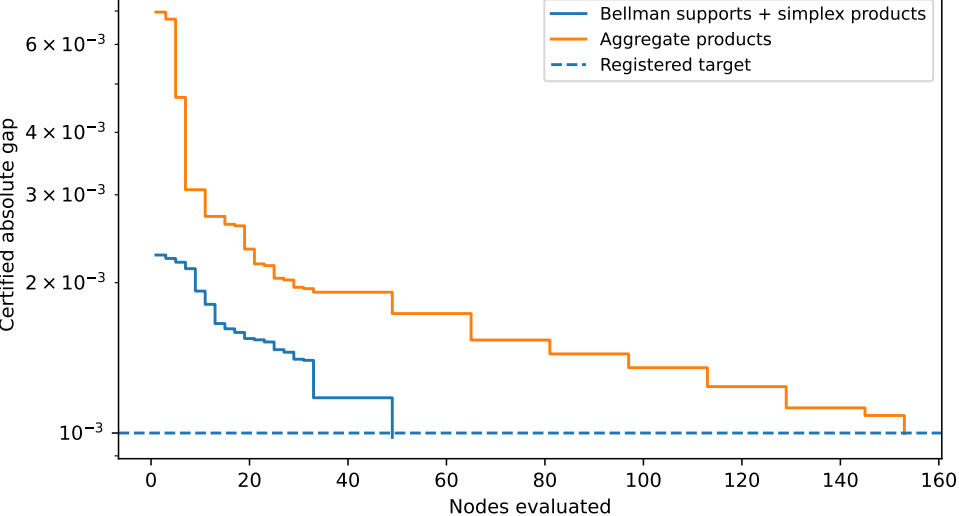


FIGURE 1.—Q1: full-cover certified gap versus nodes. The main article reports the corresponding time trace.

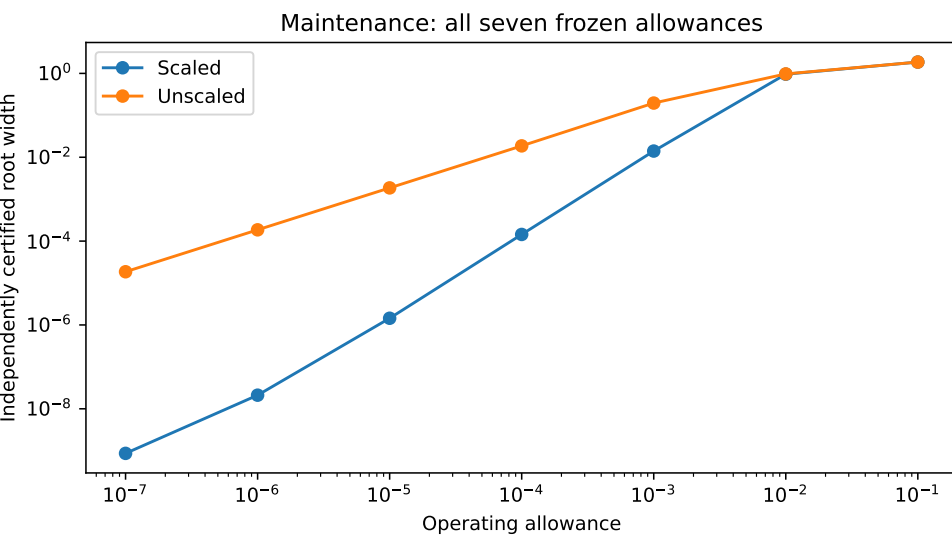


FIGURE 2.—Maintenance: every frozen allowance point. These are measured certificate widths, not the exact-LP em bound.

Inventory: all seven frozen allowances

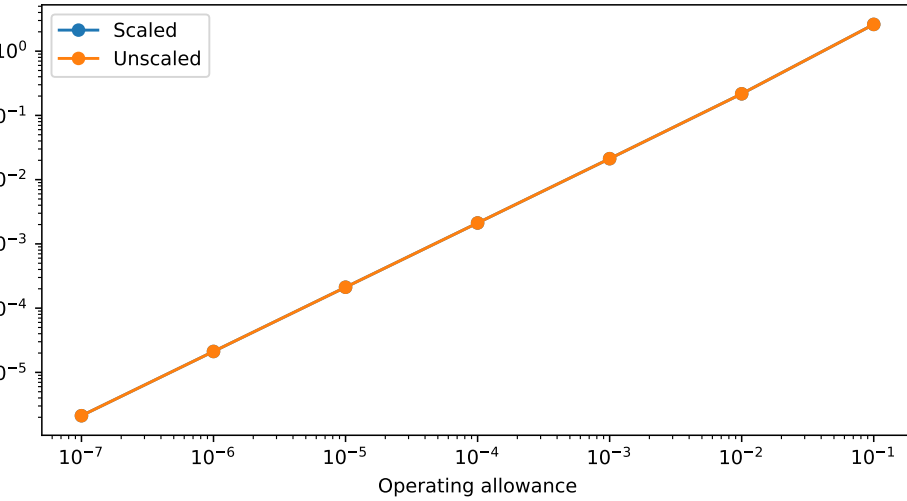


FIG. 3.—Inventory: every frozen allowance point. These are measured certificate widths, not the exact-LP bound.

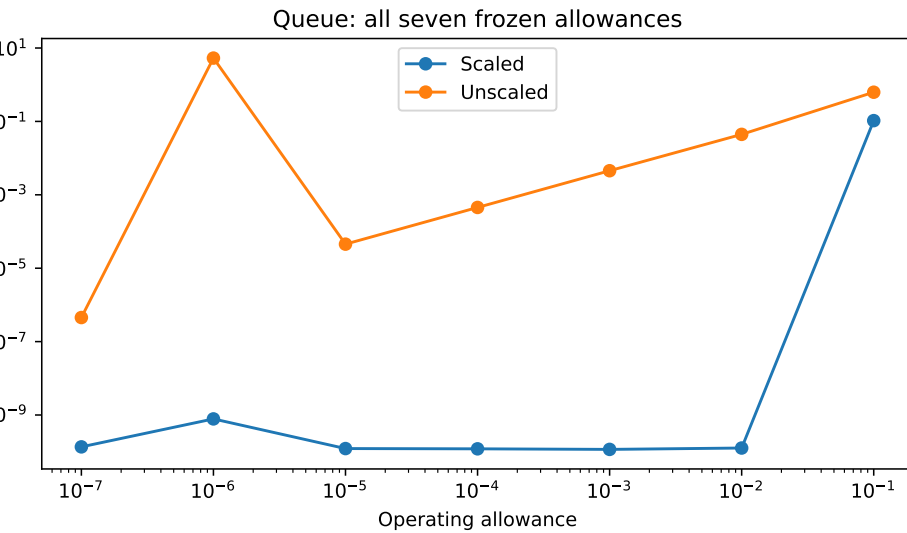


FIG. 4.—Queue: every frozen allowance point. These are measured certificate widths, not the exact-LP theorem

TABLE XI: Eight historical root pairs: individual widths and their best valid intersection

Case	Unscaled	Scaled	Intersection	Relative (%)	Target
00	0.0908	0.0591	0.0591	2.1807	No
01	0.3359	0.0684	0.0684	1.3056	No
02	0.5289	0.5289	0.5289	9.8157	No
03	1.8087	0.9000	0.9000	4.6349	No
04	0.1311	0.1014	0.1014	1.8478	No
05	0.3039	0.0452	0.0452	0.2781	No
06	28.6138	28.6138	28.6138	100.0000	No
07	15.7973	15.7973	15.7973	100.0000	No

TABLE XII: Historical individual root proposals, not global solutions

Case	Method	LP status	Lower	Upper	Width	LP (s)
00	unscaled	0	2.6281	2.7189	0.0908	0.006
00	scaled	0	2.6530	2.7121	0.0591	0.004
01	unscaled	0	4.9481	5.2840	0.3359	0.050
01	scaled	0	5.1676	5.2360	0.0684	0.072
02	unscaled	0	4.8596	5.3885	0.5289	0.340
02	scaled	0	4.8596	5.3885	0.5289	0.367
03	unscaled	0	17.9216	19.7303	1.8087	9.077
03	scaled	0	18.5169	19.4169	0.9000	27.514
04	unscaled	0	5.3647	5.4959	0.1311	2.182
04	scaled	0	5.3844	5.4858	0.1014	2.032
05	unscaled	0	15.9744	16.2783	0.3039	2.328
05	scaled	0	16.2134	16.2586	0.0452	1.935
06	unscaled	1	0	28.6138	28.6138	30.052
06	scaled	1	0	28.6138	28.6138	30.052
07	unscaled	1	0	15.7973	15.7973	30.078
07	scaled	1	0	15.7973	15.7973	30.077

TABLE XIII: Historical action gaps and savings-ratio extrema

Case	$\underline{d}$	λ_-	λ_+
00	0.003709	-63.6061	539.1951
01	0.000610	-1120.4240	75.1573
02	0.000116	-11155.2283	1228.3552
03	0.000331	-4632.9338	421.3481
04	2.203×10^{-5}	-9787.4264	75078.7316
05	0.000156	-268.5077	10054.3144
06	6.200×10^{-5}	-901.1248	29032.7987
07	7.439×10^{-5}	-22114.4443	3180.9620

proposal in either formulation, using zero dual multipliers and the operating-optimal ϵ for a valid but uninformative $[0, U]$ interval. The other six pairs returned numerical s 's, not solutions of the constrained nonlinear problem. The original protocol amendment and the original observations is retained in the R43 history and is not relabeled as a code-frozen observation for future evaluation.

6. ORIGINAL RANDOMIZED SAVINGS AND CONTINUUM EVIDENCE

Following signed savings are $s = L_D - U_R$, where L_D bounds every feasible deterministic cost and U_R is a feasible randomized cost. A positive value proves a policy-class separation; a negative value does not prove that deterministic policies are better. The percentages are computed from the preserved six-decimal display inputs: $100s/(U_R + s)$ and $100s/U_R$. They are approximate display summaries, not new exact certificates. All forty-two rows are retained, including twelve zero-cost rows and three negative signed comparisons. Exactly twenty-seven rows show positive signed savings.

TABLE XIV: Every original signed savings comparison, with normalization

T	Installed	ϵ	U_R	s	$100s/L_D$	$100s/U_R$
4	defer	0.01	4.730105	0.206584	4.18	4.37
4	defer	0.05	3.890120	1.041948	21.13	26.78
4	neural31001	0.01	0.089277	0.021898	19.70	24.53
4	neural31001	0.05	0.030155	0.044177	59.43	146.50
4	neural31002	0.01	0.125168	0.024289	16.25	19.41
4	neural31002	0.05	0.054026	0.058588	52.03	108.44
4	neural31003	0.01	0.099085	0.022055	18.21	22.26
4	neural31003	0.05	0.037227	0.047069	55.84	126.44
4	occupancy_stress	0.01	4.020718	0.186083	4.42	4.63
4	occupancy_stress	0.05	3.248042	0.941433	22.47	28.98
4	spline129	0.01	0.000000	0.000000	—	—
4	spline129	0.05	0.000000	0.000000	—	—
4	spline33	0.01	0.000000	0.000000	—	—
4	spline33	0.05	0.000000	0.000000	—	—
8	defer	0.01	10.637940	0.172786	1.60	1.62
8	defer	0.05	9.854394	0.896320	8.34	9.10
8	neural31001	0.01	0.402312	0.066330	14.15	16.49
8	neural31001	0.05	0.140458	-0.015946	-12.81	-11.35
8	neural31002	0.01	0.277716	0.058701	17.45	21.14
8	neural31002	0.05	0.059511	-0.015198	-34.30	-25.54
8	neural31003	0.01	0.653902	0.211403	24.43	32.33
8	neural31003	0.05	0.100439	0.244576	70.89	243.51
8	occupancy_stress	0.01	9.365726	0.125235	1.32	1.34
8	occupancy_stress	0.05	8.565982	0.682582	7.38	7.97
8	spline129	0.01	0.000000	0.000000	—	—
8	spline129	0.05	0.000000	0.000000	—	—
8	spline33	0.01	0.000000	0.000000	—	—
8	spline33	0.05	0.000000	0.000000	—	—
12	defer	0.01	15.466905	0.145915	0.93	0.94
12	defer	0.05	14.724122	0.747177	4.83	5.07
12	neural31001	0.01	3.360150	0.133784	3.83	3.98
12	neural31001	0.05	2.023532	0.379048	15.78	18.73
12	neural31002	0.01	2.385046	0.151782	5.98	6.36

12	neural31002	0.05	1.455145	0.309646	17.55	21.28
12	neural31003	0.01	1.010276	0.217318	17.70	21.51
12	neural31003	0.05	0.142107	-0.001624	-1.16	-1.14
12	occupancy_stress	0.01	13.655059	0.044193	0.32	0.32
12	occupancy_stress	0.05	12.877517	0.365025	2.76	2.83
12	spline129	0.01	0.000000	0.000000	—	—
12	spline129	0.05	0.000000	0.000000	—	—
12	spline33	0.01	0.000000	0.000000	—	—
12	spline33	0.05	0.000000	0.000000	—	—

a feasible lottery costs U_R and deterministic costs are at least L_D , an additional nonnegative implementation charge bounded by F preserves the strict comparison whenever $F < L_D - U_R$. For a stagewise charge bounded by f , the sufficient condition is $f \sum_{t=0}^{T-1} \beta^t < L_D - U_R$. These are conditional governance budgets in the model's implementation units. No empirical administration-cost estimate or institutional permission to randomize is inferred. The maximum relative positive-cost randomized widths remain 14.61%, 32.29%, and 34.23% for horizons 4, 8, and 12. Their medians are 5.24%, 14.33%, and 15.75%. The repair-density program addresses the same controlled Borel object, but has not been executed here as a new optimizer of these thirty positive-cost rows.

TABLE XV

RETAINED NONLINEAR TWO-STATE INTERVALS AT ALLOWANCE 0.5

T	N	Lower	Upper	Width/upper (%)	Independent check
8	64	0.9438	3.9348	76.01	Yes
16	128	4.1908	7.6644	45.32	No
32	64	6.5389	16.7884	61.05	Yes
32	128	8.1264	12.5226	35.11	Yes (R40)
32	256	8.8718	11.8492	25.13	Yes (R41)

At allowance 0.05 the retained exercise is lower-only: no candidate has a matching all-state feasible upper certificate. The available lower-only summary does not identify every failure as either true policy infeasibility or enclosure conservatism. No such diagnosis is manufactured. The distinction requires an additional candidate-specific certificate or a witnessed violation, neither of which is newly supplied here.

The observable-fiber example remains one fixed-covariate family. Its finest, sixteen-cell interval is approximately $[10.074215, 10.184476]$, with width about 0.110261. Approximately 4982 comes from finite endpoint gaps and 0.035279 from partition/value variation. More partition cells cannot remove the former. The new conditional initial-law argument and full selection proof extend the analytic identity, but do not constitute a new nontrivial-covariate experiment. Exact exogenous service-economy examples are retained as aggregation tests with economically ancillary continuous coordinates.

6.1. Nested fiber work and primitive errors

The cell reference is its left endpoint on $z \in [0, 1]$. In the original example $r_z = (1 + z)r$, $g_z = (1 + z)g$, $k_z = (1 + z/2)k$, and the finite regime matrix is independent of z . For M equal cells, valid uniform primitive errors are $d_r = \max |r|/M$, $d_k = \max k/(2M)$, $d_g = \max |g|/M$,

ase model. They are generic error inputs, not the sharper monotonicity partition term obtain the interval above.

TABLE XVI: Cellwise primitive errors in the original fiber example

Cells	d_r	d_k	d_g	α
2	0.50000000	0.50000000	0.25000000	0
4	0.25000000	0.25000000	0.12500000	0
8	0.12500000	0.12500000	0.06250000	0
16	0.06250000	0.06250000	0.03125000	0

TABLE XVII: Nested fiber width and the actual needed endpoint work

Cells	Endpoints	Width	Finite part	Partition part	LP (s)	Build/arith. (s)	Check (s)
2	3	0.342810	0.056917	0.285893	0.0762	0.1548	0.1625
4	5	0.208049	0.066491	0.141558	0.1291	0.2591	0.2708
8	9	0.142796	0.072193	0.070603	0.2293	0.4684	0.4854
16	17	0.110260	0.074981	0.035279	0.4326	0.8939	0.9152

work sums the unique stored endpoints needed for each nested partition. These are not shly executed mesh experiments, and their times must not be added as independent rk. The finest construction uses seventeen endpoint solves. No tighter finite search or ity label evolution is newly executed in this calculation.

7. VERIFICATION, REPRODUCIBILITY, AND ANALYTIC CONTRACTS

resent recheck covers all 32 complete custom trees, all 16 repaired SCIP candidate and all 42 scaling root proofs. All 90 passed. Eleven frozen scientific-source identities en primitive identities were also matched. The local interpreter is Python 3.13.5; these s do not remeasure the original solver benchmark on Python 3.13.15. The global replay an resume previously completed checks, so its coordinator elapsed time must not be ted as a fresh single-run timing of every object.

rchived adversarial suite is kept separately from the new stopping-modulus tests. It mutated proof structures and selected support identities. Source and primitive hashes identity, not correctness. Arithmetic readers rebuild finite recursions, constraints, cover e, and residual bounds; they do not machine-prove measurable selection, density, ana- gregation, or the economic assumptions. The machine-readable `ASSUMPTIONS.json` these boundaries for every headline result.

generate the tables without an optimizer, run `replication/tabulate.py` R45 directory using the supplied workspace. To replay certificates, run `replication/recheck.py` and `replication/recheck_scaling.py`. To ce the exact new property test and precision calculations, run `replication/chec rch_modulus.py`. The plotting script uses only the archived traces and scaling rows. mand paths are supplied in the root review guide.

the principal article contains the model, central results, complete proofs of the new stopping density results, the global comparison, and the economic conclusions. This numerical supplement contains all newly tabulated rows, including unsuccessful outcomes. The response document addresses all seventeen blocking findings and all thirty additional comments. The historical volume reproduces the full R43 article and supplement together with the inherited R42 historical volume. Their original PDFs and source files remain available as distinct unchanged

. The preservation manifest records 461 inherited files in the supplied working package. The proposed repository update is add-only under a new R45 directory and new R45 entry points. It neither deletes other branches nor rewrites the larger remote historical archive. The local stage is not advertised as a new complete clone of every historical branch or as containing the entire remote 349-page annex. Unmaterialized remote history remains untouched by the only publication procedure.

1. REVISION AND EVIDENCE BASIS

We thank the referee for identifying the difference between valid endpoint certificates and effective global method for the economic object. We have retained the unrestricted commonomized Markov-policy target, strengthened the analytic construction, and rebuilt the article and the certification argument. The revision does not respond by replacing the original rolled-state problem with an exogenous or observable-type problem.

This response addresses the report dated September 25, 2026, at review commit 554cae0c865498cf3fd6def18a80b75a1ee1abb3, on manuscript commit 60fd5b5f37906920511a1c62fc20300c886c3. It also incorporates the newer en R44 sources and outcomes already present in the repository and its workflow artifacts. source freeze is 88314815c10f197e6cac10fc503658dbdbe16bd4; the execution workflow run 36093502294. These experiments are not represented as newly designed R45 outs after observation.

The principal additions are a gap-free Bellman support construction, a precision-dependent global stopping proof, and a repair-density theorem for unrestricted controlled Borel cics. The latter supplies an upper-approximation mechanism, not a paired convergence rem. The global evaluation now includes complete three- and four-action trees and a SCIP parison. We independently replayed 90 preserved certificate objects and added 256 exact ping-modulus tests. All passed their stated checks. The historical paper and all inherited are preserved separately.

The requested accuracy on the original thirty positive-cost continuum cases, a new nonlinear ed computation, and a difficult approximate operating-oracle experiment have not been vered. These are stated explicitly below, rather than converting a theorem, a small example, n arithmetic replay into evidence for an unperformed computation. The document therefore nguishes a response to each finding from a claim that every requested empirical extension een completed.

2. BLOCKING FINDINGS

2.1. *R43-F1: The original controlled atomic problem*

Referee concern. The report asks for a validated convergent scheme for the original action-dependent condition process, rather than a substitution of aggregation or observable-type models.

Revision. The unrestricted Borel common-policy target and the original maintenance cohort ain in the article. Theorem 4 and Appendix B supply a new repair-density result for arbitrary controlled Borel kernels. The action-summed measures dominate every policy occupancy; onal simple proposals approximate a feasible policy in those measures; backward repair ores every restart inequality, including measure-zero states. The repaired implementation es converge to the unrestricted infimum. For finite affine atomic successors and finitely fied incumbent regions, the repaired operating values are piecewise affine and the probabies and cost values are piecewise rational, yielding an atom-preserving upper-evaluation e. Section 4.2 also supplies reference-free Bellman support lower witnesses evaluated at the inal successor maps.

at a paired convergence theorem. A finite support-price portfolio is not asserted to be a dual. The thirty original positive-cost randomized intervals were not newly optimized and remain open. The requested closure of a substantial part of that cohort is therefore not achieved by this revision.

References: Main 4.1–4.2, Theorem 4 and Appendix B; Main 6; Supplement 6.

2.2. R43-F2: Actual complete search rather than root-only ablation

Issue concern. The report requests a global algorithm using the strengthened relaxation, node widths, time, prescribed targets, and successful nonbinary cases.

Resolution. The article now incorporates all frozen covering trees in the newer repository of sixteen cases under Bellman-support/disaggregated search (B) and aggregate-product search (A). Every node shares the same policy variables across restarts, keeps a rational lower bound, and evaluates a repaired lower-corner policy independently of LP success. At the end of the run with absolute width 0.001, B closes four cases and A closes five. Q1 is a genuine non-root node after 49 B nodes and 153 A nodes; M0 additionally closes after 13 A nodes. All cases are solved in three or four actions. The main table includes every case, with node/time traces for Q1; Supplement 6 gives all root widths, final widths, node counts, stop reasons, proof sizes, and running times.

Comments of the response. Several hits occur at the root or have zero cost, and the larger cases are generally close. These limitations are recorded rather than treating every target hit as a branching success. B does not uniformly outperform A.

References: Main 3.3 and 5.1–5.2; Supplement 2.

2.3. R43-F3: Conditionality of the quadratic theorem

Issue concern. The fixed-model inverse-gap rate must not be interpreted as a uniform optimality theorem when beta, horizon, or action separation changes.

Resolution. The strict-gap theorem remains explicitly a specialization in Section 3.2, with the advantage, ratio-spread, accumulated product error, rare-departure, and repair constants. It achieves an exactly optimizing LP point. Theorem 2 instead constructs Bellman support bounds without dividing by an action gap. Proposition 1 permits an affine intercept for near-optimal effects. Theorem 3 proves a separate stopping bound $H(w,b)$ depending on policy-box width and arithmetic precision, without a minimum action gap. This establishes validity and model completeness at ties while preserving the useful strictly separated result.

Comments of the response. The gap-free bound is conservative and exponential in the number of policy coordinates. It is not a uniform efficient complexity statement for state refinements as unit factors approaching one.

References: Main 3.1–3.3 and Appendix A; Supplement 3.

2.4. R43-F4: Complete allowance diagnostics

Issue concern. The report asks for more allowances, more models, complete bound diagnostics, and a distinction between theoretical and implemented widths.

queueing, seven allowances from 0.1 to 0.0000001, and two formulations. The tables each measured width, width divided by allowance squared, and the exact-LP theoretical and where its assumptions hold. Companion tables report accumulated operating and savings, implementation perturbation constant, probability cap, repair fraction, and repair term. Rate plots retain every point, including the unfavorable unscaled queueing result at 0.000001. Maintenance has an intermediate near-quadratic range; the tied inventory family and queueing metric floor provide distinct counterexamples to a universal empirical slope.

Scope of the response. An independently certified LP optimality-loss scalar was not recorded for these numerical proposals. It remains explicitly unquantified rather than being set to zero and on solver status. Direct rational residual checking establishes the endpoints, but does not identify every contribution to the distance from the exact LP optimum. Thus this particular nested decomposition remains incomplete.

Locations: Main 3.2 and 5.4; Supplement 4; `scaling_retabulation.json`.

2.5. R43-F5: Near ties and switching limits

Referee concern. Action gaps can vanish or reference selectors can change under economic perturbations and state refinement.

Revision. Four frozen controlled near-tie variants use perturbations 0, 0.00000001, 0.0001, 0.01 with different transition and implementation consequences. Their complete trees, rates to meet the target, and external comparisons are retained. The new support representation uses an arbitrary bounded potential, and its cost lower expression eliminates that potential. Section 4.2 bounds its change through operating and support-value witness errors, not through identity of an argmax selector. Proposition 1 separates near-tie costs with an affine intercept, the finite stopping theorem avoids inverse gaps entirely.

Scope of the response. This gives a robust analytic alternative, not a demonstrated uniform ϵ -resolution rate. The four tie variants are related sensitivity cases, not independent economic cases, and none reaches the registered target.

Locations: Main 3.1, 3.3, 4.2, 5.3; Supplement 3.

2.6. R43-F6: Continuous coupling beyond recoverable labels

Referee concern. The observable-fiber theorem removes cross-label coupling, whereas the activating actions move the continuous state.

Revision. The revised article now distinguishes two constructions rather than extending a label beyond its assumptions. The repair-density theorem allows arbitrary action-dependent kernel kernels, noninvertible maps, and stochastic innovations, and the support witnesses are equated on the same original controlled kernel. Neither construction assumes an autonomous variate. The observable-fiber identity remains useful under its actual bijection assumptions and is strengthened by allowing the initial regime distribution to depend measurably on the observed label, with a complete Borel-selection proof.

Scope of the response. The general controlled-state upper approximation and global lower bounds have not been proved to converge together. The numerical fiber example remains a bivariate example; it is not relabeled as an interacting continuous-state computation.

Locations: Main 4 and Appendices B–C.

concern. Partition refinement alone cannot remove the residual finite root gaps; joint refinement and multiple covariate dynamics are requested.

on. The retained fiber certificate is presented with its two distinct width components, rather than a single apparently convergent mesh error. At sixteen cells its width is about 0.035279; approximately 0.074982 from finite endpoint gaps and 0.035279 from partition/value gaps. The complete nested two-, four-, eight-, and sixteen-cell accounting is preserved, with endpoint work reported in the supplement. The extended conditional-law identity and its bound state an additional term that any joint refinement must control.

of the response. No new jointly tightened finite/fiber run, multidimensional label evolution, or nontrivial autonomous label evolution was executed here. Those requested conditional extensions remain absent. New finite trees in different environments do not exist for them.

as: Main 4.3 and 6.2; Supplement 6; inherited controlled_fibers.tex.

2.8. R43-F8: Economic role of the aggregation examples

concern. The service examples have continuous coordinates that do not enter economically relevant primitives.

on. The article now identifies those examples explicitly as exact exogenous-optimization tests with economically ancillary continuous coordinates. Rewards, implementation and regime transitions are finite-regime objects, so the optimization reduces to an ordinary finite linear program. The original exactness proofs, endpoint certificates, and perturbation calculations are retained without using their continuous-coordinate countenance of high-dimensional controlled optimization. The controlled maintenance object is separately defined and visible.

of the response. The exact aggregation examples are not promoted into a calibrated condition or a benchmark for action-dependent continuous condition dynamics.

as: Main 4.3 and 6; Supplement 6; historical volume.

2.9. R43-F9: Completeness versus computational usefulness

concern. The report asks for global progress and larger problems rather than a minimization theorem presented as practical performance.

on. Complete nonbinary trees, all unsuccessful stops, proof-producing work, and the baseline now accompany the theorem. The stopping proof includes an additional issue resolved by a formal exact-arithmetic statement: fixed denominator 2^{44} leaves a positive integer arithmetic floor. Theorem 3 states the precision-dependent $H(w,b)$, and Appendix A through restricted Bellman minima and lower-corner repair. Sixteen model-specific precision calculations show sufficient widths at the registered target; the associated worst-case depth bounds are reported as analytic bounds, never as executed depths. A new exact 256-box test checks the proposed inequality at ties.

the fact that A beats B in each largest family member remain in the main comparison. No nominal or generally scalable solver claim is made.

utions: Main 3.3, 5, and Appendix A; Supplement 2–3.

The final implementation audit also separates the 40-bit deployed-probability grid and its tie guard from 44-bit node arithmetic. Corollary 1 and Appendix A charge both in the element-repair stopping bound. A second exact implementation passes on the same 256 cases; these are not additional independent environments. All sixteen sufficient frozen-grids remain below 0.001.

2.10. *R43-F10: External global-optimization comparison*

referee concern. Internal U-versus-S ablation is insufficient as a competitive baseline.

revision. The complete frozen experiment includes SCIP on the same rational common-objective and constraints, root domains, Bellman support information, tolerance, seed, one-thread setting. Per-case comparisons use the same runner. SCIP receives the strengthening information rather than being artificially weakened. Native status, gap, nodes, and times are reported alongside independently repaired candidate costs. SCIP reports eight native numerical target hits and frequently tighter native lower bounds. Every proposed upper policy is independently checked, and the best valid intersection is displayed.

scope of the response. SCIP native nonlinear lower bounds are not independently rationally verified by this adapter. They are identified as numerical baseline output, not discarded, but are inserted into the rigorous intersection. The custom method does not claim universal speed superiority or that SCIP lacks other verified-solving facilities.

utions: Main 5.1–5.2; Supplement 2.

2.11. *R43-F11: Prospective identity and environment counts*

referee concern. The prior arithmetic amendment and repeated endpoint objects must not be presented as untouched independent environments.

revision. The R43 amendment after partial observation remains documented. The newer R44 situation has a pre-execution scientific-source freeze and primitive manifest; the present replay checked all eleven scientific source identities and all sixteen model identities. The response article explicitly identifies the results as inherited frozen executions, not as R45 experiments proposed after seeing them. The design is twelve family cases and four tie variants; the advance exercise has three model families, seven allowances, and two formulations. The older objects are decomposed into 16 paired finite roots, 8 allowance roots, and 17 endpoints of the fiber family.

scope of the response. Source freezing establishes prospective identity, not statistical independence, correctness, or generalization to an empirical population. The three families are designed known-model environments.

utions: Main 5.1 and 7; Supplement 1, 7; PROVENANCE.json.

2.12. *R43-F12: Difficult approximate operating oracle*

referee concern. The experiments use exact finite operating dynamic programming and do not charge a difficult approximate operating solution.

alized-support witness errors, expectation-error propagation, and the need to charge policy repair as well as lower support witnesses. Atom displacement is handled by test-expectation bounds where justified, not an invalid total-variation claim. These formulas are missing input contract concrete for a difficult operating problem.

of the response. No new high-dimensional approximate-oracle experiment, action-time charge, or informative end-to-end interval with that bottleneck was added. The frozen global evidence isolates policy optimization with exact operating dynamic timing. It does not answer this empirical request.

as: Main 4.2; ASSUMPTIONS.json; retained witness-driven theorem.

2.13. *R43-F13: Nonlinear two-state optimality evidence*

concern. The nonlinear intervals are wide and the tighter-allowance exercise lacks a upper certificate.

ion. All selected nonlinear rows remain visible, with explicit independent-check indicator. The finest retained interval at allowance 0.5 is approximately [8.8718,11.8492], a relative 25.13 percent; the other wide rows are not hidden. The allowance-0.05 exercise remains fully. The new repair-density argument is applicable at the abstract controlled-kernel level, not presented as an executed box-solver closure or a proof of convergence of the historical boxes.

of the response. No new paired nonlinear certificate or nontrivial improvement of numerical interval was executed. Available summary records alone do not support every candidate failure to true infeasibility versus enclosure conservatism; the response invent that diagnosis.

as: Main 6; Supplement 6; historical technical evidence.

2.14. *R43-F14: Economic interpretation and administrative costs*

concern. The methods evidence is synthetic, without calibration, monetary welfare, institutional model of randomized implementation.

ion. The economic target is kept intact and stated before the numerical construction. It distinguishes departures from an installed rule, physical action switching, and fixed administration. It states that the Markov lottery is redrawn conditional on the observed date, without hidden persistent types. For each positive signed deterministic-versus-sized separation, an extra bounded administration charge preserves the comparison if it is more than the certified saving. Both percentage normalizations of all original signed comparisons are supplied, explicitly labeled as display approximations. Three different controlled families and an external solver improve the methods evidence.

of the response. The models remain designed and known, not estimated or calibrated. Monetary interpretation, sampling uncertainty, or empirical claim of lottery feasibility is not created. The conditional administration budget is a sensitivity result, not a measured implementation cost.

as: Main 2 and 6; Supplement 6.

referee concern. The theorem-level contribution must be distinguished from classical Cormick inequalities, dynamic programming, and ordinary LP reductions.

revision. The principal article is reorganized around one common-policy certification theorem: repair, gap-free Bellman supports, and a precision-dependent global stopping theorem. A new controlled-Borel repair-density theorem makes a separate exact statement about upper-policy approximation and is proved in full. The strictly separated quadratic bound is retained as a specialization, not an unqualified headline. The literature discussion identifies constrained-LP compatibility across restarts, policy recovery, weak dual bounds, product relaxations, and dependent verification, and expressly disclaims novelty for their established ingredients. The new comparison supplies a direct strong computational alternative.

scope of the response. The manuscript does not claim a priority result over every possible formulation in parametric nonconvex control. The mathematical statements, assumptions, and comparison objects are made explicit for referee evaluation.

mutations: Main 1.1, 3–4, Appendices A–C; references.

2.16. R43-F16: What independent checking does and does not establish

referee concern. Finite arithmetic replays and a few mutations do not validate the analytic conditions or economic primitives.

revision. All 32 complete trees, 16 repaired external policies, and 42 scaling roots were replayed independently of numerical optimization, with all 90 passing. The frozen adversarial game has 17 rejected mutations, including altered discounts and disadvantages, changed labels, transformed boxes, forged policies, and incomplete or cyclic covers. Its finite transformation checks cover 145 policies and 19,720 support inequalities. R45 adds a separate exact stopping-mulus implementation on 256 tied three-action boxes, plus 16 precision-budget calculations. MUTATIONS.json maps each headline theorem to its policy class, primitive assumptions, convergence type, and actual executable coverage.

scope of the response. These finite tests and replays are not a machine-checked proof of optimality, measurable selection, aggregation, or economic appropriateness. Fiber-weight mutation coverage is not claimed where no new test was executed. General analytic validity is supported by the supplied mathematical proofs, not by a proof-object count.

mutations: Main 7; Supplement 7; independent_recheck.json; scaling_recheck.json; check_modulus_tests.json.

2.17. R43-F17: Coherent article with preservation of research history

referee concern. The report asks for a focused article rather than an omnibus of separate articles and revision history.

revision. The main manuscript now develops the economic object and the central certification chain, keeps all new proofs needed for that chain in its own appendices, and presents the complete global comparison and the original economic conclusions. The numerical supplement supplies exhaustive tables and implementation accounting. A separate historical volume reproduces the previous full article, previous full supplement, and inherited R42 historical volume. Earlier drafts, examples, failures, and source files are not silently deleted. The publication overlay adds new files only, and the inherited-file manifest checks byte preservation.

belong in the current journal article. The local bundle does not claim to contain every
ialized remote historical annex; the proposed add-only update leaves them untouched.
as: Main complete; SUPP_R45; HISTORY_R45; preservation manifest.

3. ADDITIONAL TECHNICAL AND PRESENTATION COMMENTS

3.1. *Comment 1: Abstract failure counts*

Abstract now states both the new certified and native target counts and the earlier eight-case
ure record, including two fallback intervals.

3.2. *Comment 2: Absolute and relative widths*

main global table reports absolute widths and targets; the supplement reports absolute
tive intersection widths and retains individual historical root rows.

3.3. *Comment 3: Best valid intersection*

xteen new global cases and eight historical root pairs have an intersection column. Only
dently valid lower bounds and feasible upper policies enter it.

3.4. *Comment 4: LP status, availability, and fallback*

new global root table gives status, primal and dual availability, and dimension. The full
ain the exact multiplier/residual certificate. Old cases 06 and 07 are explicitly identified
multiplier operating-reference fallbacks. Missing optimizer-specific decomposition fields
fabricated.

3.5. *Comment 5: Use of the word solved*

numerical LP return is called a proposal; a constrained problem is described as reaching
cribed interval target, not as solved exactly.

3.6. *Comment 6: Registered target indicator*

target is 0.001 and every global/intersection row has its target status. Exact fractions, not
table values, determine the Boolean.

3.7. *Comment 7: Quadratic bound components*

ull allowance tables include accumulated regret error, savings error, cost constant,
ity cap, repair fraction, and theoretical sum. The independently certified numerical LP
ty loss is unavailable and is labeled unquantified rather than zero.

3.8. *Comment 8: All action-gap and ratio constants*

supplement reports the constants for all eight old controlled cases and all sixteen new
ases. Tied cases are flagged and excluded from the inverse-gap theorem.

our frozen variants with perturbations 0, 1e-8, 1e-4, and 0.01 are included with complete
ch and SCIP records; all target misses remain visible.

3.10. *Comment 10: Reference action and rational rounding*

he reference is selected by exact rational comparisons of the stored model. It is not
ned stable under perturbations across a tie. The reference-free support lower expression and
ctation-error contract avoid relying on that identity.

3.11. *Comment 11: Exact-LP rate versus implemented rate*

he abstract, strict-gap section, diagnostic discussion, and conclusion distinguish the theorem,
sured widths, and fixed-precision effects.

3.12. *Comment 12: Conditional regime law*

he initial regime vector may depend measurably on the observable label. Main Appendix C
es the identity and gives a cost error bound for approximating this conditional law.

3.13. *Comment 13: Full Borel selection*

ain Appendix C supplies a finite rational-candidate, first-minimizer, backward-repair con-
ction with uniform primitive bounds, rather than leaving the selection step implicit.

3.14. *Comment 14: Fiber primitive errors*

he supplementary fiber accounting reports representative-cell reward, cost, and terminal
rs from the actual scaling identities; the finite regime-matrix error is zero. The sharper
tonicity certificate is kept distinct from these generic perturbation bounds.

3.15. *Comment 15: Fiber time versus width*

he nested 2, 4, 8, and 16 cell table sums the actual stored LP, arithmetic/construction, and
k work for the required subset of endpoints. These are re-tabulated endpoint costs, not new
pendent mesh executions.

3.16. *Comment 16: Nontrivial autonomous evolution*

he theorem allows Borel bijective autonomous evolution, but no new nonidentity-covariate
erical run is supplied. The fixed-covariate example is not presented as that experiment.

3.17. *Comment 17: Ancillary continuous coordinates*

he article explicitly states that service-economy continuous coordinates do not enter rewards,
s, or regime transitions.

3.18. *Comment 18: Aggregation terminology*

hose examples are described as exact aggregation tests with economically ancillary atomic
dinates, not general solved continuous-state applications.

main historical-evidence discussion retains that the stronger McCormick formulation
ed interval search on the difficult binary pairs.

3.20. *Comment 20: Search progress*

main article includes Q1 certified gap versus time; the supplement adds gap versus nodes.
complete trace is retained in the machine-readable results.

3.21. *Comment 21: Median original widths*

main article includes both median and maximum positive-cost randomized width by
all thirty positive-cost intervals remain open.

3.22. *Comment 22: Normalized deterministic separations*

erty-two signed rows, including the twenty-seven positive ones, now give both requested
izations. The percentages are explicitly approximate because they use the preserved
mal display inputs; exact source certificates remain authoritative.

3.23. *Comment 23: Meaning and cost of randomization*

model uses a fresh conditional state-date action draw. Fixed administration and gover-
nments are separate primitives; the signed-savings margin gives a conditional overhead

3.24. *Comment 24: Nonlinear independent-check indicator*

selected nonlinear table marks the 16-period,128-cell row as not independently checked
early identifies the independently checked rows.

3.25. *Comment 25: Tight nonlinear upper failures*

desired feasible upper certificate is available at allowance 0.05. The retained summary does
not support a per-candidate distinction between true violation and conservative enclosure; this
certificate remains unavailable rather than being guessed.

3.26. *Comment 26: Machine-readable theorem matrix*

ASSUMPTIONS.json supplies assumptions, policy class, dynamics, guarantee, and executed
evidence for every headline claim.

3.27. *Comment 27: Broader mutations*

frozen archive includes 17 rejected mutations, covering changed disadvantages, labels,
boxes, repair rows, and missing or cyclic covers. R45 adds 256 exact stopping-
checks. No unexecuted fiber-weight mutation is claimed.

the main argument no longer proceeds through revision history. Exhaustive historical content is reserved separately, with byte-identity checks rather than deletions.

3.29. *Comment 29: Asymptotic versus attained accuracy*

The conclusion distinguishes mathematical convergence statements, actual target hits, and the original intervals.

3.30. *Comment 30: Decomposition of object counts*

Every total is decomposed: 12 primary global environments plus 4 tie variants; 32 custom plus 16 external candidate policies; 3 by 7 by 2 scaling roots; historical 16 plus 8 plus 17 point objects.

4. FILES FOR RENEWED REVIEW

CTA_R45.pdf is the revised article; SUPP_R45.pdf contains complete numerical tables and verification accounting; HISTORY_R45.pdf preserves the previous complete article, supplement, and inherited historical volume. The source, exact results, assumptions matrix, this response in machine-readable form, and preservation audit are included in the review package. All work is confined to R45 paths. The attached publication procedure selects a new branch from the observed R44 base and does not overwrite prior revisions or review branches.

1. PRESERVATION MAP

This volume reproduces three previously existing documents in full: the R43 principal article, R43 supplement, and the inherited R42 historical volume. No mathematical statement, table, successful experiment, or historical caveat is edited in these embedded documents. Their original PDF files and corresponding source trees remain separately available with their original licenses. This combined PDF is a reading copy, not a byte-identical replacement for the three source PDFs.

The current scientific argument and new proofs are in `ECTA_R45.pdf`. Full numerical counting is in `SUPP_R45.pdf`, and point-by-point responses are in `RESPONSE_R45.pdf`. The preservation manifest records the inherited source and evidence files in the local review package. An add-only publication overlay leaves other repository history and branches unaltered. This local reading volume is not claimed to include every remote historical branch or entire unmaterialized remote annex.

performance after every state and date. One randomized Markov continuation
 parts. Backward feasibility repair yields complete finite-state search, including
 on extension without discounting. For controlled finite models with strictly
 acting actions, disadvantage and savings coordinates contract the bilinear relax-
 ed root-node certificate has a quadratic small-allowance error bound at fixed
 observable-covariate disintegration further gives convergent certificates for a
 lled atomic continuous-state models without smoothing their transitions. Exact
 regation and a verified density-transfer exercise provide complementary routes.
 ons retain all registered solver stops, same-object relaxation comparisons, and
 ational endpoint checks. In the original maintenance cohort, 27 lotteries beat
 lower bounds, while all 30 positive-cost randomized intervals remain open. The
 uish structural rates, complete search, and actual numerical closure without
 common-policy objective.

S: Dynamic programming, Policy revision, Constrained Markov control, Global
 Verified computation.

1. INTRODUCTION

ay wish to improve an installed rule without replacing every decision it
 on cost then depends both on the decisions changed today and on the states
 ed rule tomorrow. A feasible revision must preserve the operating objective,
 ive revision must economize on intervention along its own induced state
 o requirements create a constrained dynamic problem. Pointwise selection
 es need not solve it.

uation must also be specified. Two histories that meet at the same date and
 continuation under a Markov rule. Allowing a different continuation for each
 ch remembered randomization can reduce costs, but changes the economic
 quiring a deterministic rule can increase implementation cost even when
 issible. We therefore take a common randomized Markov rule as the
 e. Performance is protected after every date and state; implementation cost
 stated initial distribution. A performance guarantee at every restart is not a
 ty at every restart.

on is a constructive connection between these requirements and a complete
 cy that violates every operating constraint by at most δ can be repaired
 ding a persistent random seed or a history variable. The repair improves
 ery state. Its additional implementation cost is bounded explicitly by a
 with δ . Strict discounting provides a constant slack, while a finite-horizon
 o permits undiscounted decisions. At zero tolerance an exact restriction
 izing actions replaces the repair. No global duality assumption is used.
 error bound for rational policy lattices. For binary actions it also gives a
 the interval Bellman branch-and-bound procedure used in the experiments.

preference mass. These observations yield an explicit $O(\varepsilon^2)$ root-
 positive operating-action separation and strict discounting. The claim
 al for the original constrained optimum, not only approximation of an
 value. Constants and action ties remain visible.
 analysis is also extended beyond exogenous transitions. When an ob-
 n invertible autonomous evolution, it labels finite controlled problems
 integrated exactly in principle and enclosed by verified covariate par-
 all regime transition probabilities. The policy uses only the current
 no hidden memory is introduced. Uniform finite-matrix perturbation
 er solves provide convergence at a fixed positive allowance. This
 m replacing atomic successors by smooth reset densities.
 tension separates operating approximation error from policy-search
 global bound. It supplies a positive-margin check and a constructive
 the exact optimum to the repair routine. A further perturbation result
 operating and cost allowances; strict randomized separations also yield
 n charges. These statements extend the numerical method rather than
 objective.
 es a different part of the general problem computationally tractable.
 exogenous, pointwise aggregation and a compatible family of within-
 reserve the optimum over unrestricted measurable common policies.
 for nonlinear atomic transitions even when total-variation reset ap-
 sulting all-restart regret program is linear. Twelve designed service
 e transition structures, up to 64 regimes and 64 periods, and three or
 pendently verified intervals below 9×10^{-8} . An unreduced Bellman
 classical comparator; we do not conceal its advantage on some small
 s finite-state certificates to a continuous state space. Under a certified
 tion of the transition kernel by a within-cell reset kernel, cell averag-
 asurable Markov policy to a finite-state policy. Lifting reverses this
 imitive errors give an explicit operating-tolerance adjustment and a
 ir modulus controls the remaining difference between the relaxed and
 ma. The theorem makes the approximation assumption operational
 eat weak convergence of an atomic transition law as total-variation
 not silently replace discrete shocks by continuous noise.
 s separates three questions. First, does complete search return sound
 Markov problem, and how do those intervals compare with a direct
 programming formulation? Second, does randomization change the con-
 ntinuous-state maintenance configurations? Third, can a nontrivial
 rive nonlinear, two-state dynamics and nonzero operating-witness
 tains the same objective on the two sides of its interval. The inherited
 -state environments and their two solver implementations were fixed
 ts were generated. The additional controlled, three- and four-action

changes the zero lower bound into a positive bound; its finest horizon-
 [11.8492]. That interval is informative but not tight. In the prospective
 on, both methods face the same node allowance and wall-clock limit.
 relaxation often improves the lower endpoint, but longer horizons retain
 intervals. The theory supplies completeness, not a claim that complete search

1.1. *Relation to the literature*

ov decision processes provide the natural starting point (Altman, 1999).
 s are particularly effective when the constraints are expectations under a
 . Our formulation additionally requires every restarted problem to meet
 e using one shared continuation. A separate occupation measure for each
 compatibility restrictions: at a shared date and state, its action probabilities
 ich restart generated the occupancy. Dropping that restriction produces a
 s. Our finite formulation imposes it directly through a single policy variable
 te. This observation does not preclude occupation-measure methods; it
 they must preserve in this problem.

ons to constrained control are not new. Saldi (2018) proves value conc-
 and average-cost constrained problems, and develops approximately
 rates under additional continuity conditions. Saldi, Linder, and Yüksel
 systematic treatment of quantized stochastic control. Our transfer result
 n, all-restart constraints and combines the within-cell reset identity with an
 bility repair. It states the total-variation condition needed to average unre-
 policies rather than assuming that a convenient policy sieve is automatically
 ed optimum. The theorem is not a replacement for those approximation
 d object is the common-continuation restart constraint and its computable
 ulus.

methods also solve constrained expected-return problems. Ding, Zhang,
 c (2020) establishes global convergence guarantees for natural policy-
 discounted constrained MDPs. Ni and Kamgarpour (2025) studies safe
 cumulative constraints. Those learning problems have information and fea-
 different from our known-model, all-restart certification problem. Safe
 t (Laroche, Trichelair, and Tachet des Combes, 2019) and approximation-
 ed dynamic programming (Munos and Szepesvári, 2008, Scherrer, 2013)
 model uncertainty, policy approximation, and optimization error. We claim
 comparison against methods given less information.

ation ingredients are classical. Convex envelopes of factorable products
 ick (1976); the numerical LP proposals here use a modern simplex im-
 ed with Huangfu and Hall (2018). We do not attribute novelty to bilinear
 d search, or weak duality. The contribution is a common-policy economic
 cal approximation errors can be repaired into an admissible deployment,

side whole-cell inequalities and isolated switching points. Information-
 / another route to dynamic-programming lower and upper bounds
 (2010, Brown and Smith, 2014); they are related to, but not identical
 functions used below. The support functions remain useful accelerators
 the set is asserted to exhaust a global dual.

tion is adjustment rather than unconstrained operating optimization.
 s long emphasized model structure, approximation error, and state-
 (1998, Brumm and Scheidegger, 2017). Empirical replacement models
 pipeline operating primitives with data. Our revision cost is instead the
 installed decision rule, not a cost paid whenever the previous physical
 physical switching costs would require including the previous action in
 ntenance environments are designed, known-model economies. Their
 implementation units are not estimated monetary welfare, and their
 rates of empirical prevalence. The convergence and exact-reduction
 imed calibration, carry the methodological argument. The density
 the reset error chain; the original endogenous-transition models remain
 than being relabeled as instances of the exogenous theorem.

2. THE ECONOMIC PROBLEM

rel space, $A = \{1, \dots, m\}$ a finite action set, and $t = 0, \dots, T$. Rewards
 costs $k_t(x, a)$, and terminal operating payoff $g(x)$ are bounded Borel
 kernels P_t^a are known Borel probability kernels. Write $|r_t| \leq R$,
 Unless otherwise indicated, $0 < \beta < 1$ and $\varepsilon > 0$. Define

$$(x, a) + \beta P_t^a v(x), \quad V_T = g, \quad V_t = \max_{a \in A} T_t^a V_{t+1}. \quad (1)$$

Borel and permits a Borel maximizing selector, for example the lowest
 on.

policy is a sequence of Borel kernels $p_t(\cdot | x)$ on A . Its values satisfy

$$J_t^p = \sum_a p_t(a | x) T_t^a J_{t+1}^p, \quad (2)$$

$$C_t^p = \sum_a p_t(a | x) \{k_t(x, a) + \beta P_t^a C_{t+1}^p(x)\}. \quad (3)$$

I histories reaching (t, x) . Randomization is newly drawn conditional
 ate; a persistent hidden coin indexing different future policies is not

y law ν , the central object is

$$\inf_{\mathcal{F}_R(\varepsilon)} \int C_0^p d\nu, \quad \mathcal{F}_R(\varepsilon) = \{p : J_t^p(x) \geq V_t(x) - \varepsilon \forall t, x\}. \quad (4)$$

$$k_t(x, a) = w_t(x) \mathbf{1}\{a \neq a_t(x)\}, \quad w_t \geq 0. \quad (5)$$

the action actually drawn. Its expectation under a lottery is not an indicator probable action differs from the installed action. The unchanged installed on cost even when it is operating-infeasible. Consequently, dividing an its revision cost is undefined, not an informative relative-gap statistic. We then $U > 0$, and report zero-cost exactness separately.

the development economy weights all normalized initial conditions equally. ence distribution, not an estimated stationary law. We also integrate exactly ensities $2x$ and $2(1-x)$. These calculations change the initial objective -state operating constraints and installed rule unchanged. A pointwise cover arbitrary initial laws, including atoms; an almost-everywhere cost

FEASIBILITY REPAIR AND COMPLETE POLICY SEARCH

$$D_C = K \sum_{s=0}^{T-1} (s+1) \beta^s, \quad (6)$$

$$D_J = \max_{1 \leq h \leq T} \left\{ 2R \sum_{s=0}^{h-1} (s+1) \beta^s + 2Gh\beta^h \right\}. \quad (7)$$

remaining horizons in D_J matters: the terminal term $h\beta^h$ need not increase n between action distributions is one half of their ℓ_1 distance.

perturbations: *If $\sup_{t,x} \text{TV}(p_t(\cdot | x), q_t(\cdot | x)) \leq \theta$, then*

$$\sup_{t,x} |J_t^p - J_t^q| \leq D_J \theta, \quad \sup_{t,x} |C_t^p - C_t^q| \leq D_C \theta.$$

an action's remaining cost lies in $[0, K \sum_{s=0}^{h-1} \beta^s]$ at a state with h periods its distribution changes the conditional expectation by at most the length θ . Adding β times the next-date value difference and iterating gives For operating rewards use the interval length $2R$ for each stage and $2G$ d. A reward at distance s can be affected by $s+1$ action changes, and the changes. Taking the maximum over h gives (7). The same conclusions horizon coupling, but do not require one. *Q.E.D.*

mon-continuation repair: *Suppose $p \in \mathcal{F}_R(\varepsilon + \delta)$, $\delta \geq 0$. There is a Borel policy $\hat{p} \in \mathcal{F}_R(\varepsilon)$ with $J_t^{\hat{p}} \geq J_t^p$ for every (t, x) and*

$$\rho_\varepsilon(\delta) := \frac{\delta}{(1-\beta)\varepsilon + \delta}, \quad \int C_0^{\hat{p}} d\nu \leq \int C_0^p d\nu + D_C \rho_\varepsilon(\delta). \quad (8)$$

action a_* and set

$$\hat{p}_t = (1 - \alpha)p_t + \alpha e_{a_*}, \quad \alpha = \frac{b - q_p}{q_* - q_p}.$$

$\delta]$, the denominator is at least $(1 - \beta)\varepsilon + d$. Thus $\alpha \leq \rho_\varepsilon(\delta)$ and iterations are Borel; the division is used only where the denominator is non-zero. The construction uses only t and x and the already fixed next-date continuation. Apply the construction to arbitrarily near-minimizers at tolerance $\varepsilon + \delta$ to obtain (9). *Q.E.D.*

convex combination of whole policies. Such a combination, if selected, could enlarge the information state. Here the mixture is selected anew for each repaired continuation. The slack $(1 - \beta)\varepsilon$ explains why the theorem requires a positive tolerance. No vanishing modulus at $\beta = 1$ or $\varepsilon = 0$ is possible.

Certified search without an exact operating solve

The preceding statements is not an oracle that the method must possess. It supplies the slack needed by the repair. This matters when operating with a policy search alone, is expensive. Let $L_t \leq V_t \leq U_t$ be bounded by g . Define checked scalar bounds

$$\begin{aligned} & \max_{t < T} \|U_t - L_t\|_\infty, \quad \sigma = (1 - \beta)\varepsilon - d, \\ & \max \left\{ 0, \sup_{t < T, x} \left[U_t(x) - \max_a T_t^a U_{t+1}(x) \right] \right\}. \end{aligned} \quad (10)$$

Let $m = |A|$, let $M \geq 1$ be an integer, and set $\theta_M = \min\{1, (m - 1)/M\}$. The same primitive transition and reward functions are used for the fitness check and the repair. Error in a different operator must first be bounded. A small residual is not, by itself, a certificate that $U \geq V$; the witness condition are also required.

Repair-driven repair and global error: *Suppose $\sigma > 0$ and $\delta \geq 0$. If a common policy satisfies $J_t^p \geq U_t - \varepsilon - \delta$ for all $t < T$, the backward construction produces a common policy $\hat{p}$ such that*

$$J_t^{\hat{p}} \geq V_t - \varepsilon, \quad J_t^{\hat{p}} \geq J_t^p, \quad \sup_{t, x} \text{TV}(\hat{p}_t, p_t) \leq \frac{\delta}{\sigma + \delta}. \quad (11)$$

($\sigma + e + D_J \theta_M$)
, and $d \rightarrow 0$ give convergence to the actual constrained optimum using
her than an exact operating solve. All displayed errors and the stopping
or rational represented finite-state primitives.

paired continuation and put $q_a = T_t^a J_{t+1}^{\hat{p}}$, $q_p = \sum_a p_t(a | x) q_a$, and $b =$
decision date $J_T^{\hat{p}} = U_T$; at every earlier date the induction hypothesis gives
itivity and (10) imply

$$= \max_a q_a \geq \max_a T_t^a U_{t+1} - \beta \varepsilon \geq U_t - d - \beta \varepsilon = b + \sigma.$$

ement gives $q_p \geq J_t^p \geq b - \delta$. Where $q_p < b$, set $\alpha = (b - q_p)/(q_* - q_p)$
least-index maximizing action. For $z = b - q_p \in (0, \delta]$, one has $\alpha \leq$
). Otherwise leave p_t unchanged. This proves (11); the cost bound follows
action maximization and division on a strictly positive-denominator Borel
ibility.

ment, round any genuinely feasible policy as in Theorem 3. Its rounded
 $- D_J \theta_M \geq L - \varepsilon - D_J \theta_M$, and its cost increases by at most $D_C \theta_M$. This
point. A lattice minimizer is at worst $e + D_J \theta_M$ below the upper-witness
at assertion with that value of δ to obtain the upper endpoint and (13).
tify clipping the lower endpoint at zero. Q.E.D.

means that this particular witness pair has not certified the repair margin;
infeasibility of the original problem. Witness refinement and complete
roles in (13). Refining the witnesses alone does not close the remaining
nversely, an increasingly fine policy lattice does not erase a fixed operating-
ict-value theorem is recovered by $L = U = V$ and $d = e = 0$.

3.2. A complete finite-state construction

s. For an integer $M \geq 1$, let $\mathcal{P}_M$ be the policies whose action probabil-
 $1/M$. Put $\theta_M = \min\{1, (m - 1)/M\}$ and $\delta_M = D_J \theta_M$. Evaluate every
mon Bellman equations and let

$$= \min \left\{ \int C_0^p d\nu : p \in \mathcal{P}_M, J_t^p \geq V_t - \varepsilon - \delta_M \forall t \right\}.$$

these comparisons, values, and policy probabilities use rational arithmetic.

vergence to the constrained optimum: *Repair a minimizer defining v_M*
the cost by U_M . Then

$$- D_C \theta_M \leq K_R(\nu, \varepsilon) \leq U_M, \quad U_M - L_M \leq D_C \{\theta_M + \rho_\varepsilon(\delta_M)\}. \quad (14)$$

dist plus $D_C\theta_M$. Hence $v_M - D_C\theta_M \leq K_R$. Theorem 1 applied to the states $U_M \leq v_M + D_C\rho_\varepsilon(\delta_M)$. The weak-composition count gives the at each date and state. The displayed error tends to zero. *Q.E.D.*

distance to (4), not distance to a one-step relaxation. Its count is); it is a completeness construction, not a polynomial-time claim. The problem can instead be solved by evaluating its m^{nT} vertices with the randomized repair is neither required nor valid as a deterministic-

3.3. The executed spatial search

te p_{ti} for the probability of action 1. A policy box $Q = \prod_{ti} [\ell_{ti}, u_{ti}]$ states across all restarts. Backward interval Bellman evaluation gives $\bar{J}$ and unconstrained implementation lower values $\underline{C}$ for that box. can contract its coordinate intervals: if q_0^+, q_1^+ are next-date operating le p must satisfy

$$(1 - p)q_0^+ + pq_1^+ \geq V_t(i) - \varepsilon. \quad (15)$$

with $[\ell_{ti}, u_{ti}]$. Empty intersections or inconsistent directed operating . Every removal is justified by a necessary inequality. Feasible policies e a sufficient candidate check fails.

lower bound is $\int \underline{C}_0 d\nu$. Repairing the midpoint of the contracted box policy. The algorithm selects a live box with the smallest lower bound maximizing $\beta^t(u_{ti} - \ell_{ti})$. It preserves both children and every live leaf. is the minimum of all live or cost-pruned leaf lower bounds and the stops when their difference is no greater than the requested tolerance, interval when the preassigned budget is exhausted.

pping bound for interval Bellman search: Let $w = \max_{ti}(u_{ti} - \ell_{ti})$ Its midpoint-repair cost minus its interval Bellman lower bound is at

$$H(w) = D_C\{w + \rho_\varepsilon(D_J w)\}. \quad (16)$$

cap, the best-bound bisection procedure reaches every prescribed many steps. In particular, choose $w_* > 0$ with $H(w_*) \leq \eta$. A sufficient h is

$$d \left\lceil \log_2 \frac{1}{\beta^{T-1} w_*} \right\rceil, \quad d = nT, \quad (17)$$

by zero when its argument is at most one. The associated full binary- ential in this depth. Adding valid McCormick lower bounds cannot aarantee.

and refinement while the global gap exceeds η : its repaired cost is already η above the global incumbent or higher. On the other hand, weighted longest-side bisection makes every physical width at most w_* by depth d , where there are finitely many nodes up to that depth. Valid stronger lower bounds are obtained by enclosing boxes. The supplement details the effect of the necessary contractions; the algorithm maintains the boxes and preserves every feasible point. *Q.E.D.*

This theorem for the implemented search rule, not an inference from favorable results, explains why a 60-second run can end with a large gap: a finite worst-case bound is not always attainable. The numerical results report that distinction directly.

3.4. Finite-horizon repair without strict discounting

Theorem 1 supplies the constant margin in Theorem 1, but it is not necessary for finite-horizon repair. A date-dependent allowance supplies a different margin. This distinction is useful for applications that assess a finite service commitment without discounting its value. In this subsection only, allow $0 < \beta \leq 1$; the constants D_J, D_C retain their

Finite-horizon slack ladder: Let $\varepsilon > 0$, $0 < \delta < \varepsilon$, and let p satisfy $J_t^p \geq V_t - \varepsilon$ at date t . Choose $z \in (0, 1)$ with $\delta \leq \varepsilon z^T$ and put

$$\gamma_t = \varepsilon z^{T-t}, \quad \tau_t = \varepsilon - \gamma_t, \quad 0 \leq t \leq T. \quad (18)$$

At each date t , choose an action maximizing the repaired operating continuation, with current continuation τ_t . Adopt a common Borel Markov policy q satisfying

$$-\tau_t \geq V_t - \varepsilon, \quad J_t^q \geq J_t^p, \quad \sup_{t,x} \text{TV}(q_t, p_t) \leq \frac{2z}{1+z}.$$

At $t=0$, $J_0^p + D_C 2z/(1+z)$ pointwise. At $\delta = 0$ retain p ; at $\delta \geq \varepsilon$ an operating-bound D_C . With $z = (\delta/\varepsilon)^{1/T}$ the resulting modulus vanishes as $\delta \downarrow 0$,

Minimal allowance is zero. Given an already repaired continuation, write $J_{t+1}^q, q_p = \sum_a p_t(a | x) q_a$, and $q_* = \max_a q_a$. Inductive improvement

$$q_* - b \geq s_t := \tau_t - \beta \tau_{t+1} \geq \gamma_{t+1} - \gamma_t > 0.$$

At date t , choose q_p to maximize s_t . Otherwise the mixture share $\alpha = (b - q_p)/(q_* - q_p)$ is at most $\delta + \gamma_t$, so

$$\alpha \leq \frac{\delta + \gamma_t}{s_t + \delta + \gamma_t} \leq \frac{\delta + z\gamma_{t+1}}{\delta + \gamma_{t+1}} \leq \frac{2z}{1+z}.$$

implementation need not evaluate an irrational root: a dyadic z with a rational construction. For $\beta < 1$, either valid repair can be used; their error is less than a presumed ordering of their worst-case constants, determine the bound.

Use operating witnesses. For checked $L_t \leq V_t \leq U_t$, $U_T = g$, let d_t be the deficit from above, and choose $0 \leq \tau_t \leq \varepsilon$, $\tau_T = 0$, with

$$\sigma_t = \tau_t - \beta\tau_{t+1} - d_t > 0.$$

The same proof with target $U_t - \tau_t$ bounds the current mixture by $(\delta + \varepsilon - \tau_t)$. This is a checked, date-specific condition at $\beta = 1$, not a claim that every date has adequate slack. In a lattice bracket the deficit is $\delta = e + D_J\theta_M$. The deficit and policy-rounding terms remain separate.

Zero tolerance: At $\varepsilon = 0$, define $A_t^*(x) = \arg \max_a T_t^a V_{t+1}(x)$. The minimum cost is the ordinary cost Bellman recursion restricted to $A_t^*(x)$. A deterministic selector attains its minimum at every state and date; randomization is not needed for finite T and $0 < \beta \leq 1$.

A policy has $J_t^p = V_t$ everywhere, since no policy exceeds V_t . Thus the minimum is in A_t^* at each date, by the operating Bellman equality. Conversely, a policy in these correspondences is operating-optimal by backward induction. The minimum cost on each nonempty finite correspondence is an unconstrained minimum problem with Borel selectors. *Q.E.D.*

When the optimal-action ties are known exactly; a small numerical Bellman algorithm with an exact tie. It addresses zero tolerance without taking a singular repair bound.

CERTIFICATES FOR STRUCTURED ATOMIC STATE DYNAMICS

Complete but potentially expensive. A distinct question is when the problem can be reduced exactly, without searching over probability boxes or approximations by densities. The following conditions answer that question for Markov processes. They are restrictive and testable: neither action independence alone suffices.

Homogenous aggregation identities: At each date let $\{E_{ti}\}_{i=1}^{n_t}$ be a finite set of states; let μ_i be a probability measure supported in E_{ti} . The rewards and implementation constraints on each cell for each action; the terminal payoff is constant on each cell. The transition matrix P_t does not depend on the action. For a stochastic matrix $\bar{P}_t$,

$$\sum_i \nu_i \mu_{0i}.$$

pointwise identity, including cell boundaries. Equation (20) is an identity only on partition cells. It allows each $P_t(x, \cdot)$ to be finitely atomic. In be a reset kernel.

act common-policy aggregation: *Under Assumption 1, for finite T , $0 < \epsilon$ optimum over all Borel common randomized Markov policies equals ϵ with primitives $(\bar{r}, \bar{k}, \bar{g}, \bar{P}, \nu)$. A finite optimal policy lifted constantly ϵ every original state and date and attains that continuum optimum. No ϵ -constant restriction is imposed on the original competitor policies.*

ating optimum is cellwise constant by (19) and backward induction. For an p , define $\bar{p}_t(a | i) = \int p_t(a | x) \mu_{ti}(dx)$. Action independence is essential: $\bar{p}_t$ is $P_t J_{t+1}^p$, because $\sum_a p_t(a | x) = 1$. Integrating the Bellman equation (19) shows that the averages of J^p obey the finite Bellman equations for the $\bar{p}_t$. The argument applies to C^p . Every pointwise operating constraint therefore $\bar{p}_t$ counterpart after averaging, while the initial cost is unchanged by the mixture $\bar{p}_t$. The finite infimum is no larger than the original infimum. Conversely lift any $\bar{p}_t$ on cells. Equation (19) makes its original values equal to the finite values $\bar{p}_t$. $\bar{p}_t$ feasibility and the reverse inequality. The finite optimum is attained by $\bar{p}_t$. *Q.E.D.*

original policy can average to a nondegenerate lottery. The theorem therefore $\bar{p}_t$ equality between finite and continuum deterministic optima. It cannot by itself $\bar{p}_t$ lower bound for a continuum model. The deterministic separations in the $\bar{p}_t$ cohort continue to use their own whole-domain deterministic certificates.

An exact linear program, rather than a bilinear search

$\bar{r}_{tib} - \bar{r}_{tia} \geq 0$ and let $w_0 = \nu$, $w_{t+1} = w_t \bar{P}_t$ be the exogenous state $\bar{r}_{tib}$ and the continuation is action-independent, operating regret $D_t = V_t - J_t^p$ $\bar{r}_{tib}$ version. The continuum optimum in Theorem 5 is exactly the program

$$\begin{aligned} \min_{p, D} \quad & \sum_{t, i, a} \beta^t w_{ti} \bar{k}_{tia} p_{tia}, \\ D_{ti} = \quad & \sum_a \ell_{tia} p_{tia} + \beta \sum_j \bar{P}_{tij} D_{t+1, j}, \\ \sum_a p_{tia} = 1, \quad & p_{tia} \geq 0, \quad 0 \leq D_{ti} \leq \epsilon, \quad D_T = 0. \end{aligned} \tag{21}$$

the flow y_{tia}^s for each restart $s = (u, j)$. With $z_{ti}^s = \sum_a y_{tia}^s$, it imposes usual transition flow, and

$$z_{tia}^s = z_{ti}^s p_{tia} \quad \text{for every restart } s \text{ and } t \geq u. \quad (22)$$

$\sum_{t \geq u, i, a} \beta^{t-u} r_{tia} y_{tia}^s + \beta^{T-u} \sum_i g_i z_{Ti}^s \geq V_u(j) - \varepsilon$. A further flow implementation cost. Zero-occupancy states cause no division: (22) is equality. With controlled transitions these are genuine common-policy dependent restart occupation measures omit them. With exogenous, so compatibility becomes linear. Equation (21) eliminates the by retaining their regret recursion. This is the structural reason the bears in this class.

ons and their convergence rates are established tools in constrained o-Rumeau, 2013, Saldi, 2018). The claim here is the exact preservation a-policy constraint under the two identities above, including atomic repair-based certification framework. It is not priority for occupation on, or linear programming.

Arithmetic certificates and economic charges

ed bounded LP $\min\{c'z : Az = b, l \leq z \leq u\}$ has, for every rational ate

$$= y'b + \sum_j \min\{(c - A'y)_j l_j, (c - A'y)_j u_j\}. \quad (23)$$

the objective and minimize each remaining term over its interval to exact dual feasibility or optimizer status is required. In (21), probabilities $[0, \varepsilon]$. Rationally rounded LP probabilities are evaluated and, where rd; their actual cost is an upper certificate. An independent checker ts from primitives, probabilities, and multipliers. It imports neither the ctor.

ics make the implementation economics particularly transparent. A ion-dependent administration charge that leaves operating payoffs and pply adds to $\bar{k}_{tia}$. Program (21) remains exact, rather than requiring a imination. An overhead that changes rewards must also alter ℓ and the verhead that changes the transition law can invalidate exogeneity and neral common-policy program. Implementation units in the following signed adjustment units, not estimated monetary welfare.

TAILED GLOBAL BOUNDS WITH ENDOGENOUS TRANSITIONS

bove requires exogenous transitions. A different structure is available ure state probabilities. Tight operating allowances limit how often

$$d_{tia} = V_{ti} - r_{tia} - \beta P_t^a V_{t+1}(i), \quad (24)$$

$$g_{tia} = H_{ti} - k_{tia} - \beta P_t^a H_{t+1}(i). \quad (25)$$

while $d_{tia} > 0$ for the other actions. Put $\underline{d} = \min_{t,i,a \neq a_{ti}^*} d_{tia}$. With a fixed λ , this is a checkable separation condition, not a uniform assertion over all policies. If the maximizing actions become tied.

For any policy p , write $D^p = V - J^p$ and $S^p = H - C^p$. Their exact recursions are

$$D_{t+1}^p(i) = d_{tia} + \beta P_t^a D_{t+1}^p(i), \quad S_{t+1}^p(i) = \sum_a p_{tia} (g_{tia} + \beta P_t^a S_{t+1}^p(i)), \quad (26)$$

values. Let

$$\lambda_- = \min \left\{ 0, \min_{t,i,a \neq a_{ti}^*} \frac{g_{tia}}{d_{tia}} \right\}, \quad \lambda_+ = \max \left\{ 0, \max_{t,i,a \neq a_{ti}^*} \frac{g_{tia}}{d_{tia}} \right\}.$$

Advantage and savings cones: *Every feasible common policy satisfies*

$$D_{ti} \leq \varepsilon, \quad \sum_{a \neq a_{ti}^*} p_{tia} \leq \min\{1, \varepsilon / \underline{d}\}, \quad \lambda_- D_{ti}^p \leq S_{ti}^p \leq \lambda_+ D_{ti}^p. \quad (27)$$

$$\lambda_- \varepsilon \leq S^p \leq \lambda_+ \varepsilon.$$

Feasibility and optimality gives $D^p \geq 0$. The first recursion in (26) and feasibility gives the second. Iterating the two recursions gives bounds on the discounted expected sums of d and g under the same policy and state-action pair. The inequalities $\lambda_- d \leq g \leq \lambda_+ d$ also hold at the maximizing action, where both d and g are nonnegative. The last bounds use the signs of the expectations. *Q.E.D.*

Let $\Delta P_t^a = P_t^a - P_t^{a^*}$. The two recursions become

$$D_{t+1}^a(i) = \beta P_t^{a^*} D_{t+1}(i) + \sum_{a \neq a^*} p_{tia} \{d_{tia} + \beta \Delta P_t^a D_{t+1}(i)\}, \quad (28)$$

$$S_{t+1}^a(i) = \beta P_t^{a^*} S_{t+1}(i) + \sum_{a \neq a^*} p_{tia} \{g_{tia} + \beta \Delta P_t^a S_{t+1}(i)\}. \quad (29)$$

Use the probability simplexes, the first inequality in (27), the savings cone, and nonnegative-cost bounds. Replace each remaining product $p_{tia} D_{t+1,j}$ by its certified rectangle. All unrestricted variables are nonnegative. The construction preserves one probability simplex per state and action; it does not relax common continuations into restart-specific

continuation obeys $\hat{J}_{t+1}^{\hat{p}} \geq V_{t+1} - \varepsilon$. Since a_{ti}^* maximizes the original on, its value with that continuation is at least $V_{ti} - \beta\varepsilon$. It therefore lies whenever a repair is needed. The proof of Theorem 1 applies with this the mixing fraction by $\rho_\varepsilon(\delta)$. The total-variation change is exactly that at most $\chi\rho_\varepsilon(\delta)$. Operating value still improves at every backward step. *Q.E.D.*

tic small-allowance relaxation width: Let $L_\varepsilon^{\text{LP}}$ be the exact optimum ation. Set $e_{D,T-1} = 0$ and, for $t < T - 1$, set

$$\sum_{i \neq a_{ti}^*} \frac{\|\Delta P_t^a(i, \cdot)\|_1}{d_{tia}}, \quad e_{S,t} = (\lambda_+ - \lambda_-)e_{D,t}, \quad e_{S,T-1} = 0.$$

and $E_{D,t} = e_{D,t} + \beta E_{D,t+1}$, $E_{S,t} = e_{S,t} + \beta E_{S,t+1}$. Write $\bar{E}_D = \{1, \varepsilon/d\}$. An exact minimizing LP policy can be repaired to a feasible hat

$$U_\varepsilon^{\text{rep}} - L_\varepsilon^{\text{LP}} \leq E_{S,0} + D_C \chi_\varepsilon \frac{\bar{E}_D}{(1 - \beta)\varepsilon + \bar{E}_D}. \quad (31)$$

with a strictly positive operating-action gap and fixed $\beta < 1$, the right

nt in a McCormick rectangle, the discrepancy between its product product of its two factor coordinates is at most one quarter of the lengths. Here a nonreference probability has length at most ε/d_{tia} . length ε , and a savings coordinate has length at most $(\lambda_+ - \lambda_-)\varepsilon$. equality to (28)–(29) gives one-step errors e_D, e_S . Recombining the e terms gives the stochastic kernel P^p ; its sup-norm is one. Backward nds actual regret by $\varepsilon + \bar{E}_D$ and the actual policy cost by $L_\varepsilon^{\text{LP}} + E_{S,0}$. age inequality bounds its nonreference mass by χ_ε . Lemma 3 proves $E_D, E_S = O(\varepsilon^2)$ and $\chi_\varepsilon = O(\varepsilon)$; the repair ratio is $O(\varepsilon)$. *Q.E.D.*

t bound the time required to solve an arbitrary LP, nor the size of Its gain is quantitative: shrinking the operating allowance contracts rare-departure repair prevents a first-order feasibility penalty from bound. An exact operating tie, a vanishing action gap, or a discount invalidate the uniform rate. Such a case retains the ordinary finite-state search; it is not discarded. For a nonoptimal feasible LP proposal, t be added to (31). In computation we instead report independently

them alone does not justify replacing a continuous model by a grid. The conditions permit such a transfer without restricting the original policy to π .

Certified reset approximation: *For a finite Borel partition $(E_i)_{i=1}^{n_h}$, let μ_i be measures supported on E_i and write $\nu = \sum_i \nu_i \mu_i$. There are cellwise rewards $r[\cdot]$, and terminal values g_h , and a kernel*

$$Q_{h,t}^a(x, dy) = \sum_j P_{h,t}^a(i, j) \mu_j(dy), \quad x \in E_i,$$

errors d_r, d_k, d_g and α_h satisfying

$$|k - k_h| \leq d_k, \quad |g - g_h| \leq d_g, \quad |(P_t^a - Q_{h,t}^a)f(x)| \leq \alpha_h \|f\|_\infty$$

over f . The approximation rewards obey the same bounds R, G as the original. R, G are enlarged to cover both models. All finite transition entries and rewards are putable with verified directions.

ν uses the operator convention for total variation; it is not the half- ℓ_1 norm of the probabilities above. Define $M_{J,t} = R \sum_{s=0}^{T-t-1} \beta^s + G \beta^{T-t}$ and $M_{C,t} =$

$$d_g, \quad a_t = d_r + \beta a_{t+1} + \beta \alpha_h M_{J,t+1}, \quad (32)$$

$$0, \quad b_t = d_k + \beta b_{t+1} + \beta \alpha_h M_{C,t+1}, \quad (33)$$

$$\max_t a_t, \quad b_h = b_0.$$

maximum of the finite common-Markov model at operating allowance s and

1-to-end transfer: *Under Assumption 2, if $\varepsilon > 2a_h$, then*

$$v_h(\varepsilon + 2a_h) - b_h \leq K_R(\nu, \varepsilon) \leq v_h(\varepsilon - 2a_h) + b_h. \quad (34)$$

Approximation returns intervals $[L_+, U_+]$ and $[L_-, U_-]$ for these relaxed and original widths at most η_+ and η_- . A valid global interval is

$$[\max\{0, L_+ - b_h\}, U_- + b_h], \quad (35)$$

ic objective.

urable Markov policy p , not necessarily cellwise constant. Backward and reset-model Bellman equations gives (32)–(33). Bellman computing error for optimal values. In the reset model, averaging the $= \int p_t(a | x) \mu_i(dx)$, averages its value exactly: after every transition in μ_j , and stage rewards and costs are cellwise constant. This identity lifting the common continuation. An original feasible policy has reset the reset optimum minus $\varepsilon + 2a_h$ pointwise, hence also after cell averaging differs from the original initial cost by at most b_h . Taking an infimum. Conversely, lift a finite policy feasible at $\varepsilon - 2a_h$ by using its action each cell. The reset value is constant in each cell; the two operating original policy feasible, and its cost differs by at most b_h . This proves the apply Theorem 1 in the finite model with target allowance $\varepsilon - 2a_h$ and the two finite-computation widths and the cost errors. *Q.E.D.*

6.1. Where the transfer theorem applies

a transition density with computable uniform continuity in the source proximal density in the destination variable can furnish the reset reference measures must also represent the initial law as in Assumption 2. ergodic-uniform cell measures and a uniform initial law, a uniformly ergodic gives a computable $\alpha_h = O(h)$ on a shape-regular mesh. Cellwise approximations give $d_r, d_k, d_g = O(h)$ when their moduli are Lipschitz. In an installed rule can be retained as exact partition boundaries. In $(\eta_+ + \eta_-)$ at fixed positive ε and $\beta < 1$, with explicitly displayed cells still scales as $h^{-\dim X}$; the theorem does not remove the curse of

lock model and the nonlinear two-state model below have atomic and their successor atoms into cells generally leaves the total-variation zero. We therefore do *not* apply Theorem 7 to those models by asserting reported intervals are direct whole-domain bounds for the unchanged main valid without a convergence claim for that particular spatial function prevents the approximation theorem from being used as an for closing an atomic-model gap.

CERTIFICATES THAT ACCELERATE THE GLOBAL PROBLEM

exact structural reduction, and reset transfer provide different routes to under its stated assumptions. The inherited one-sided certificates are agnostic tools, but are not substituted for that argument.

action satisfies the test. Its graph is Borel, since it is a finite union of backward minimization of $k_t + \beta P_t^a B_{t+1}$ over this set yields a Borel lower index minimizer is a Borel selector. Thus the general-state recursion does not measurable-selection assertion.

action outer class, not the original constrained problem. A selected simulate losses that violate later operating requirements. Equality between upper bound and a separately feasible policy's cost certifies $K_D(\nu, \varepsilon)$. Equality certifies more. Tightening L, U alone does not force equality; complete is the remaining structural gap.

Randomized support floors and a connected local program

and supersolution W_t^λ of

$$W_T^\lambda \geq \lambda g, \quad W_t^\lambda \geq \max_a \{ \lambda r_t - k_t + \beta P_t^a W_{t+1}^\lambda \}$$

$(\varepsilon) - W_t^\lambda$ for every feasible policy, including randomized Markov policies. the maximum of zero and finitely many such terms, using L_t in place of V_t in a valid next-date lower function B_{t+1} , define

$$(x) = \left\{ p \in \Delta(A) : \sum_a p_a T_t^a U_{t+1}(x) \geq L_t(x) - \varepsilon \right\}.$$

$$\left\{ F_t(x), \min_{p \in \mathcal{D}_t(x)} \sum_a p_a \{ k_t(x, a) + \beta P_t^a B_{t+1}(x) \} \right\}, \quad B_T = 0. \quad (38)$$

distribution of any admissible policy belongs to the local constraint, and its minimizes B_{t+1} . Induction proves that (38) is a global randomized cost lower bound. We prove that the local optimizer can be combined into a feasible policy or that we close a global duality gap.

and one local operating inequality, the local polytope has only pure feasible on boundary mixtures. Our new primary calculation evaluates these vertices arithmetically on directed whole-cell bounds. Unlike the earlier diagnostic which uses the nonzero F_t at every date and is paired with a feasible policy for a global randomized objective. It improves that lower endpoint in 23 of the 42 prior adaptive local-LP theorem and all of its experimental records remain invariant; their target is still the local relaxation, not (4).

7.3. Whole-domain randomized deployments

construction supplies feasible upper policies without changing the atomic feasible next-date value J_{t+1} , write $q_0 = T_t^{a_0} J_{t+1}$, $q_* = \max_a T_t^a J_{t+1}$, and

mixing probability can be a ratio of affine functions. Cost is therefore
 ly represented as an exact piecewise-affine function.
 olated switching points in every operating identity. Directed cost boxes
 e absolutely continuous initial laws. All affine transition slopes are
 e set of switching points and its finite-horizon preimages have zero
 ws. That fact justifies almost-everywhere cost enclosures; it does not
 a restart operating constraint or using the same cost enclosure for a

ified direct mathematical-programming comparator

formulation minimizes $\sum_i \nu_i C_{0i}$ subject to (2)–(3), $J_{ti} \geq V_{ti} - \varepsilon$,
 simplexes. The constraints are bilinear in policy probabilities and
 McCormick relaxation on each probability box provides a classical
 he baseline uses the same incumbent repair, contraction rule, box
 , initial law, and all-restart constraints as interval Bellman search. The
 nger LP bound and an LP-proposed feasible candidate after repair.
 us is not a certificate. For a minimization relaxation $c'z$ subject to
 ery rational vector $\lambda \geq 0$ gives

$$c'b + \sum_j \min\{(c + A'\lambda)_j \ell_j, (c + A'\lambda)_j u_j\} \leq \min c'z. \quad (40)$$

$c'(Az - b) \leq 0$ and minimizing the resulting affine function over the
 e solver marginals are evaluated in (40) using exact rational arithmetic.
 the interval Bellman lower bound; a floating infeasibility declaration
 box. A separate verifier rebuilds every relaxation and every search-tree
 ve model.

CONTROLLED ATOMIC MODELS WITH OBSERVABLE COVARIATES

and not preclude a complete continuum certificate. They do preclude
 t of an atom as a small total-variation error. This section uses a dif-
 erent covariate labels finite controlled problems, and its label remains
 ent state. The regime transition is allowed to depend on the action.

observable finite fibers: *The state is $(i, z) \in \{1, \dots, n\} \times Z$, where Z
 ate t , the next state is $(j, \Phi_t(z))$ with probability $P_t^a(i, j; z)$. Each
 ection with Borel inverse, independent of i, j, a . Primitive rewards,
 and regime probabilities are bounded Borel functions satisfying the
 z has law ξ and the regime has a fixed conditional law q , independent*

$$K_R(\nu, \varepsilon) = \int_Z v(\zeta, \varepsilon) \xi(d\zeta). \quad (41)$$

left ranges over all Borel common randomized Markov policies, not only tant on covariate cells. No exogeneity of the regime transition is required.

policy induces the fiber policy $p_t^\zeta(a \mid i) = p_t(a \mid i, F_t(\zeta))$. The Bellman ry date and regime. An all-state feasible policy consequently has fiber cost ry ζ , proving the lower inequality after integration.

quality, apply the rational-lattice construction of Theorem 3 separately in d denominator, there are finitely many candidate probability arrays. Their xed-feasibility tests, lowest-index minimizers, and repaired policies are rimitive bounds make the lattice and repair cost errors uniform in ζ . Thus supplies a Borel family of exactly feasible policies whose costs approach also establishes measurability of v as a limit of Borel lower endpoints. ing $\zeta = F_t^{-1}(z)$ at the current date. Its continuation is common to every z ; it remembers no unobserved initial condition. Integrate its uniform and let the denominator tend to infinity. *Q.E.D.*

–Computable covariate-cell enclosures: *Partition Z into finitely many oose a representative finite fiber for each cell. Suppose primitive errors ounded by $d_{r,c}, d_{k,c}, d_{g,c}$ and the finite regime matrices satisfy*

$$\sup_{t, i, a, \zeta \in B_c} \sum_j |P_t^a(i, j; F_t(\zeta)) - P_{c,t}^a(i, j)| \leq \alpha_c.$$

32)–(33), using these bounds. If $\varepsilon > 2a_c$ for every cell, and finite computa- ppoints L_c^+ at allowance $\varepsilon + 2a_c$ and feasible-policy upper endpoints U_c^- then

$$B_c) \max\{0, L_c^+ - b_c\} \leq K_R(\nu, \varepsilon) \leq \sum_c \xi(B_c)(U_c^- + b_c). \quad (42)$$

ls and finite-solver widths tend uniformly to zero under refinement, these the original Borel-policy optimum at the fixed allowance ε .

r of fibers has the same finite regime coordinates. Their matrix ℓ_1 bound of arbitrary bounded functions on these coordinates. Standard Bellman operating and cost perturbation bounds, for every policy, and gives the same ptimal values. The representative model at the relaxed allowance supplies ach true fiber; its tightened feasible policy lifts to a feasible policy in every grate these inequalities, using Theorem 8. The tolerance-to-cost modulus ither with the finite-solver widths, proves convergence. Nonnegativity ch lower endpoint at zero. *Q.E.D.*

9. NUMERICAL EVIDENCE

model class, information, initial distribution, and operating allowance reports every configuration; no failed or budget-stopped case is removed. Records proof-object sizes, memory, wall time, the exact environment, an independent arithmetic checks and shared-code validation.

Controlled transitions and regret-scaled relaxation

es a condition regime $i = 0, \dots, n - 1$, three or four actions, and reward all rational price perturbation. More intensive actions shift transition toward improved condition and replacement. The installed rule, rights, uniform initial regime law, and terminal reward are fixed by a ve contract. Thus the transition law is genuinely endogenous, unlike onomies below. The primitives are designed, not estimated.

red before evaluation, together with solver and verifier source hashes. on, accumulating rational expressions with unrelated denominators ic unnecessarily expensive; an integer-string limit also interrupted a imulates outward dyadic enclosures of individual summands. Original lment, and both remote hash registrations remain in the computation g allowance, action set, transition generator, or LP proposal rule was lared-case experiment with a disclosed arithmetic amendment, not an oldout.

reference-eliminated Bellman formulation, identical operating-regret ps, the same LP engine and tolerances, and the same exact candidate dinary implementation-cost bounds. Method S additionally uses the cted savings bounds. This is a same-object root-relaxation ablation; in this experiment. The 30-second cap applies to the LP call, not proof checking. All those costs are recorded separately. A stopped fallback bound and the operating-optimal feasible reference, not an estimate.

terially reduce the certified interval in 5 of the eight pairs (Table I); a absolute width is not counted. The eight-regime, 32-period, four-action ves [16.213382, 16.258594], a relative width of 0.279%. Other cases 4 LP calls reach their cap. In particular, a stronger mathematical a stronger *computed* certificate within a fixed time limit. Intersecting intervals is permitted, but the table retains the individual methods rather e run behind a portfolio.

scaled positive-cost interval has width at most 0.000008657, compared unscaled relaxation. This calculation illustrates why tight allowances n formulation even though they worsen the generic repair denominator.

3	U	5.187687	5.288978	1.888	Solved
3	S	4.859588	5.388508	9.816	Solved
4	U	17.921643	19.730330	9.168	Solved
4	S	18.516903	19.416854	4.635	Solved
3	U	5.364739	5.495882	2.387	Solved
3	S	5.384431	5.485796	1.848	Solved
4	U	15.974376	16.278277	1.867	Solved
4	S	16.213382	16.258594	0.279	Solved
4	U	0.000000	28.613836	100.000	Cap
4	S	0.000000	28.613836	100.000	Cap
3	U	0.000000	15.797290	100.000	Cap
3	S	0.000000	15.797290	100.000	Cap

led. Endpoints are outward decimal displays of independently checked rational endpoints. The sound fallback interval. These are root bounds for the same constrained common-policy point LP objective intervals. The supplement reports all allowances, dimensions, timings, memory, proof sizes, and worst-case constants.

TABLE II
TIMES OF ALLOWANCE REFINEMENT IN ONE FIXED EIGHT-PERIOD MODEL

U width	S width	S width/ ε^2	S total (s)	S check (s)
181082557	2.181330051	218.13	0.049	0.035
194637170	0.174925073	1749.3	0.045	0.035
030918109	0.004397164	4397.2	0.044	0.035
009521686	0.000008657	865.63	0.043	0.035

regimes and three actions. U and S are defined in Table I. Total time includes primitive arithmetic, and constructor arithmetic; the independent check is additional. Widths are displayed as floating point: the theorem is a fixed-model upper rate, not an assertion that every pair of observed widths has slope two.

the action-separation constants are uniformly favorable. The supplement reports cone slopes, and each term of (31); these sufficient bounds can be much tighter than the sound fallback interval. None of the eight larger-state cases is relabeled as having met the sound fallback interval because its relative gap is small.

4. An executed controlled atomic continuum certificate

Let z be the maximum of observable establishment sizes $z \in [0, 1]$, fixed over the decision horizon. The system evolves according to the four-regime, three-action controlled maintenance contract; the horizon is eight and $\beta = 0.95$. Operating rewards are multiplied by $s(z) = 1 + z$, while revision costs are multiplied by $1 - z$. The initial regime and the initially uniform regime are independent, and size is uniform. The

and all four weighted sums are independently checked. “Finite widths” is the sum of with exact cell weights. The remainder is the displayed total interval width minus that ge and differences between adjacent lower endpoints, not an independently estimated quadrature error.

01 for every size, condition, and restart. Every transition is atomic in nate. Actions nonetheless change the condition transition probabilities. finite-model optimum, Theorem 8 gives

$$K_R = \int_0^1 (1 + z/2) v_0 \left(\frac{0.01}{1 + z} \right) dz. \quad (43)$$

identity, not an assumed restriction to a parametric policy. On a cell in its allowance supplies a lower endpoint from the finite solve at le upper policy from the tighter solve at $0.01/(1 + b)$. Multiply these positive weight $(b - a) + (b^2 - a^2)/4$ and sum. The upper policy uses very date and retains common continuations. [10.074215, 10.184476], with relative width 1.083%. It is a paired accountable-state controlled atomic model, not a density approximation se policy tests. Partition refinement reduces the displayed interval, but ain material. Arbitrarily refining size cells alone does not force this width: the finite fibers must also be solved more accurately, as required ginal action-dependent continuous-condition maintenance models are n their existing status.

Closed atomic continuous-state service economies

on selecting one of m service intensities $a/(m - 1)$ in demand regime i . $= (2i + 1)/(2n)$, operating reward is $-(q_i - a/(m - 1))^2$, and terminal ed rule is one intensity below the lowest-index reward-maximizing Keeping that rule has zero implementation cost; a realized departure $n - 1$). Demand is exogenous to service intensity. Initial regime masses Banded, cyclic, and dense transition matrices are specified exactly in e designed known-model economies, not calibrated staffing data. a finite regime. It is $(i, u, v) \in \{0, \dots, n - 1\} \times [0, 1)^2$. Conditional $\rightarrow j$, the continuous coordinates move deterministically according to

$$(u + v^2) \bmod 1, \quad v' = (v + (u')^2) \bmod 1. \quad (44)$$

on one circle conditional on the other coordinate and hence preserves composition does too. Taking uniform measures on the two-dimensional tion identities exactly. Every transition law is finitely atomic, including

B	0.99	0.05	59.128133587	59.128133588	4.618×10^{-8}
C	0.99	0.05	12.174612806	12.174612832	2.535×10^{-8}
C	0.99	0.05	23.440340924	23.440340965	4.034×10^{-8}
C	0.99	0.05	43.707336818	43.707336878	5.966×10^{-8}
D	0.95	0.01	18.290633115	18.290633156	3.952×10^{-8}
B	1	0.01	15.605690050	15.605690075	2.431×10^{-8}
B	1	0.01	32.013733752	32.013733787	3.352×10^{-8}
C	1	0.05	50.441701001	50.441701067	6.497×10^{-8}
B	0.95	0.01	20.009468627	20.009468717	8.820×10^{-8}

tward to nine decimal places; the last column bounds the unrounded arithmetic gap. B, C, cyclic, and dense regime laws. Both continuous coordinates remain atomic under (44).
 rn finite arithmetic; measure preservation and the aggregation theorem are analytic proofs.

he modulo representation. The theorem applies to arbitrary policies that both continuous coordinates; it does not assume them away by restricting payoffs depend on the regime, so this is a structured continuum application of general high-dimensional numerical performance.

ere registered on the new branch before execution, including the 60-second certificate target 10^{-4} . Source code was not frozen by that protocol commit; consequently distinguished from the earlier fully code-frozen prospective moved or replaced. Table IV reports all twelve. Every pair of endpoints constructed in rational arithmetic. All costs are positive, and every interval target.

um has 16,384 variables and 8,192 equalities, corresponding to 4,096plexes. The supplement separates model assembly, LP proposal, exact cerndependent checking, and process memory. It also reports the same-object LP with ordinary presolve. Neither formulation branches on probabilities. is faster on several larger rows, whereas the unreduced LP is faster on some ervations establish tractable closure for this class; they do not demonstrate best classical LP formulation or over the earlier McCormick code on its ndent problems.

. An executed nonzero total-variation transfer budget

continuous-state model keep the service economy but replace (44) by the

$$v) = \bar{P}_{ij}[1 + \zeta f(u)f(v)] dv, \quad f(u) = 2u_1 - 1, \quad 0 \leq \zeta \leq 1.$$

re a point of $[0, 1)^2$. This density is nonnegative and integrates to one. form source cell gives a uniform destination cell, so the exact aggregation ndependent structural reference optimum. The reset approximation sets rror, in the convention of Theorem 7, is $\alpha_h = \zeta/2$; its probability total

directly checked finite lower and feasible upper policies at allowances $\varepsilon + 2a_h$ and the reset condition is not economic infeasibility: the exact structural reference remains available.

implementation, and terminal approximation errors are all zero. The same uniform reference measures.

$\beta = 0.95$, and $\varepsilon = 0.05$, we construct every primitive and propagated trusted finite problems whenever $\varepsilon > 2a_h$. Table V retains the failed as well as the successful transfers. The supplement gives the two finite-repair-cost bound separately. In particular a small measured transfer used with a small worst-case repair bound.

restates the reset theorem rather than substituting a mesh trend for its asymptotic conditioning: the largest density perturbation defeats the prescribed one gives a wide interval. The exact aggregation result does not incur the result is applied to the original action-dependent maintenance kernels, few exogeneity assumption.

Prospectively fixed common-Markov problems

new environments, we committed their seeds, joint parameter changes, cap 4,096, wall-clock cap 60 seconds per solver and case, and absolute The freeze is commit 13fd3ee; it follows development seed 390001 evaluated seeds. Both solvers see the same known model. The two-state high condition; four-state models use a pair of condition indicators their transition probabilities. Horizons are 3, 4, 8, and 12; discount allowances are 0.01 or 0.05; operating prices, transition persistence, initial masses vary jointly. The exact formulas and resulting rational argument and proof objects.

eight pairs, including unfinished searches. A positive absolute-gap certify a small relative-gap certificate when the optimum is small. We linear relaxation often improves the lower endpoint substantially, but four-state longer-horizon cases expose the combinatorial burden: the hidden by a success count. The comparison does not establish that dominates direct global programming. Instead it identifies the value and variation under matched information and caps.

The unchanged continuous-state development cohort

ance model has $x \in [0, 1]$, three actions, two shocks with probabilities stage reward $x - c_a$ with $c = (0, 3/20, 9/20)$, terminal payoff $x/2$, c . Its two affine successors for actions 0, 1, and 2 respectively are

$$(17x/20 + 1/10, 17x/20 + 7/50), \quad (x/10 + 4/5, x/10 + 17/20).$$

8	I	7.2534	9.0046	19.45	17.19
8	M	8.5290	9.0046	5.28	60.41
12	I	7.1419	7.5553	5.47	16.32
12	M	7.4446	7.5553	1.46	60.40
3	I	2.6157	2.6217	0.23	10.82
3	M	2.6206	2.6217	0.04	35.99
4	I	0.0233	0.0244	4.00	0.47
4	M	0.0235	0.0244	3.13	2.60
8	I	4.6631	7.8481	40.58	60.04
8	M	6.1990	7.8481	21.01	60.41
12	I	4.8936	6.6439	26.34	60.04
12	M	5.8606	6.6439	11.79	60.43

M: verified McCormick search. Bounds concern the identical common randomized Markov reported times include model construction and the operating solve; the search’s 60-second cap means, not by interrupting a proof-producing node. A zero upper cost is an exact zero-cost case. I: stop reasons, node counts, and memory appear in the supplement.

TABLE VII

RANDOMIZED MARKOV CERTIFICATES ON THE DEVELOPMENT COHORT

Cost exact	Strict gain over K_D	Improved lower	Maximum gap/upper (%)
4	10	8	14.61
4	8	7	32.29
4	9	8	34.23

Identified randomized upper cost is below the deterministic lower bound for the same initial law means that the randomized optimum is attained by that candidate. Improved lower compares support bound with the prior randomized support lower bound. All 42 uniform-initial rows and both nonuniform sensitivity tables are retained.

Horizons, horizons 4, 8, and 12, and allowances 0.01, 0.05 are unchanged. Neural network fixed computational incumbents, not estimated economic policies. Exact solution inexpensive in this model and their computation is charged. We do not use performance when operating approximation itself is difficult.

Deterministic comparison has 12 zero-cost and 12 positive-cost integrated positive-cost equalities occur at horizon 4, two at horizon 8, and none at maximum deterministic relative gaps at each horizon are reported alongside supplement. In the horizon-8 neural31003 case at $\varepsilon = 0.05$, the two initial identical pointwise, but the arrays differ at later restart dates. Thus initial equality at every restart. The new diagnostic records the full rational

to certify strict improvement over *every* feasible deterministic policy in 27 initial configurations (Table VII). The horizon-12 defer case at allowance

among positive-cost intervals. Savings are $L_D - U_R$, not a difference between lower and upper bounds. All 42 signed savings are in the supplement. Exact re-tabulations of retained certificates, not new optimization runs.

TABLE IX

SELECTED NONLINEAR COMMON-MARKOV INTERVALS

axis	Lower	Upper	Gap/upper (%)	Independent check
64	0.9438	3.9348	76.01	full
128	4.1908	7.6644	45.32	not independent
64	6.5389	16.7884	61.05	full
128	8.1264	12.5226	35.11	full R40
256	8.8718	11.8492	25.13	full R41

square and operating allowance $1/2$. Inherited full independent scalar checks cover R40 additionally replays the complete (32, 128) object, including the candidate upper bound. R41 subsequently independently rebuilt the complete (32, 256) object; this verifies that successful arithmetic audit with its source and proof hashes.

randomized interval $[15.3172, 15.4670]$ and a positive deterministic interval as the economically relevant policy-class difference directly, without history-conditioned frontier. The 12 zero-cost cases are already nonnegativity and are not counted as nontrivial search successes. All reported intervals retain a reported nonzero width; the narrowest and widest are distinguishable in the full table.

Sensitivity to $2x$ yields 28 strict deterministic separations; changing it to x are sensitivity results on the same developed environments, not new. The connected local program improves 23 uniform lower endpoints, 10 horizon cases; it thus affects the primary global interval rather than a secondary. It still does not establish global strong duality or close every

9.7. Nonlinear two-state bounds

The model has $(x, y) \in [0, 1]^2$, four actions, bilinear transitions, two shocks, terminal payoff $(x + y)/4$, and revision weight $1 + (x + y)/2$. Its exact solution is reproduced in the supplement. Whole-box operating witnesses use outward arithmetic. The new lower construction uses 11 global nonnegative local multipliers, including zero, at every date. It is paired with feasible upper policies. We run no new upper-portfolio competition or dual proposal success fractions as evidence of optimality.

The endpoint was not forced by the definition of revision cost: the operating cost departs from the installed rule. Nonzero support supersolutions now

4) bytes before temporary arrays and interpreter overhead; the machine
ual payload and process high-water mark.
2 belongs to the inherited exploratory nonlinear design. It is not made
mply by normalizing state variables: the operating scale and remaining
ditionally compute lower-bound diagnostics at $\varepsilon = 0.05$ on four stated
is paired with a policy certified only at $1/2$. Without a newly feasible
ghter-tolerance rows remain lower-bound-only results. A failed sufficient
led uncertified, not infeasible. A box crossing an installed-rule threshold
revision weight conservatively; it does not assert that every realized state
ction.

VERIFICATION, ECONOMIC INTERPRETATION, AND LIMITS

ree distinct burdens. Its mathematical reduction must preserve the target
arithmetic must check the asserted inequalities with the correct direction.
correspond to the checked objects. Hashes address the third burden, not
ndent arithmetic addresses part of the second burden and can expose a
but is not a proof that every economic assumption is appropriate.
rifier uses SymPy rationals and imports neither search implementation
ebuilds the common Markov equations, primitive contracts, every policy-
ssary contraction, the LP residual bounds, and the global leaf minimum.
verifier independently checks the new lotteries, whole-cell costs, support
ocal primal vertices, using the prior independent primitive parser rather
piecewise-affine operations. The nonlinear verifier uses symbolic rational
arbitrary-precision scalar integer induction instead of the constructor's
tions. Its recorded coverage now includes the complete finest (32, 256)
elected rows. The R42 structural checker separately reconstructs every
dual certificate without importing its constructor. The verification report
nich objects passed; it never substitutes cross-run equality or a mutation
proof.

erves all prior scientific material. Deterministic bounds, history-conditioned
iod analytic randomization example, adaptive local-LP analysis, 66 eco-
iled runs, and the stopped-control derivations remain available in full. The
lume preserves the prior article and technical supplement verbatim; the
pplies a direct map of the new results. The older 349-page historical annex
s an archival object rather than a prerequisite for reading this article. The
-portfolio problem is not relabeled solved: its inherited full-domain bound
818 does not meet its 0.01 target. A killed Bellman comparison supports
ut supplies neither missing action coverage nor a diffusion-discretization
(and Dupuis, 2001).

clusion established by the new primary calculation is precise. When random-
nt is admissible and costs attach to realized departures, common Markov

uncertainty and the economic margin for randomization

is translated into a robustness statement without treating designed estimated data. Fix a nominal model, initial law, and policy class. The set of alternative models has verified uniform bounds

$$|J_t^{p,0}| \leq a, \quad |V_t^M - V_t^0| \leq a, \quad |C_0^{p,M} - C_0^{p,0}| \leq b \quad (45)$$

for all p , state, and date, and every M in the set. Superscript 0 denotes the nominal model. The recursion in Section 6 supplies such bounds from uniform reward, transition, and enclosures on a common state space; no reset aggregation is needed for the robustness statement. Uniformity over policies is essential. A fitted residual for a proposed policy does not establish (45).

Verifiable operating and cost intervals: Let $\varepsilon > 2a$. A nominal policy p is ε -feasible at allowance ε in every alternative model. If L_+ is the nominal cost at allowance $\varepsilon + 2a$ and U_- is the nominal cost of such a tightened policy, the model-specific common-policy optimum belongs to

$$[\max\{0, L_+ - b\}, U_- + b].$$

Proof: The optimum that minimizes worst-case implementation cost over all alternative models. In the latter assertion the upper endpoint uses a common policy for each model.

Proof: For a proposed policy, $J_t^{p,M} \geq J_t^{p,0} - a \geq V_t^0 - \varepsilon + a \geq V_t^M - \varepsilon$. Conversely, if model M satisfies $J_t^{p,0} \geq J_t^{p,M} - a \geq V_t^M - \varepsilon - a \geq V_t^0 - \varepsilon - 2a$. Its cost in M is at least $L_+ - b$. Take infima for the model-specific problem, the first implication supplies one policy for all models and a robustly feasible policy in each model. Taking the worst-case cost $Q.E.D.$

The lottery also has an economically interpretable administration-cost certified cost of a feasible common lottery and L_D a lower bound for the lottery policy, for the same operating constraint and initial law. Suppose the lottery is unchanged by administering the lottery. A nonnegative extra cost at date t bounded by h_t increases its expected discounted cost by at most

$$U_R + H < L_D, \quad (46)$$

cheaper than every deterministic feasible policy even after this additional cost. The lottery does not require administration at every visited state; charging the

	0.01	10.81672	9.85794	0.133151
er	0.05	10.75071	9.85440	0.133151
al31001	0.01	0.46864	0.40232	0.009853
er	0.01	15.61281	15.46691	0.015872
er	0.05	15.47129	14.72413	0.081278
al31001	0.01	3.49393	3.36015	0.014553

nged development economy. Endpoints and the threshold are rounded outward. A positive
y smaller constant per-date overhead preserves a strict cost advantage over every feasible
26 rule/horizon/tolerance/initial-law margins are retained in the machine-readable results,
including zeros.

conservative. Deterministic rules are assumed not to pay the randomization-

date ceiling h , the certified break-even threshold is

$$h_* = (L_D - U_R)_+ \bigg/ \sum_{t=0}^{T-1} \beta^t. \quad (47)$$

es the strict ranking. Equality at h_* yields no strict conclusion. For model
erministic nominal lower bound at $\varepsilon + 2a$ and a nominal lottery feasible
ce the numerator by $(L_{D,+} - U_{R,-} - 2b)_+$. The current development
0: they calculate economic margins under known designed primitives, not
radii.

11. CONCLUSION

requirement becomes a global revision certificate only when continuation
cost of restoring feasibility are controlled together. Backward repair sup-
A date-dependent allowance ladder covers finite undiscounted problems,
action restrictions handle zero tolerance. These extensions retain the same
Markov objective rather than replacing it by a necessary-action relaxation.
te-state dynamics, the operating-action disadvantage provides a probability
ntation savings obey a regret cone. Their combination with reference-
a quadratic small-allowance bound for a root relaxation at fixed action
structural rate, not a polynomial guarantee for the unrestricted global
billed audit, tighter relaxation geometry can materially improve a certificate,
ount cases and solver stops remain in the evidence.
te disintegration supplies a separate controlled atomic continuum route. It
common policy by using a label recoverable from the current state, and its
verge when the finite fiber solves and primitive errors are refined. Exact
on remains a tractable complementary class; its twelve executed service
ly closed positive-cost intervals. The density exercise retains all primitive,
ns, including a failed sufficient tightening condition.

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ment gives exact primitive contracts, complete computational accounting, the
t, a discounted infinite-horizon tail result, and a theorem-to-checker map. The
d technical supplement are retained without modification in a separate historical
erical proposal, independent arithmetic verification, structural mathematical
and publication provenance are distinguished throughout.

LED-MODEL PRIMITIVE CONTRACT AND COMPLETE ACCOUNTING

s use the exact generator in `regret.py`, independently reconstructed by
are $i = 0, \dots, n - 1$. For each regime and action, four integer weights are
from

$$\} + u_1, 36 + u_2, 12 + 7a + u_3, 2 + 11a + u_4), \quad u_j \in \{0, 1, 2, 3\}.$$

nts and assign them to successors $\max(0, i - 1)$, i , $\min(n - 1, i + 1)$, and
ncident successors. The seeded standard-library pseudorandom generator
/action order. The stage reward is

$$c_a - \frac{av}{100000}, \quad c = (0, 9/100, 8/25, 11/20), \quad v \in \{1, \dots, 19\},$$

mediately after the four transition weights. The installed action is zero for
seed modulo two otherwise. The implementation cost is $(1 + i/(n - 1))$
f a realized departure. The terminal operating value is $i/[2(n - 1)]$, and
form. Transitions, rewards, and costs are constant over dates; Bellman
lly are not. All primitives and every declared seed are retained in the
sed proof objects.

ration is `commit 18cbcf41a4c35660be824f6984f68a1d38288a9`. The
amendment is `950b1f1ac2ebaf974d4ef528add8b2c2e4ce72b`. The
ll cases and changes only certificate accumulation and its theoretical
erical constructor, independent reader, and runner hashes are checked
es. This sequence is not called a fully code-frozen holdout. The old source
record remain in the R43 history directory.

I: Controlled audit: complete proposal and verification work

Coordinates	Variables	Nonzeros	Build	LP	Arithmetic	Check	Bytes
24	144	1102	0.005	0.006	0.011	0.015	1380
24	144	1150	0.005	0.004	0.010	0.017	1290
128	1040	8611	0.039	0.050	0.081	0.123	6804
128	1040	8867	0.040	0.072	0.084	0.130	7484
384	3408	29018	0.176	0.340	0.290	0.456	19833

68	6488	57088	0.307	2.328	0.576	0.895	37114
68	6488	58112	0.315	1.935	0.632	0.955	40608
536	13720	121698	0.852	30.052	1.000	1.951	2491
536	13720	123746	0.870	30.052	1.002	1.955	2495
448	19472	169793	1.635	30.078	1.608	3.468	3575
448	19472	173889	1.641	30.077	1.673	3.541	3583

ordinates are $nT(m-1)$, not just nT . Each row uses one root relaxation, and improvement is a one-node result; this is not a node-count scaling. 30-second cap. Model construction, exact constructor arithmetic, and are additional. Process high-water memory is recorded in the raw rows. Initial, that memory includes earlier allocations and cannot identify a y advantage. Comparing isolated peak memory would require separate comparison is asserted.

RTIFICATE ARITHMETIC AND INDEPENDENT RECONSTRUCTION

on as $\min c'z$ subject to $Ez = e$, $Az \leq b$, and $\ell \leq z \leq u$. Any rational $\mu \geq 0$ give the lower bound

$$b + \sum_j \min\{h_j \ell_j, h_j u_j\}, \quad h = c - E'y + A'\mu.$$

st $\nu' H_0$ and truncate at zero. A floating-point LP success flag is neither for this inequality. Numerical equality marginals and sign-corrected merely proposals for y, μ .

act rational denominators before rounding is unnecessarily expensive. and constructor instead use denominator 2^{40} . They round each constant for each residual coefficient, they accumulate dyadic lower and upper . The safe contribution is the downward-rounded minimum of

$$h_j^- \ell_j, h_j^- u_j, h_j^+ \ell_j, h_j^+ u_j.$$

s necessary when a savings coordinate can be negative. This procedure posed multipliers with the required sign, and keeps denominator growth ete lower endpoint, including reference cost, is retained as a rational

t, the constructor repairs backward toward the reference action. It margin before rounding nonreference probabilities down to denominator y is assigned to the reference action. At a fixed repaired continuation, regret change is at most $(m-1)2^{-40} \max_a |q_a - q_{a^*}|$. If the reserved reference slack, the constructor uses the feasible reference action.

1/100	0.00000001	0.00000001	0.00000001	0.00000001	0.00000001	0.00000001	0.00000001
1/1000	0.000156	1.03e+04	0.000715	4.99	376	767	767
1/20	6.2e-05	2.99e+04	4.73	1.27e+05	1.28e+05		
1/100	7.44e-05	2.53e+04	0.129	3.18e+03	4.04e+03		

mal approximations of retained rational quantities. Error recursions are rounded upward to
cient bound concerns an exact minimizing LP policy and exact repair; a numerical proposal
c losses are not included in this theoretical column. Actual certified widths in the main table
n be larger when a solver stops. Constants are not suppressed when they are uninformative.

choices. The independent reader ignores the repair routine and checks
ilities by the original operating and implementation Bellman equations.
nonnegative and sums to one, and every date/regime satisfies the original,
.

ndpoint reader imports neither the constructor nor NumPy, SciPy, or any
ucts the operating optimum, reference policy, disadvantage and savings
le bounds, all product envelopes, and every cone and Bellman constraint.
t audit reconstructs the declared random primitive sequence and checks
against its registered parameters. It also independently recomputes the
sums. These checks do not mechanically prove the analytic repair, rate, or
ns. Four retained mutations test rejection of a forged endpoint, negative
negative inequality multiplier, and invalid transition mass.

3. ALL CONTROLLED CONTINUUM ENDPOINTS

ersistent observed size $z \in [0, 1]$. Regime transitions are controlled but
Operating primitives are multiplied by $1 + z$ and implementation costs by
te allowance is $0.01/(1 + z)$; it is not 0.01 after rescaling. The following
endpoints used by every displayed partition.

II: Every finite endpoint used in the atomic continuum certifi-

Allowance	Lower	Upper	Width	Check (s)
0.01000000	7.765997	7.883084	0.117085733	0.054
0.00941176	7.822697	7.929826	0.107128306	0.054
0.00888888	7.872910	7.971275	0.098364582	0.053
0.00842105	7.918003	8.007721	0.089717171	0.053
0.00800000	7.958346	8.040570	0.082223429	0.054
0.00761904	7.994660	8.070364	0.075703281	0.053
0.00727272	8.027518	8.097495	0.069976171	0.054

00571428	8.173805	8.218688	0.044882388	0.054
00551724	8.192044	8.234190	0.042144623	0.054
00533333	8.209073	8.248540	0.039466645	0.054
00516129	8.224969	8.260548	0.035578105	0.054
00500000	8.239809	8.273341	0.033531010	0.054

s to $z = j/16$. Allowances are printed downward for display only; the use the exact rational allowance. Lower endpoints are rounded down widths up. The 2-, 4-, 8-, and 16-cell calculations reuse subsets of these to new model or initial law is substituted when the partition changes. The stable-fiber theorem, the bijectivity of the covariate evolution has an model that selects a continuation is recoverable from the current observed s can merge different labels. Selecting a different continuation for add memory and would not prove the stated common-Markov result. ident continuous-condition transition is not covered merely because it ors. The original affine and nonlinear condition models are therefore -width evidence.

4. EXACT SERVICE-ECONOMY CONTRACTS

$i = 0, \dots, n-1$. Demand, rewards, installed actions, costs, and initial main article. Ties in reward maximization use the lowest action index. three regime laws are as follows. In the banded law, $P_{i,i+1} = P_{i+1,i} = \rho_{ii}$ is the remaining row mass; other entries are zero. The cyclic law has $\rho_{in} = 1/4$. The dense law has $P_{ij} = 1/(2n) + (1/2)\mathbf{1}\{i = j\}$. Each dent of the action. These definitions, including the installed rule and constructed independently by the checker. space consists of n disjoint copies of $[0, 1]^2$ equipped with their Borel en convention assigns each modulo boundary a unique image. If (u, v) , then for every bounded Borel h , Fubini's theorem and translation give $\int h(S_1(u, v)) du dv = \int h(u, v) du dv$. The analogous identity $(u, v) \mapsto (u + u^2 \bmod 1, v)$. Their composition preserves uniform measure. Condi-gime successor, this is precisely the pushforward identity required by theorem. Its proof concerns arbitrary bounded Borel test functions, not Pointwise regime transition probabilities give the other aggregation om an individual state is nonetheless a finite sum of Dirac masses. alternative is $1 + \zeta f(u)f(v)$, where $f(u) = 2u_1 - 1$ on $[0, 1]^2$. Since it is a probability density for $|\zeta| \leq 1$, and its source average is the or a fixed source, its L_1 difference from the reset density is $|\zeta f(u)|/2$. ces is $|\zeta|/2$ (a supremum need not be attained on the half-open cell). n over test functions bounded by one. Probability total variation is

Variables	Nonzeros	LP (s)	Full LP (s)	Exact (s)	Check (s)	Bytes
256	538	0.0032	0.0036	0.0057	0.0065	2233
1024	2226	0.0063	0.0140	0.0266	0.0355	5648
4096	9058	0.0191	0.1127	0.1546	0.2239	17114
16384	36546	0.2618	1.2414	0.9772	1.6081	56086
256	496	0.0032	0.0036	0.0051	0.0063	1976
1024	2016	0.0079	0.0119	0.0248	0.0333	5123
4096	8128	0.0326	0.0753	0.1440	0.2201	15921
1024	5376	0.0074	0.0227	0.0391	0.0394	5135
256	538	0.0032	0.0038	0.0057	0.0064	2188
1024	2226	0.0064	0.0140	0.0267	0.0348	5616
4096	8128	0.0325	0.0781	0.1515	0.2075	15251
1536	3250	0.0079	0.0154	0.0321	0.0416	6577

ated regret program; “Full LP” retains operating and implementation the same common policy. Both use HiGHS with presolve and the same tolerances and 60-second solver limit. They solve the same rational mathe- their floating-point objective estimates are not the certificates. The exact rately evaluated rational policy and rational equality-multiplier residual ward allowance is a proposal device only. Policy and dual multipliers are ator 10^{12} ; the checker always uses the original declared allowance. Every ction was zero in the recorded execution; the feasibility assertion follows , not from that observation or the floating-point solver status.

Additionally reports model preparation time, solver iterations, original ons, and process high-water resident memory. Memory is measured per s and includes Python, NumPy, SciPy, and the optimizer; it is not isolated The independent checker runs in a separate optimizer-free process. Table descriptive timings, not statistical speed estimates. For the largest sparse ecker is more expensive than the LP proposal. This cost is included rather ublication overhead.

ram, order the equality multipliers as s_{ti} for simplex equations and d_{ti} for n $\ell_{ia} = \max_b r_{ib} - r_{ia}$, the reduced costs are

$$\beta^t w_{ti} k_{ia} - s_{ti} + \ell_{ia} d_{ti}, \quad h_{ti}^D = -d_{ti} + \beta \sum_j P_{ji} d_{t-1,j},$$

date term is zero at $t = 0$. The lower certificate is

$$\times \left\{ 0, \sum_{ti} s_{ti} + \sum_{tia} \min(0, h_{tia}) + \varepsilon \sum_{ti} \min(0, h_{ti}^D) \right\}.$$

that uses $R = \max_{i,a} |r_{ia}|$, $K = \max_{i,a} k_{ia}$, and $G = 0$. Starting at the error and zero payoff envelope, backward induction computes

$$b_t = \beta b_{t+1} + \beta \alpha_h B_{t+1}^C, \quad B_t^J = R + \beta B_{t+1}^J, \quad B_t^C = K + \beta B_{t+1}^C.$$

are the maxima over dates. The two finite problems have allowances the first supplies a lower endpoint; the second supplies a feasible lifted. Independently checked arithmetic widths are η_+ and η_- . No finite solve is tightening is nonpositive. The repair term is $D_C 4a_h / [(1 - \beta)(\varepsilon - 2a_h) +$

TABLE V

TESTED TERMS IN THE SUFFICIENT TRANSFER-WIDTH BOUND

η_+	η_-	$2b_h$	Repair bound	Total bound
$\times 10^{-8}$	2.291×10^{-8}	0.058461	73.581005	73.639466
$\times 10^{-8}$	2.291×10^{-8}	0.000058	1.153667	1.153725

al is computed from rational values, with decimal displays only for d interval width in the main article is appreciably smaller than this d. For $\zeta = 1/10$, a_h exceeds the permissible tightening threshold; that tual error budget and a failed sufficient condition. The exact structural because its two aggregation identities hold at that parameter as well. particular error bound from failure of the economic model or of the

TAIL BRIDGE FOR DISCOUNTED INFINITE-HORIZON REVISION

ual experiments are finite horizon. There is also a simple infinite-explicit terminal trust boundary. Consider time-homogeneous bounded and $0 < \beta < 1$. Let V^∞ be the exact operating fixed point and let a^* be izing selector. Allow common Markov policies to be time dependent, $(x) - \varepsilon$ for every date and state. Denote this infinite-horizon minimum ite-horizon revision minimum with terminal operating payoff V^∞ and tion cost.

enclosure: *For any initial law and $\varepsilon \geq 0$,*

$$v_T \leq K_R^\infty \leq v_T + \frac{K\beta^T}{1 - \beta}.$$

$_T, U_T]$ gives the infinite-horizon certificate $[L_T, U_T + K\beta^T/(1 - \beta)]$.

does not give a stationary-policy optimum: the admissible prefix is time-varying. We do not treat an approximate terminal value as exact, or an average-cost problem as exact. A numerical use requires a certified terminal fixed point and tail policy, and a bound on errors. No infinite-horizon execution is claimed in the tables.

8. DATA-DEPENDENT REPAIR ACCOUNTING

For finite-horizon common policies p, q , define the implementation action values $V_t^p + \beta P_t^a C_{t+1}^p(x)$. Their exact performance difference under the state law

$$J_0^p - J_0^q = \sum_{t=0}^{T-1} \beta^t \mathbb{E}_q \left[\sum_a (q_t(a | X_t) - p_t(a | X_t)) Q_{k,t}^p(X_t, a) \right].$$

Use Bellman equations and iterate the resulting q -transition recursion to prove $J_0^p - J_0^q \leq \alpha_t p_t + \alpha_t \delta_{a_t}$, its additional cost is at most the discounted q -occupancy of the action span of $Q_{k,t}^p$. For a given finite repaired policy this can be bounded by $\sup \alpha_t$. The generic stopping theorem still needs a uniform bound over all policies. A favorable realized repair cost does not replace that requirement. The new algorithm computes exact primal evaluation, and do not charge an unnecessary worst-case repair cost per endpoints.

9. ALL SIGNED SAVINGS IN THE RETAINED PRIMARY COHORT

We compute an exact re-tabulation of R40 rational endpoints. It does not change any primary optimizer. A positive signed saving $L_D - U_R$ proves a gain over the baseline policy; a nonpositive value establishes no such separation. None of the endpoints is equality of L_R and U_R .

Table VI: All 42 uniform-initial primary certificates and signed savings

Rule	ε	L_R	U_R	$L_D - U_R$	Width (%)
	0.01	4.617419	4.730105	0.206584	2.38
	0.05	3.664042	3.890120	1.041948	5.81
1001	0.01	0.085102	0.089277	0.021898	4.68
1001	0.05	0.026223	0.030155	0.044177	13.04
1002	0.01	0.120480	0.125168	0.024289	3.75

	0.01	0.000000	0.000000	0.000000	0.00
	0.05	0.000000	0.000000	0.000000	0.00
	0.01	0.000000	0.000000	0.000000	0.00
	0.05	0.000000	0.000000	0.000000	0.00
	0.01	10.503551	10.637940	0.172786	1.26
	0.05	9.449120	9.854394	0.896320	4.11
1	0.01	0.343445	0.402312	0.066330	14.63
1	0.05	0.104088	0.140458	-0.015946	25.89
2	0.01	0.225489	0.277716	0.058701	18.81
2	0.05	0.040296	0.059511	-0.015198	32.29
3	0.01	0.552828	0.653902	0.211403	15.46
3	0.05	0.086348	0.100439	0.244576	14.03
stress	0.01	9.173627	9.365726	0.125235	2.05
stress	0.05	7.983805	8.565982	0.682582	6.80
	0.01	0.000000	0.000000	0.000000	0.00
	0.05	0.000000	0.000000	0.000000	0.00
	0.01	0.000000	0.000000	0.000000	0.00
	0.05	0.000000	0.000000	0.000000	0.00
	0.01	15.317224	15.466905	0.145915	0.97
	0.05	14.142037	14.724122	0.747177	3.95
1	0.01	2.770956	3.360150	0.133784	17.53
1	0.05	1.584778	2.023532	0.379048	21.68
2	0.01	2.051734	2.385046	0.151782	13.98
2	0.05	1.195731	1.455145	0.309646	17.83
3	0.01	0.772482	1.010276	0.217318	23.54
3	0.05	0.093464	0.142107	-0.001624	34.23
stress	0.01	13.420033	13.655059	0.044193	1.72
stress	0.05	12.158413	12.877517	0.365025	5.58
	0.01	0.000000	0.000000	0.000000	0.00
	0.05	0.000000	0.000000	0.000000	0.00
	0.01	0.000000	0.000000	0.000000	0.00
	0.05	0.000000	0.000000	0.000000	0.00

REM-TO-CHECKER MAP AND EVIDENCE CLASSIFICATION

	a proof over arbitrary Borel polynomials; exact shear pushforward identity	arbitrary controlled atomic kernels or approximate lumpability.
Inter-	Independent rational probabilities, all-restart regrets, actual cost, and residual-corrected lower bounds for every new row	No reliance on solver success as a proof; no assertion about the continuum deterministic optimum.
	Nonzero operator error and full propagated budget, plus independent finite certificates at both adjusted allowances	No assertion that every error budget is useful; the failed tightening row is retained.
near	Complete independent scalar rebuild of the retained (32,256,1/2) object in the previously successful recorded run	Not a new upper policy, tighter operating tolerance, narrower optimum interval, or local re-run in R42.
ica-	Pre-execution case registration; content hashes and additive preservation checks	The source was not frozen in the case-registration commit; hashes do not verify a mathematical inequality.
eta-	Known-model adjustment costs and all-restart operating protection	No estimated primitives, empirical confidence sets, or monetary welfare estimates.

audit reports 92,274,688 support-action inequalities and 2,097,152 state-
 fully reproduces upper cost and lower bound

$$= \frac{1667622841980339}{140737488355328}, \quad L = \frac{2497217525995195}{281474976710656},$$

47483439/4294967296 < 1/2. Its arithmetic replay took approximately
 manifest retains the source commit, verifier hash, proof hash, successful
 and retrieved result. R42 does not misclassify this recorded prior run as a
 n.

11. UNCHANGED EARLIER TECHNICAL MATERIAL

ry files remain intact. The separate file HISTORY_R42.pdf combines
 cle and technical supplement without editing their pages; the latter already
 cle and supplement. Revision-relative labels in that archive are historical.
 d this supplement state R42's additions directly. The original primary and
 s, failed sufficient certificates, adaptive local-LP work, model sensitivities,

certificates for every state and date. A common randomized Markov continuation restarts. We construct a backward feasibility repair with an explicit cost modulus. A given form avoids an exact operating solve. It yields convergent global policy revision, two-state problems and an end-to-end approximation bound for continuous-state problems. Fitting certified total-variation aggregation. Exact rational spatial search and a heuristic comparator target the same constrained objective. A prospectively fixed budget experiment reports all budget stops. On the original 42 continuous-state configurations, whole-domain lotteries certify strict improvement over every policy in 27 cases. Nonlinear two-state lower bounds become positive, but gaps remain material. Independent checks distinguish arithmetic validity from assumptions. The results identify when local operating guarantees can be global revision certificates without changing the policy class.

Keywords: Dynamic programming, Policy revision, Constrained Markov control, Global optimization, Verified computation.

1. INTRODUCTION

One may wish to improve an installed rule without replacing every decision it makes. The cost then depends both on the decisions changed today and on the states reached by the rule tomorrow. A feasible revision must preserve the operating objective, and a cost-effective revision must economize on intervention along its own induced state transitions. These requirements create a constrained dynamic problem. Pointwise selection of improvements need not solve it.

A continuation must also be specified. Two histories that meet at the same date and state have the same continuation under a Markov rule. Allowing a different continuation for each state reached by remembered randomization can reduce costs, but changes the economic problem. Requiring a deterministic rule can increase implementation cost even when it is not necessary. We therefore take a common randomized Markov rule as the baseline. Performance is protected after every date and state; implementation cost is bounded by the stated initial distribution. A performance guarantee at every restart is not a requirement at every restart.

There is a constructive connection between these requirements and a complete characterization of policy that violates every operating constraint by at most δ can be repaired by adding a persistent random seed or a history variable. The repair improves performance at every state. Its additional implementation cost is bounded explicitly by a function of δ . Strict discounting and a positive operating allowance provide a natural regularity assumption is used. This repair gives an error bound for rational actions. For stationary actions it also gives a stopping theorem for the interval Bellman procedure used in the experiments. Unlike convergence to a necessary-action problem, the target object is the original common-continuation constrained optimum.

A new extension separates operating approximation error from policy-search error. It supplies a positive-margin check and a constructive global bound. It supplies a positive-margin check and a constructive

imitive errors give an explicit operating-tolerance adjustment and a prior modulus controls the remaining difference between the relaxed and original. The theorem makes the approximation assumption operational and guarantees weak convergence of an atomic transition law as total-variation error goes to zero, not silently replace discrete shocks by continuous noise. The paper separates three questions. First, does complete search return sound results for a Markov problem, and how do those intervals compare with a direct dynamic programming formulation? Second, does randomization change the conclusions about state-maintenance configurations? Third, can a nontrivial revision be made for linear, two-state dynamics and nonzero operating-witness error? Each question has an objective on the two sides of its interval. The eight new finite-state solver implementations were fixed before those environments were released. The 12 configurations are explicitly a development cohort, not a prospective cohort. The results are economically consequential. In the development cohort, 27 out of 28 deployments cost strictly less than a certified lower bound for every policy under the same uniform initial law. This is a statement about the optimization, not about a neural label attached to the installed rule. The results change the zero lower bound into a positive bound; its finest horizon is [0.8492, 0.8492]. That interval is informative but not tight. In the prospective cohort, both methods face the same node allowance and wall-clock limit. The randomization often improves the lower endpoint, but longer horizons retain the same bounds. The theory supplies completeness, not a claim that complete search is optimal.

1.1. Relation to the literature

Decision processes provide the natural starting point (Altman, 1999). The paper is particularly effective when the constraints are expectations under a prior. The prior formulation additionally requires every restarted problem to meet the same shared continuation. A separate occupation measure for each problem meets the continuity restrictions: at a shared date and state, its action probabilities are the same as the restart generated the occupancy. Dropping that restriction produces a prior-free finite formulation imposes it directly through a single policy variable. This observation does not preclude occupation-measure methods; it only requires they must preserve in this problem. The paper's treatment of constrained control are not new. Saldi (2018) proves value constraints for average-cost constrained problems, and develops approximately optimal policies under additional continuity conditions. Saldi, Linder, and Yüksel develop a systematic treatment of quantized stochastic control. Our transfer result combines the restart constraints and combines the within-cell reset identity with an average-cost repair. It states the total-variation condition needed to average unrestarted policies rather than assuming that a convenient policy sieve is automatically available.

different from our known-model, all-restart certification problem. Safe t (Laroche, Trichelair, and Tachet des Combes, 2019) and approximation- d dynamic programming (Munos and Szepesvári, 2008, Scherrer, 2013) model uncertainty, policy approximation, and optimization error. We claim comparison against methods given less information.

ation ingredients are classical. Convex envelopes of factorable products ick (1976); the numerical LP proposals here use a modern simplex im- ed with Huangfu and Hall (2018). We do not attribute novelty to bilinear d search, or weak duality. The contribution is a common-policy economic al approximation errors can be repaired into an admissible deployment, d stopping bound. We use the direct McCormick formulation as a serious or, not as a deliberately weak price portfolio. A rational residual correction multipliers into lower bounds even when floating-point dual feasibility is ration of numerical proposal and proof follows the broader program of (Rump, 2010).

nization emphasizes the difference between checking finitely many con- ing them on a region (Hettich and Kortanek, 1993). Our continuous-state nclude whole-cell inequalities and isolated switching points. Information- pply another route to dynamic-programming lower and upper bounds un, 2010, Brown and Smith, 2014); they are related to, but not identical rt functions used below. The support functions remain useful accelerators price set is asserted to exhaust a global dual.

ivation is adjustment rather than unconstrained operating optimization. s has long emphasized model structure, approximation error, and state- dd, 1998, Brumm and Scheidegger, 2017). Empirical replacement models discipline operating primitives with data. Our revision cost is instead the an an installed decision rule, not a cost paid whenever the previous physical ng physical switching costs would require including the previous action in maintenance environments are designed, known-model economies. Their and implementation units are not estimated monetary welfare, and their timates of empirical prevalence. The general convergence results, rather ation, carry the methodological argument.

2. THE ECONOMIC PROBLEM

Borel space, $A = \{1, \dots, m\}$ a finite action set, and $t = 0, \dots, T$. Rewards ion costs $k_t(x, a)$, and terminal operating payoff $g(x)$ are bounded Borel tion kernels P_t^a are known Borel probability kernels. Write $|r_t| \leq R$, K . Unless otherwise indicated, $0 < \beta < 1$ and $\varepsilon > 0$. Define

$$= r_t(x, a) + \beta P_t^a v(x), \quad V_T = g, \quad V_t = \max_{a \in A} T_t^a V_{t+1}. \quad (1)$$

$$C_t = \sum_a p_t(a | x) \{k_t(x, a) + \beta F_t C_{t+1}(x)\}. \quad (3)$$

histories reaching (t, x) . Randomization is newly drawn conditional state; a persistent hidden coin indexing different future policies is not

by law ν , the central object is

$$\inf_{\mathcal{F}_R(\varepsilon)} \int C_0^p d\nu, \quad \mathcal{F}_R(\varepsilon) = \{p : J_t^p(x) \geq V_t(x) - \varepsilon \forall t, x\}. \quad (4)$$

restricts p_t to vertices of the simplex and defines K_D . An operating-
e, so both classes are nonempty. On a finite state space, the product
impact, J^p and C^p are continuous, and the feasible set is closed. The
ained. We use infimum, not an unproved function-space minimum, in
on.

(x) , the principal cost specification is

$$c_t(x, a) = w_t(x) \mathbf{1}\{a \neq a_t^0(x)\}, \quad w_t \geq 0. \quad (5)$$

action actually drawn. Its expectation under a lottery is not an indicator
table action differs from the installed action. The unchanged installed
cost even when it is operating-infeasible. Consequently, dividing an
revision cost is undefined, not an informative relative-gap statistic. We
 $U > 0$, and report zero-cost exactness separately.

development economy weights all normalized initial conditions equally.
distribution, not an estimated stationary law. We also integrate exactly
ties $2x$ and $2(1 - x)$. These calculations change the initial objective
e operating constraints and installed rule unchanged. A pointwise
er arbitrary initial laws, including atoms; an almost-everywhere cost

ASIBILITY REPAIR AND COMPLETE POLICY SEARCH

$$D_C = K \sum_{s=0}^{T-1} (s+1) \beta^s, \quad (6)$$

$$D_J = \max_{1 \leq h \leq T} \left\{ 2R \sum_{s=0}^{h-1} (s+1) \beta^s + 2Gh\beta^h \right\}. \quad (7)$$

ning horizons in D_J matters: the terminal term $h\beta^h$ need not increase
between action distributions is one half of their ℓ_1 distance.

7. Adding β times the next-date value difference and iterating gives For operating rewards use the interval length $2R$ for each stage and $2G$ d. A reward at distance s can be affected by $s + 1$ action changes, and the changes. Taking the maximum over h gives (7). The same conclusions horizon coupling, but do not require one. *Q.E.D.*

Common-continuation repair: Suppose $p \in \mathcal{F}_R(\varepsilon + \delta)$, $\delta \geq 0$. There is a Borel policy $\hat{p} \in \mathcal{F}_R(\varepsilon)$ with $J_t^{\hat{p}} \geq J_t^p$ for every (t, x) and

$$\rho_\varepsilon(\delta) := \frac{\delta}{(1 - \beta)\varepsilon + \delta}, \quad \int C_0^{\hat{p}} d\nu \leq \int C_0^p d\nu + D_C \rho_\varepsilon(\delta). \quad (8)$$

$$0 \leq K_R(\nu, \varepsilon) - K_R(\nu, \varepsilon + \delta) \leq D_C \rho_\varepsilon(\delta). \quad (9)$$

backward, with $J_T^{\hat{p}} = g$. Given a repaired continuation, put $q_a = T_t^a J_{t+1}^{\hat{p}}$, $q_* = \max_a q_a$, and $b = V_t - \varepsilon$. By the induction hypothesis, $q_p \geq J_t^p \geq V_{t+1} - \varepsilon$ implies $q_* \geq V_t - \beta\varepsilon$. If $q_p \geq b$, retain p_t . Otherwise choose action a_* and set

$$\hat{p}_t = (1 - \alpha)p_t + \alpha e_{a_*}, \quad \alpha = \frac{b - q_p}{q_* - q_p}.$$

$\varepsilon \in (0, \delta]$, the denominator is at least $(1 - \beta)\varepsilon + d$. Thus $\alpha \leq \rho_\varepsilon(\delta)$ and all operations are Borel; the division is used only where the denominator is positive. The construction uses only t and x and the already fixed next-date continuation. Apply the construction to arbitrarily near-minimizers at tolerance $\varepsilon + \delta$ to obtain (9). *Q.E.D.*

convex combination of whole policies. Such a combination, if selected, would enlarge the information state. Here the mixture is selected anew from the repaired continuation. The slack $(1 - \beta)\varepsilon$ explains why the theorem requires a positive tolerance. No vanishing modulus at $\beta = 1$ or $\varepsilon = 0$ is present.

1.1. Certified search without an exact operating solve

the preceding statements is not an oracle that the method must possess. The oracle supplies the slack needed by the repair. This matters when operating is more expensive than policy search alone, is expensive. Let $L_t \leq V_t \leq U_t$ be bounded

ent, let $m = |A|$, let $M \geq 1$ be an integer, and set $\theta_M = \min\{1, (m - 1)/M\}$. The same primitive transition and reward function check and the repair. Error in a different operator must first be fixed. A small residual is not, by itself, a certificate that $U \geq V$; the witness condition are also required.

Lemma 4 (Backward-driven repair and global error): *Suppose $\sigma > 0$ and $\delta \geq 0$. If a common policy $\hat{p}$ satisfies $J_t^{\hat{p}} \geq U_t - \varepsilon - \delta$ for all $t < T$, the backward construction produces a common policy $\hat{p}$ such that*

$$J_t^{\hat{p}} \geq V_t - \varepsilon, \quad J_t^{\hat{p}} \geq J_t^p, \quad \sup_{t,x} \text{TV}(\hat{p}_t, p_t) \leq \frac{\delta}{\sigma + \delta}. \quad (11)$$

Theorem 5 (Cost bound for backward-driven repair): *On a finite state space, let $\hat{p}$ be a common policy among rational-lattice policies satisfying $J_t^{\hat{p}} \geq L_t - \varepsilon - D_J \theta_M$. Let U be the common upper-witness, and let its cost be $U_{M,L,U}$. Then*

$$\max\{0, v_{M,L} - D_C \theta_M\} \leq K_R(\nu, \varepsilon) \leq U_{M,L,U}, \quad (12)$$

$$\max\{0, v_{M,L} - D_C \theta_M\} \leq D_C \left\{ \theta_M + \frac{e + D_J \theta_M}{\sigma + e + D_J \theta_M} \right\}. \quad (13)$$

Lemma 6 (Convergence to the actual constrained optimum using backward-driven repair): *As $d \rightarrow 0$ give convergence to the actual constrained optimum using backward-driven repair. All displayed errors and the stopping time are bounded by the cost of the backward-driven repair.*

Lemma 7 (Backward continuation and put): *Let $q_a = T_t^a J_{t+1}^{\hat{p}}$, $q_p = \sum_a p_t(a | x) q_a$, and $b = U_t$. On date $J_T^{\hat{p}} = U_T$; at every earlier date the induction hypothesis gives $q_a \geq U_t$ and (10) imply*

$$\max_a q_a \geq \max_a T_t^a U_{t+1} - \beta \varepsilon \geq U_t - d - \beta \varepsilon = b + \sigma.$$

Lemma 8 (Backward continuation and put): *Let $q_p \geq J_t^p \geq b - \delta$. Where $q_p < b$, set $\alpha = (b - q_p)/(q_* - q_p)$. Let $z = b - q_p \in (0, \delta]$, one has $\alpha \leq 1$. Otherwise leave p_t unchanged. This proves (11); the cost bound follows from maximization and division on a strictly positive-denominator Borel measurable function.*

Lemma 9 (Rounded repair): *Let $\hat{p}$ be a common policy as in Theorem 3. Its rounded cost $J_t^{\hat{p}} \theta_M \geq L_t - \varepsilon - D_J \theta_M$, and its cost increases by at most $D_C \theta_M$. This is a lattice minimizer is at worst $e + D_J \theta_M$ below the upper-witness cost. A lattice maximizer is at worst $e + D_J \theta_M$ below the upper-witness cost. Clipping the lower endpoint at zero. $Q.E.D.$*

3.2. A complete finite-state construction

es. For an integer $M \geq 1$, let $\mathcal{P}_M$ be the policies whose action probabilities are multiples of $1/M$. Put $\theta_M = \min\{1, (m-1)/M\}$ and $\delta_M = D_J\theta_M$. Evaluate every common Bellman equations and let

$$v_M = \min \left\{ \int C_0^p d\nu : p \in \mathcal{P}_M, J_t^p \geq V_t - \varepsilon - \delta_M \forall t \right\}.$$

For these comparisons, values, and policy probabilities use rational arithmetic.

Convergence to the constrained optimum: *Repair a minimizer defining v_M by the cost by U_M . Then*

$$v_M - D_C\theta_M \leq K_R(\nu, \varepsilon) \leq U_M, \quad U_M - L_M \leq D_C\{\theta_M + \rho_\varepsilon(\delta_M)\}. \quad (14)$$

Positive absolute accuracy is reached at a finite M . *At most $\binom{M+m-1}{m-1}^{nT}$ repairs are needed at that level. Running maxima of lower bounds and running minima of upper bounds are nested certificates without changing their validity.*

For the first $m-1$ probabilities of any feasible policy down to multiples of $1/M$, the remainder to the last action. The total-variation error is at most $(m-1)/M$, and Lemma 1 shows that the rounded policy belongs to the relaxed finite set and has total cost plus $D_C\theta_M$. Hence $v_M - D_C\theta_M \leq K_R$. Theorem 1 applied to the relaxed set gives $U_M \leq v_M + D_C\rho_\varepsilon(\delta_M)$. The weak-composition count gives the number of repairs at each date and state. The displayed error tends to zero. *Q.E.D.*

controls distance to (4), not distance to a one-step relaxation. Its count is $\binom{M+m-1}{m-1}^{nT}$; it is a completeness construction, not a polynomial-time claim. The state problem can instead be solved by evaluating its m^{nT} vertices with Lemma 1. The randomized repair is neither required nor valid as a deterministic.

3.3. The executed spatial search

write p_{ti} for the probability of action 1. A policy box $Q = \prod_{ti} [\ell_{ti}, u_{ti}]$ contains coordinates across all restarts. Backward interval Bellman evaluation gives lower values $\bar{C}$ and unconstrained implementation lower values $\underline{C}$ for that box. The search can contract its coordinate intervals: if q_0^+, q_1^+ are next-date operating costs, the possible p must satisfy

$$(1-p)q_0^+ + pq_1^+ \geq V_t(i) - \varepsilon. \quad (15)$$

stops when their difference is no greater than the requested tolerance, interval when the preassigned budget is exhausted.

opping bound for interval Bellman search: *Let $w = \max_{ti}(u_{ti} - \ell_{ti})$. Its midpoint-repair cost minus its interval Bellman lower bound is at*

$$H(w) = D_C\{w + \rho_\varepsilon(D_J w)\}. \quad (16)$$

cap, the best-bound bisection procedure reaches every prescribed many steps. In particular, choose $w_ > 0$ with $H(w_*) \leq \eta$. A sufficient h is*

$$d \left\lceil \log_2 \frac{1}{\beta^{T-1} w_*} \right\rceil, \quad d = nT, \quad (17)$$

by zero when its argument is at most one. The associated full binary- ential in this depth. Adding valid McCormick lower bounds cannot guarantee.

rectangular probability restrictions, ordinary Bellman maximization operating value simultaneously at every state. It is a common Markov incompatible histories. A surviving contracted box has its operating ε at every date and state. Each such maximizing policy differs from tal variation at most w . Lemma 1 therefore makes the midpoint $D_J w$ - ne same rectangular Bellman argument for minimum cost gives a lower dpoint cost minus $D_C w$. Repair and (8) imply (16). A box satisfying nfinement while the global gap exceeds η : its repaired cost is already so its lower bound is within η of the global incumbent or higher. On ighted longest-side bisection makes every physical width at most w_* by re finitely many nodes up to that depth. Valid stronger lower bounds boxes. The supplement details the effect of the necessary contractions; and preserve every feasible point. *Q.E.D.*

theorem for the implemented search rule, not an inference from favorable s why a 60-second run can end with a large gap: a finite worst-case y large. The numerical results report that distinction directly.

ERROR BUDGET FOR CONTINUOUS-STATE APPROXIMATION

alone does not justify replacing a continuous model by a grid. The itions permit such a transfer without restricting the original policy to

errors d_r, d_k, d_g and α_h satisfying

$$|k - k_h| \leq d_k, \quad |g - g_h| \leq d_g, \quad |(P_t^a - Q_{h,t}^a)f(x)| \leq \alpha_h \|f\|_\infty$$

rel f . The approximation rewards obey the same bounds R, G as the R, G are enlarged to cover both models. All finite transition entries and putable with verified directions.

uses the operator convention for total variation; it is not the half- ℓ_1 probabilities above. Define $M_{J,t} = R \sum_{s=0}^{T-t-1} \beta^s + G \beta^{T-t}$ and $M_{C,t} =$

$$d_g, \quad a_t = d_r + \beta a_{t+1} + \beta \alpha_h M_{J,t+1}, \quad (18)$$

$$0, \quad b_t = d_k + \beta b_{t+1} + \beta \alpha_h M_{C,t+1}, \quad (19)$$

$$\max_t a_t, \quad b_h = b_0.$$

mum of the finite common-Markov model at operating allowance s and

l-to-end transfer: Under Assumption 1, if $\varepsilon > 2a_h$, then

$$v_h(\varepsilon + 2a_h) - b_h \leq K_R(\nu, \varepsilon) \leq v_h(\varepsilon - 2a_h) + b_h. \quad (20)$$

utation returns intervals $[L_+, U_+]$ and $[L_-, U_-]$ for these relaxed and widths at most η_+ and η_- . A valid global interval is

$$[\max\{0, L_+ - b_h\}, U_- + b_h], \quad (21)$$

st

$$\eta_+ + \eta_- + 2b_h + D_C \frac{4a_h}{(1 - \beta)(\varepsilon - 2a_h) + 4a_h}. \quad (22)$$

e errors tend to zero and both finite-solver widths tend to zero, this interval al measurable-policy optimum. Its measured width supplies a termination nomic objective.

measurable Markov policy p , not necessarily cellwise constant. Backward ginal and reset-model Bellman equations gives (18)–(19). Bellman com- operating error for optimal values. In the reset model, averaging the $t \mid i) = \int p_t(a \mid x) \mu_i(dx)$, averages its value exactly: after every transition again μ_j , and stage rewards and costs are cellwise constant. This identity cluding the common continuation. An original feasible policy has reset

4.1. Where the transfer theorem applies

a transition density with computable uniform continuity in the source variable and a computable density in the destination variable can furnish the reset measure. Reference measures must also represent the initial law as in Assumption 1. Shape-regular uniform cell measures and a uniform initial law, a uniformly bounded density gives a computable $\alpha_h = O(h)$ on a shape-regular mesh. Cellwise approximations give $d_r, d_k, d_g = O(h)$ when their moduli are Lipschitz. An installed rule can be retained as exact partition boundaries. In $(\eta_+ + \eta_-)$ at fixed positive ε and $\beta < 1$, with explicitly displayed cells still scales as $h^{-\dim X}$; the theorem does not remove the curse of

block model and the nonlinear two-state model below have atomicity. Putting their successor atoms into cells generally leaves the total-variation error zero. We therefore do *not* apply Theorem 4 to those models by asserting that the reported intervals are direct whole-domain bounds for the unchanged model. This is valid without a convergence claim for that particular spatial discretization. The function prevents the approximation theorem from being used as an accelerator for closing an atomic-model gap.

ACCELERATORS THAT ACCELERATE THE GLOBAL PROBLEM

and the transfer theorem provide the convergence argument. The inherited accelerators are useful accelerators and diagnostic tools, but are not substituted for that

Deterministic outer restrictions and measurability

bounded Borel operating witnesses, including the terminal date. Every deterministic policy must select an action in

$$\mathcal{N}_t^{L,U}(x) = \{a : L_t(x) - T_t^a U_{t+1}(x) \leq \varepsilon\}. \quad (23)$$

continuation cannot exceed $V_{t+1} \leq U_{t+1}$. The set is nonempty because the action satisfies the test. Its graph is Borel, since it is a finite union of sets. Forward minimization of $k_t + \beta P_t^a B_{t+1}$ over this set yields a Borel lower bound. A minimizer is a Borel selector. Thus the general-state recursion does not require a measurable-selection assertion.

restriction outer class, not the original constrained problem. A selected action certifies that later losses that violate later operating requirements. Equality between the lower bound and a separately feasible policy's cost certifies $K_D(\nu, \varepsilon)$. Equality certifies more. Tightening L, U alone does not force equality; complete equality closes the remaining structural gap.

the maximum of zero and finitely many such terms, using L_t in place of V_t in a valid next-date lower function B_{t+1} , define

$$(x) = \left\{ p \in \Delta(A) : \sum_a p_a T_t^a U_{t+1}(x) \geq L_t(x) - \varepsilon \right\}.$$

$$\left\{ F_t(x), \min_{p \in \mathcal{D}_t(x)} \sum_a p_a \{k_t(x, a) + \beta P_t^a B_{t+1}(x)\} \right\}, \quad B_T = 0. \quad (24)$$

tribution of any admissible policy belongs to the local constraint, and its minimizes B_{t+1} . Induction proves that (24) is a global randomized cost lower bound. We prove that the local optimizer can be combined into a feasible policy or that there is no global duality gap.

and one local operating inequality, the local polytope has only pure feasible vertices on boundary mixtures. Our new primary calculation evaluates these vertices arithmetically on directed whole-cell bounds. Unlike the earlier diagnostic which uses the nonzero F_t at every date and is paired with a feasible policy for a global randomized objective. It improves that lower endpoint in 23 of the 42 prior adaptive local-LP theorem and all of its experimental records remain unchanged; their target is still the local relaxation, not (4).

5.3. Whole-domain randomized deployments

Our construction supplies feasible upper policies without changing the atomic feasible next-date value J_{t+1} , write $q_0 = T_t^{a_0} J_{t+1}$, $q_* = \max_a T_t^a J_{t+1}$, and retain the installed action. Otherwise mix it with a lowest-index maximizing policy

$$p_t^{\text{change}}(x) = \frac{b(x) - q_0(x)}{q_*(x) - q_0(x)}. \quad (25)$$

is positive wherever this expression is used. The resulting operating value and is feasible at every state. In the piecewise-affine model, J_t remains piecewise-affine and the mixing probability can be a ratio of affine functions. Cost is therefore also represented as an exact piecewise-affine function.

is isolated switching points in every operating identity. Directed cost boxes are three absolutely continuous initial laws. All affine transition slopes are from a finite set of switching points and its finite-horizon preimages have zero measure laws. That fact justifies almost-everywhere cost enclosures; it does not require a restart operating constraint or using the same cost enclosure for a

larger LP bound and an LP-proposed feasible candidate after repair.

us is not a certificate. For a minimization relaxation $c'z$ subject to every rational vector $\lambda \geq 0$ gives

$$c'b + \sum_j \min\{(c + A'\lambda)_j \ell_j, (c + A'\lambda)_j u_j\} \leq \min c'z. \quad (26)$$

$c'(Az - b) \leq 0$ and minimizing the resulting affine function over the solver marginals are evaluated in (26) using exact rational arithmetic. the interval Bellman lower bound; a floating infeasibility declaration box. A separate verifier rebuilds every relaxation and every search-tree ve model.

6. NUMERICAL EVIDENCE

model class, information, initial distribution, and operating allowance reports every configuration; no failed or budget-stopped case is removed. records proof-object sizes, memory, wall time, the exact environment, n independent arithmetic checks and shared-code validation.

Prospectively fixed common-Markov problems

new environments, we committed their seeds, joint parameter changes, ap 4,096, wall-clock cap 60 seconds per solver and case, and absolute The freeze is commit 13fd3ee; it follows development seed 390001 eluated seeds. Both solvers see the same known model. The two-state I high condition; four-state models use a pair of condition indicators their transition probabilities. Horizons are 3, 4, 8, and 12; discount llowances are 0.01 or 0.05; operating prices, transition persistence, initial masses vary jointly. The exact formulas and resulting rational ment and proof objects.

nt pairs, including unfinished searches. A positive absolute-gap certifi- small relative-gap certificate when the optimum is small. We therefore laxation often improves the lower endpoint substantially, but costs tate longer-horizon cases expose the combinatorial burden: the remain- by a success count. The comparison does not establish that interval s direct global programming. Instead it identifies the value and cost of der matched information and caps.

the unchanged continuous-state development cohort

nce model has $x \in [0, 1]$, three actions, two shocks with probabilities stage reward $x - c_a$ with $c = (0, 3/20, 9/20)$, terminal payoff $x/2$,

8	I	7.2534	9.0046	19.45	17.19
8	M	8.5290	9.0046	5.28	60.41
12	I	7.1419	7.5553	5.47	16.32
12	M	7.4446	7.5553	1.46	60.40
3	I	2.6157	2.6217	0.23	10.82
3	M	2.6206	2.6217	0.04	35.99
4	I	0.0233	0.0244	4.00	0.47
4	M	0.0235	0.0244	3.13	2.60
8	I	4.6631	7.8481	40.58	60.04
8	M	6.1990	7.8481	21.01	60.41
12	I	4.8936	6.6439	26.34	60.04
12	M	5.8606	6.6439	11.79	60.43

M: verified McCormick search. Bounds concern the identical common randomized Markov reported times include model construction and the operating solve; the search’s 60-second caps, not by interrupting a proof-producing node. A zero upper cost is an exact zero-cost case. I stop reasons, node counts, and memory appear in the supplement.

+ x . Its two affine successors for actions 0, 1, and 2 respectively are

$$), \quad (17x/20 + 1/10, 17x/20 + 7/50), \quad (x/10 + 4/5, x/10 + 17/20).$$

es, horizons 4, 8, and 12, and allowances 0.01, 0.05 are unchanged. Neural fixed computational incumbents, not estimated economic policies. Exact inexpensive in this model and their computation is charged. We do not use performance when operating approximation itself is difficult.

erministic comparison has 12 zero-cost and 12 positive-cost integrated positive-cost equalities occur at horizon 4, two at horizon 8, and none at imum deterministic relative gaps at each horizon are reported alongside supplement. In the horizon-8 neural31003 case at $\varepsilon = 0.05$, the two initial identical pointwise, but the arrays differ at later restart dates. Thus initial ly equality at every restart. The new diagnostic records the full rational

5) certify strict improvement over *every* feasible deterministic policy in initial configurations (Table II). The horizon-12 defer case at allowance as a randomized interval $[15.3172, 15.4670]$ and a positive deterministic esses the economically relevant policy-class difference directly, without restart history-conditioned frontier. The 12 zero-cost cases are already and nonnegativity and are not counted as nontrivial search successes. All omized intervals retain a reported nonzero width; the narrowest and widest distinguishable in the full table.

al density to $2x$ yields 28 strict deterministic separations; changing it to These are sensitivity results on the same developed environments, not ions. The connected local program improves 23 uniform lower endpoints,

randomized upper cost is below the deterministic lower bound for the same initial law
n that the randomized optimum is attained by that candidate. Improved lower compares
und with the prior randomized support lower bound. All 42 uniform-initial rows and
both nonuniform sensitivity tables are retained.

TABLE III

SELECTED NONLINEAR COMMON-MARKOV INTERVALS

axis	Lower	Upper	Gap/upper (%)	Independent check
64	0.9438	3.9348	76.01	full
128	4.1908	7.6644	45.32	not independent
64	6.5389	16.7884	61.05	full
128	8.1264	12.5226	35.11	full R40
256	8.8718	11.8492	25.13	not independent

quare and operating allowance $1/2$. Inherited full independent scalar checks cover
R40 additionally replays the complete (32, 128) object, including the candidate upper
ound. The (32, 256) result retains directed construction and structural checks, not the
independent label of adjacent rows.

izon cases; it thus affects the primary global interval rather than a
ostic. It still does not establish global strong duality or close every

6.3. Nonlinear two-state bounds

model has $(x, y) \in [0, 1]^2$, four actions, bilinear transitions, two shocks,
rminal payoff $(x + y)/4$, and revision weight $1 + (x + y)/2$. Its exact
ation is reproduced in the supplement. Whole-box operating witnesses
s use outward arithmetic. The new lower construction uses 11 global
nnegative local multipliers, including zero, at every date. It is paired
feasible upper policies. We run no new upper-portfolio competition
ual proposal success fractions as evidence of optimality.

point was not forced by the definition of revision cost: the operating
artures from the installed rule. Nonzero support supersolutions now
 $T = 32$, $N = 256$, the interval is $[8.8718, 11.8492]$, with relative width
e improvement over $[0, 11.8492]$, but it is not a near-exact optimum
alability result. Both endpoints and the all-restart operating inequality
at $T = 32$, $N = 64$, and the R40 audit extends this to the complete
Memory for the retained support arrays grows as $8N^2(T + 1)(11 + 4)$
rrays and interpreter overhead; the machine record reports the actual
-water mark.

longs to the inherited exploratory nonlinear design. It is not made
ly by normalizing state variables: the operating scale and remaining

three distinct burdens. Its mathematical reduction must preserve the target arithmetic must check the asserted inequalities with the correct direction. correspond to the checked objects. Hashes address the third burden, not ident arithmetic addresses part of the second burden and can expose a but is not a proof that every economic assumption is appropriate.

verifier uses SymPy rationals and imports neither search implementation rebuilds the common Markov equations, primitive contracts, every policy-ssary contraction, the LP residual bounds, and the global leaf minimum. verifier independently checks the new lotteries, whole-cell costs, support local primal vertices, using the prior independent primitive parser rather piecewise-affine operations. The nonlinear verifier uses symbolic rational arbitrary-precision scalar integer induction instead of the constructor's tions. It checks two complete representative objects, not merely selected n report specifies precisely which objects passed; it never substitutes a mutation test for an inequality proof.

erves all prior scientific material. Deterministic bounds, history-conditioned od analytic randomization example, adaptive local-LP analysis, 66 eco-ailed runs, and the stopped-control derivations remain available in full. es the prior article and technical supplement verbatim, with a content age historical annex remains unchanged as an archival object rather than a g this article. The stopped consumption–portfolio problem is not relabeled all-domain bound of approximately 7.1818 does not meet its 0.01 target. A arison supports witness propagation but supplies neither missing action on-discretization error bound (Kushner and Dupuis, 2001).

clusion established by the new primary calculation is precise. When random- nt is admissible and costs attach to realized departures, common Markov educe the minimum implementation cost relative to every deterministic nclusion holds on the certified configurations and initial laws. It does not tly organizations permit such lotteries, how costly they are to administer, re effect of a real deployment. Fixed randomization overhead, parameter rical calibration would require additional primitives and certificates. The dout study tests a fixed algorithm beyond its development seed, but is not a on theorem.

Model uncertainty and the economic margin for randomization

can be translated into a robustness statement without treating designed as estimated data. Fix a nominal model, initial law, and policy class. set of alternative models has verified uniform bounds

$$|J_t^M - J_t^{p,0}| \leq a, \quad |V_t^M - V_t^0| \leq a, \quad |C_0^{p,M} - C_0^{p,0}| \leq b \quad (27)$$

feasible operating and cost intervals. Let $\varepsilon > 2a$. A nominal policy at allowance $\varepsilon - 2a$ is feasible at allowance ε in every alternative model. If L_+ is at allowance $\varepsilon + 2a$ and U_- is the nominal cost of such a tightened model-specific common-policy optimum belongs to

$$[\max\{0, L_+ - b\}, U_- + b].$$

es the optimum that minimizes worst-case implementation cost over alternative models. In the latter assertion the upper endpoint uses a nt policy for each model.

osed policy, $J^{p,M} \geq J^{p,0} - a \geq V^0 - \varepsilon + a \geq V^M - \varepsilon$. Conversely del M satisfies $J^{p,0} \geq J^{p,M} - a \geq V^M - \varepsilon - a \geq V^0 - \varepsilon - 2a$. Its and its cost in M at least $L_+ - b$. Take infima for the model-specific problem, the first implication supplies one policy for all models and ry robustly feasible policy in each model. Taking the worst-case cost *Q.E.D.*

paration also has an economically interpretable administration-cost certified cost of a feasible common lottery and L_D a lower bound for tic policy, for the same operating constraint and initial law. Suppose amics are unchanged by administering the lottery. A nonnegative extra late t bounded by h_t increases its expected discounted cost by at most

$$U_R + H < L_D, \quad (28)$$

heaper than every deterministic feasible policy even after this additional does not require administration at every visited state; charging the ervative. Deterministic rules are assumed not to pay the randomization-

ceiling h , the certified break-even threshold is

$$h_* = (L_D - U_R)_+ \bigg/ \sum_{t=0}^{T-1} \beta^t. \quad (29)$$

the strict ranking. Equality at h_* yields no strict conclusion. For model inistic nominal lower bound at $\varepsilon + 2a$ and a nominal lottery feasible he numerator by $(L_{D,+} - U_{R,-} - 2b)_+$. The current development hey calculate economic margins under known designed primitives, not i.

er	0.01	10.75071	9.85440	0.133151
al31001	0.01	0.46864	0.40232	0.009853
er	0.01	15.61281	15.46691	0.015872
er	0.05	15.47129	14.72413	0.081278
al31001	0.01	3.49393	3.36015	0.014553

anged development economy. Endpoints and the threshold are rounded outward. A positive y smaller constant per-date overhead preserves a strict cost advantage over every feasible 26 rule/horizon/tolerance/initial-law margins are retained in the machine-readable results, including zeros.

8. CONCLUSION

requirement becomes a global revision method only when one can control compatibility and the cost of restoring feasibility. The backward repair for positive tolerance and strict discounting. Combined with completeuces a convergent certificate for the actual common randomized Markov with a verified reset approximation, it yields an explicit end-to-end error icted measurable-policy problem. These results retain the optimization acing it by a necessary-action or finite-price relaxation. show both the benefit and the burden of this construction. Whole-domain substantive deterministic-versus-randomized comparison on the original Connected support bounds improve the same primary objective, and over bounds replace feasibility-only intervals. A matched direct global-rator is often stronger per explored box. Longer horizons and four-state tain material gaps under fixed budgets. The stopping theorem explains nment can close them; the reported execution does not claim to have done merical challenge is to exploit economic structure strongly enough that ation becomes practical at larger state and horizon dimensions without ass or silently weakening its operating constraints.

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SUBJECTS, ASSUMPTIONS, AND VERIFICATION BOUNDARIES

ect throughout the new theory is one randomized Markov policy used at Each inequality $J_t^p(x) \geq V_t(x) - \varepsilon$ concerns that same policy. Its cost is med initial law. None of the new convergence results treats a collection of s as a solution under a dispersed initial law.

TABLE I: Assumptions and certificate scope

Target and assumptions	Computation and check
Borel state space, finite actions, bounded rewards and costs, $0 < \beta < 1$, $\varepsilon > 0$; common randomized Markov policy; any initial law	Analytic proof. Borel measurability and the model bounds are assumptions, not facts inferred from a hash.
Checked $L \leq V \leq U$, $U_T = g$, and positive residual margin $(1 - \beta)\varepsilon - d$	Analytic bracket plus 48 exact multi-action test cases. A failed margin check is not economic infeasibility.
Uniform error bounds over all common policies and every alternative model; $\varepsilon > 2a$	One tightened policy is portable across models. Error radii are assumptions, not estimated confidence sets here.
Same operating model and initial law, feasible lottery upper, valid deterministic lower, bounded non-negative overhead	All 126 exact known-model margins retained. Strict superiority requires overhead strictly below its threshold.
Finite states; same all-restart constraints and initial masses; compact policy product	Rational evaluation and verified box search. Full independent SymPy checks for 16 executed search trees.
Certified reset-kernel approximation in operator total variation; initial mixture identity; $\varepsilon > 2a_h$	Analytic end-to-end theorem. No claim that atomic benchmark kernels satisfy a vanishing reset error.

pendence; deterministic policies only	pendent verification remain retained. The deterministic lower bound is not a randomized lower bound.
Two-state bilinear dynamics; uniform square; $\varepsilon = 1/2$ for paired upper policies	Directed vector arithmetic on all 20 meshes; independent scalar checks for two inherited objects and a new complete (32, 128) replay. Four $\varepsilon = 1/20$ rows are lower-only.
A continuous one-step relaxation, not the common-policy global optimum	Prior exact whole-cell certificates retained; new connected lower recursion independently checked.
Continuous actions, boundary settlement and diffusion approximation still require coverage	Archival derivation, not a solved numerical target or external validation.

PROOF OF THE REPAIR AND COMPLETE-SEARCH ARGUMENTS

1. Policy perturbation and terminal rewards

$d(p, q) = \frac{1}{2} \sum_a |p_a - q_a|$ for action distributions. If an action-dependent interval of length D , replacing p by q changes its expectation by at most D . Thus, remaining, a stage cost at distance s can be affected by the current transitions. Thus the cost perturbation constant is $K \sum_{s=0}^{h-1} (s+1)\beta^s$. The terminal reward contributes $2Gh\beta^h$. The total cost perturbation is at most $2R \sum_{s=0}^{h-1} (s+1)\beta^s$, and the terminal reward contributes $2Gh\beta^h$. The remaining horizons is necessary: $h\beta^h$ need not be increasing in h . Taking a terminal term would not provide a uniform all-restart bound. The main result follows explicitly.

Common-continuation repair and approximate minimizers

be feasible at allowance $\varepsilon + \delta$. At each backward step, all next-date actions are fixed. The old current action probabilities evaluated against that date. The best current action gives $q_* \geq V_t - \beta\varepsilon$. If the mixture share is

$$\alpha = \frac{V_t - \varepsilon - q_p}{q_* - q_p} \leq \frac{\delta}{(1 - \beta)\varepsilon + \delta}.$$

the Borel set with a positive numerator and denominator. The first maximizer is Borel because the action set is finite. Operating value weakly

s in its coordinates, and the feasible set is closed and nonempty. There the is attained. The pure-action finite-state problem is a finite enumeration; the es not preserve deterministic policies and is not presented as a deterministic

3. *Why necessary contractions preserve completeness*

tracted next-date probability intervals. Their rectangular operating Bellman imizing Markov policy that attains its upper value at all states, by backward ality used to contract a current interval is therefore necessary for every ing in the original box. Contracting a current coordinate leaves next-date s retained endpoint set, when nonempty, still permits an operating value edding backward establishes this fact simultaneously at all dates of the ing an operating upper value by V_t does not reduce a true rectangular nconstrained operating optimum already dominates every box-restricted

n also computes lower operating enclosures and may discard a box when ceeds its upper enclosure. That is an inconsistency test, not a failure of ate. Lower operating clipping at $V_t - \varepsilon$ uses a property of every feasible al necessary tests cannot remove a feasible policy. The separate cost lower ined probability rectangle without imposing incompatible, restart-specific Ordinary cost Bellman minimization attains this rectangular lower value

est retained interval width. The midpoint differs by at most w in total tangular operating maximizer and the rectangular cost minimizer. Hence s at most $D_J w$, and its cost is at most the rectangular lower cost plus most $D_C \rho_\varepsilon(D_J w)$. This proves the stated box error, including the effect

ng count, put $a = \beta^{T-1} w_*$. Any physical width exceeding w_* has weighted qual to a . Each weighted coordinate starts at at most one and, whenever alved; contractions can only reduce it further. Along any path, after at ldings per coordinate, no weighted width can remain above a . Because coordinate is selected, no more than nT times that number of splits are sical widths are at most w_* . A positive-gap search never needs to refine a idpoint is already within the requested gap of its lower bound. There are e nodes above this depth. The count can be enormous, and does not imply xperiments exhaust it.

Rational policy lattices and a deterministic benchmark

nding down the first $m - 1$ probabilities and assigning the remainder to t total-variation error at most $(m - 1)/M$. The relaxed feasibility test at

3.1. Averaging identity

For a transition into E_j , the conditional within-cell law is exactly μ_j , the predecessor state and action. For an arbitrary Borel policy p , define $\mu_i^p(dx)$. At the terminal date the payoff is cellwise constant. If the μ_i^p coincide with those of the finite model, integrating the reset Bellman equation with action probabilities $\bar{p}_t$. Backward iteration for operating values and costs. It uses the same averaged policy at every time step.

The comparison bounds $|J^p - J_h^p|$ by a_t and $|C^p - C_h^p|$ by b_t for the very same policy. The error for optimal operating values is obtained directly from the contraction and the reset Bellman operators. An original feasible policy, when averaged, satisfies the constraints at allowance $\varepsilon + 2a_h$. A finite policy feasible at $\varepsilon - 2a_h$ satisfies the constraints in the original model and satisfies its original constraints. The initial-mixture identity holds exactly in the reset model; b_h controls the remaining original-model error. The repair modulus produces the global error budget in the

2. Verification obligations for an application

A bound for every bounded Borel test function, not merely weak convergence. A verified uniform density approximation can provide such a bound. For a compact domain with a Lipschitz density in both source and target, piecewise-constant density integration gives a computable linear error bound in volume and domain diameter included in the constant. Discontinuous approximation on a aligned partition or a separately bounded cost error. The approximation is a initial-law mixture. These conditions are sufficient, not automatically satisfied.

The benchmark kernels contain atoms. Replacing their images by uniform distributions furnish a vanishing operator-total-variation error. Their direct whole-cell approximation depend on this transfer theorem. We retain their unresolved structural error in a mesh sequence as a proof of convergence for those atomic models. The approximation would additionally need continuous-action coverage, boundary settlement, and diffusion/quadrature error bounds before the same end-to-end conclusion

CROSS-STATE TRANSFER WITHOUT A HIDDEN POLICY SIEVE

4.1. Finite-state joint perturbations

The seeds 391101–391108 after development seed 390001, with the case studies in Table II. The two-condition state rewards are $(1/5, 4/5)$. The

the source as $(i_x, i_y) \in \{0, 1\}^2$, the two destination indicators are
 dent given the source and action, with probabilities

$$i_x i_y + j_x + 37a)/120, \quad p_y = (20 + 36i_y + 5i_x i_y + j_y + 37a)/120,$$

$j_x, j_y \in \{0, \dots, 6\}$. Their product probabilities specify the four-state row.
 n initial masses; even seeds use $\nu_i = (i + 1)/\sum_{j=1}^n j$. This is a designed
 e, not a calibrated economic panel or a random sample of environments.
 mit 13fd3ee0eec78db5fa6aed34e63819c5ed546209 fixes the
 ll eight cases, 4,096 nodes, 60 seconds per solver/case, and absolute
 1000. No solver coefficient was changed after these eight environments
 bsequent CI execution is a replay of the frozen experiment, not a second
 wall-clock stopping can change the explored node count across machines.
 on summary is retained separately; tables identify the published execution

Table II: Complete finite-state work and termination records

Absolute gap	Nodes	Stop	MiB RSS	Seconds	Gap met
0.000993	1057	gap	19.8	0.99	yes
0.000981	241	gap	77.4	1.77	yes
0.000000	1	budget	19.0	0.00	yes
0.000000	1	budget	76.0	0.38	yes
1.751049	4095	budget	32.6	17.19	no
0.475442	4077	time_cap	115.7	60.41	no
0.413280	4095	budget	36.2	16.32	no
0.110602	3035	time_cap	117.2	60.40	no
0.005913	4095	budget	30.2	10.82	no
0.000999	2985	gap	97.8	35.99	yes
0.000974	159	gap	19.0	0.47	yes
0.000763	157	gap	77.8	2.60	yes
3.184929	3815	time_cap	46.3	60.04	no
1.649036	1529	time_cap	105.6	60.41	no
1.750167	2353	time_cap	41.9	60.04	no
0.783226	957	time_cap	102.1	60.43	no

interval Bellman and verified McCormick search. The memory figure is
 memory; it includes interpreter and imported libraries, not only the search
 e model construction and the operating solve. The search cap is checked
 cing iterations, so a final iteration can exceed 60 seconds. Serialization

x component (s, b, c, d)	y component (s, b, c, d)
(65, 5, 0, 3)	(70, 3, 0, 2)
(15, 3, 70, 2)	(70, 3, 0, 2)
(65, 5, 0, 3)	(15, 4, 70, 2)
(10, 2, 80, 2)	(10, 2, 80, 2).

$/50, 9/50, 8/25)$, reward is $(x + y)/2 - c_a$, and terminal reward is 3 when both coordinates are below $1/4$; otherwise it is 1 when $x < 20$, and 0 otherwise. Revision cost is $(1 + (x + y)/2)\mathbf{1}\{a \neq a^0(x, y)\}$. The algorithm separately partitions threshold-crossing boxes and verifies all transition boxes for geometry. This tests the paper-to-code model contract as well as

5. COMPLETE CONTINUOUS-STATE RESULTS

We compare identical all-restart constraints on both sides. Lower and upper bounds are rounded down and the gap column is rounded upward. Relative gaps are descriptively calculated from the exact rational interval. “Gain” means the lottery upper bound minus the deterministic lower cost for that same initial density. It does not mean the gap of an optimal randomized rule. Zero upper cost is exact by nonnegativity; zero lower cost is relabeled exact.

5.1. Uniform initial density

ε	Lower	Upper	Gap	Gap/upper (%)	Gain
0.01	4.61741	4.73011	0.11269	2.38	yes
0.05	3.66404	3.89012	0.22608	5.81	yes
0.01	0.08510	0.08928	0.00418	4.68	yes
0.05	0.02622	0.03016	0.00394	13.04	yes
0.01	0.12047	0.12517	0.00469	3.75	yes
0.05	0.04796	0.05403	0.00607	11.22	yes
0.01	0.09478	0.09909	0.00430	4.34	yes
0.05	0.03178	0.03723	0.00544	14.61	yes
0.01	3.90691	4.02072	0.11381	2.83	yes
0.05	3.02022	3.24805	0.22783	7.01	yes
0.01	0.00000	0.00000	0.00000	0.00	no
0.05	0.00000	0.00000	0.00000	0.00	no
0.01	0.00000	0.00000	0.00000	0.00	no

	0.01	0.22548	0.27772	0.05223	18.81	yes
	0.05	0.04029	0.05952	0.01922	32.29	no
	0.01	0.55282	0.65391	0.10108	15.46	yes
	0.05	0.08634	0.10044	0.01410	14.03	yes
ess	0.01	9.17362	9.36573	0.19210	2.05	yes
ess	0.05	7.98380	8.56599	0.58218	6.80	yes
	0.01	0.00000	0.00000	0.00000	0.00	no
	0.05	0.00000	0.00000	0.00000	0.00	no
	0.01	0.00000	0.00000	0.00000	0.00	no
	0.05	0.00000	0.00000	0.00000	0.00	no
	0.01	15.31722	15.46691	0.14969	0.97	yes
	0.05	14.14203	14.72413	0.58209	3.95	yes
	0.01	2.77095	3.36015	0.58920	17.53	yes
	0.05	1.58477	2.02354	0.43876	21.68	yes
	0.01	2.05173	2.38505	0.33332	13.98	yes
	0.05	1.19573	1.45515	0.25942	17.83	yes
	0.01	0.77248	1.01028	0.23780	23.54	yes
	0.05	0.09346	0.14211	0.04865	34.23	no
ess	0.01	13.42003	13.65506	0.23503	1.72	yes
ess	0.05	12.15841	12.87752	0.71911	5.58	yes
	0.01	0.00000	0.00000	0.00000	0.00	no
	0.05	0.00000	0.00000	0.00000	0.00	no
	0.01	0.00000	0.00000	0.00000	0.00	no
	0.05	0.00000	0.00000	0.00000	0.00	no

5.2. Initial density $2x$

	ε	Lower	Upper	Gap	Gap/upper (%)	Gain
	0.01	4.80430	4.91856	0.11426	2.32	yes
	0.05	3.85831	4.08355	0.22523	5.52	yes
	0.01	0.12619	0.13251	0.00632	4.77	yes
	0.05	0.03825	0.04405	0.00580	13.16	yes
	0.01	0.17588	0.18292	0.00704	3.85	yes
	0.05	0.06877	0.07759	0.00882	11.35	yes
	0.01	0.13995	0.14645	0.00650	4.43	yes
	0.05	0.04616	0.05411	0.00795	14.68	yes
ess	0.01	3.63981	3.75184	0.11203	2.99	yes
ess	0.05	2.80900	3.02583	0.21682	7.17	yes
	0.01	0.00000	0.00000	0.00000	0.00	no

0.01	0.30165	0.34589	0.04424	12.79	yes
0.05	0.11746	0.14930	0.03183	21.32	no
0.01	0.11812	0.15183	0.03371	22.20	yes
0.05	0.01393	0.02361	0.00967	40.94	no
0.01	0.59210	0.68149	0.08939	13.12	yes
0.05	0.10049	0.11544	0.01494	12.94	yes
0.01	8.90083	9.08695	0.18611	2.05	yes
0.05	7.75429	8.29459	0.54029	6.51	yes
0.01	0.00000	0.00000	0.00000	0.00	no
0.05	0.00000	0.00000	0.00000	0.00	no
0.01	0.00000	0.00000	0.00000	0.00	no
0.05	0.00000	0.00000	0.00000	0.00	no
0.01	15.50452	15.65425	0.14972	0.96	yes
0.05	14.36785	14.91248	0.54462	3.65	yes
0.01	3.00897	3.57101	0.56203	15.74	yes
0.05	1.75968	2.30639	0.54670	23.70	yes
0.01	2.29566	2.59770	0.30203	11.63	yes
0.05	1.37651	1.66452	0.28800	17.30	yes
0.01	0.70594	0.93114	0.22519	24.18	yes
0.05	0.09731	0.14941	0.05209	34.86	yes
0.01	13.14135	13.37437	0.23301	1.74	yes
0.05	11.89664	12.60188	0.70523	5.60	yes
0.01	0.00000	0.00000	0.00000	0.00	no
0.05	0.00000	0.00000	0.00000	0.00	no
0.01	0.00000	0.00000	0.00000	0.00	no
0.05	0.00000	0.00000	0.00000	0.00	no

5.3. *Initial density* $2(1 - x)$

ε	Lower	Upper	Gap	Gap/upper (%)	Gain
0.01	4.43053	4.54166	0.11113	2.45	yes
0.05	3.46976	3.69670	0.22693	6.14	yes
0.01	0.04401	0.04605	0.00204	4.42	yes
0.05	0.01419	0.01627	0.00207	12.72	yes
0.01	0.06507	0.06742	0.00235	3.47	yes
0.05	0.02714	0.03047	0.00332	10.88	yes
0.01	0.04961	0.05173	0.00211	4.07	yes
0.05	0.01741	0.02036	0.00294	14.44	yes
0.01	4.17401	4.28960	0.11559	2.69	yes

	0.01	10.31657	10.45120	0.13462	1.29	yes
	0.05	9.21257	9.66666	0.45409	4.70	yes
	0.01	0.38523	0.45874	0.07350	16.02	yes
	0.05	0.09070	0.13163	0.04092	31.08	no
	0.01	0.33285	0.40361	0.07075	17.53	yes
	0.05	0.06082	0.09542	0.03460	36.26	no
	0.01	0.51355	0.62632	0.11277	18.00	yes
	0.05	0.07220	0.08545	0.01325	15.50	yes
ess	0.01	9.44731	9.64451	0.19720	2.04	yes
ess	0.05	8.21366	8.83739	0.62372	7.06	yes
	0.01	0.00000	0.00000	0.00000	0.00	no
	0.05	0.00000	0.00000	0.00000	0.00	no
	0.01	0.00000	0.00000	0.00000	0.00	no
	0.05	0.00000	0.00000	0.00000	0.00	no
	0.01	15.12991	15.27957	0.14965	0.98	yes
	0.05	13.91621	14.53578	0.61956	4.26	yes
	0.01	2.53293	3.14930	0.61637	19.57	yes
	0.05	1.40986	1.74069	0.33082	19.00	yes
	0.01	1.80779	2.17240	0.36460	16.78	yes
	0.05	1.01494	1.24578	0.23084	18.53	yes
	0.01	0.83901	1.08942	0.25041	22.98	yes
	0.05	0.08961	0.13482	0.04521	33.53	no
ess	0.01	13.69954	13.93576	0.23621	1.69	yes
ess	0.05	12.42017	13.15316	0.73298	5.57	yes
	0.01	0.00000	0.00000	0.00000	0.00	no
	0.05	0.00000	0.00000	0.00000	0.00	no
	0.01	0.00000	0.00000	0.00000	0.00	no
	0.05	0.00000	0.00000	0.00000	0.00	no

DETERMINISTIC GAPS AND THE EXCEPTIONAL RESTART LOCUS

A deterministic headline decomposes into 12 zero-cost and 12 positive-cost positive-cost equalities, ten occur at horizon 4, two at horizon 8, and none at horizon 12. Table 1 reports every inherited absolute and relative interval and the outer operating deficit. A deficit above ε is an actual violation established by the interval representation; it is distinct from a failed sufficient nonlinear box

cost. γ has zero *revision* cost by definition. Therefore a revision gap divided by a revision cost is undefined for every row. We do not report that ratio as it would substitute its operating reward as a denominator with different units. The valid revision gap divided by a positive feasible upper revision cost; a zero upper cost is not meaningful.

	0.05	0.000000	0.00	0.05000	0.00000
	0.01	0.000000	0.00	0.01000	0.00000
	0.05	0.000000	0.00	0.05000	0.00000
ress	0.01	0.000000	0.00	0.01000	0.00000
ress	0.05	0.000000	0.00	0.05000	0.00000
	0.01	0.000000	–	0.00001	0.00000
	0.05	0.000000	–	0.00001	0.00000
	0.01	0.000000	–	0.00001	0.00000
	0.05	0.000000	–	0.00001	0.00000
	0.01	0.001622	0.01	0.01170	0.00170
	0.05	0.019446	0.18	0.09072	0.04072
	0.01	0.010990	2.29	0.01170	0.00170
	0.05	0.103474	45.39	0.09072	0.04072
	0.01	0.010990	3.16	0.01170	0.00170
	0.05	0.069936	61.21	0.07754	0.02754
	0.01	0.006434	0.74	0.01170	0.00170
	0.05	0.000000	0.00	0.07693	0.02693
ress	0.01	0.000000	0.00	0.01000	0.00000
ress	0.05	0.002381	0.03	0.05763	0.00763
	0.01	0.000000	–	0.00042	0.00000
	0.05	0.000000	–	0.00042	0.00000
	0.01	0.000000	–	0.00576	0.00000
	0.05	0.000000	–	0.00576	0.00000
	0.01	0.001359	0.01	0.01170	0.00170
	0.05	0.066102	0.43	0.12738	0.07738
	0.01	0.014120	0.40	0.01170	0.00170
	0.05	0.177733	6.89	0.09072	0.04072
	0.01	0.009537	0.37	0.01170	0.00170
	0.05	0.075788	4.12	0.07897	0.02897
	0.01	0.006662	0.54	0.01170	0.00170
	0.05	0.248472	63.88	0.11215	0.06215
ress	0.01	0.003444	0.03	0.01194	0.00194
ress	0.05	0.060221	0.45	0.06919	0.01919
	0.01	0.000000	–	0.00042	0.00000
	0.05	0.000000	–	0.00042	0.00000
	0.01	0.000000	–	0.00576	0.00000
	0.05	0.000000	–	0.00576	0.00000

agreement for horizon 8, neural31003, allowance 1/20

a stronger initial equality than the earlier verbal description suggested:
 functions, including isolated points. The failure of the all-restart equality

te 1. Open interval with endpoints 0 and 7195348168316/157454
 difference has slope 55233/2000000 and intercept 289161/500000.
 represented difference is 289161/500000.

te 2. Open interval with endpoints 11274755890307953/145920
 4598974049913359/5910038365284645. The difference has
 and intercept 373/1000.
 endpoints 36597632066686/46374770160891 and 44871290
 84445. The difference has slope 3061039/4000000 and intercept
 0.
 endpoints 4487129063774/5675886884445 and 633416277301
 22275. The difference has slope 969/2000 and intercept 627/1000.
 0307953/14592000000000000, the separately represented differ-
 22907499543/29184000000000000000.
 6686/46374770160891, the separately represented difference is
 030897/46374770160891000000.
 774/5675886884445, the separately represented difference is 2958
 734202500.

te 3. Open interval with endpoints 1288720115486/16693784
 0165032032/231550957065375. The difference has slope 1 and
 486/1669378495425, the separately represented difference is 2958
 378495425.

7. NONLINEAR BOUNDS AND WORK

lower endpoint is shown beside its connected restart improvement. All upper
 retained feasible candidates; thus the improvement is not confounded
 l portfolio. A missing upper endpoint means that no upper candidate at
 ed by the inherited sufficient test. It does not imply true infeasibility. No
 ched success fractions is used.

ort lower	Restart lower	Upper	Gap/upper (%)	Array MiB
0.0000	0.0000	2.4975	100.00	0.1
0.0005	0.0005	2.1219	99.97	0.6
0.0118	0.0118	1.7191	99.31	2.3
0.0332	0.0334	1.3503	97.53	9.4
0.0554	0.0556	0.9992	94.43	37.5

168	0.0169	–	–	0.5
7800	1.8264	–	–	2.0
1646	3.1898	8.7556	63.57	8.0
1609	4.1908	7.6644	45.32	31.9
6932	4.7174	7.3118	35.48	127.5
44443	0.5535	–	–	1.0
4755	3.5226	–	–	3.9
4932	6.5389	16.7884	61.05	15.5
0961	8.1264	12.5226	35.11	61.9
8476	8.8718	11.8492	25.13	247.5

cludes temporary continuation ranges, the interpreter, and compression memory is also saved. On a threshold-crossing box, the cost upper-gest possible realized departure cost, while the lower enclosure can d action is possible there. This conservatism concerns enclosure of the claim that the policy changes at every point of that box. The nonzero lier lower-bound weakness without removing this upper-enclosure

7.1. Operating allowance 1/20

tolerance comparable to the primary development cohort. They are ly diagnostics. A feasible policy at allowance 1/2 cannot be used as nce 1/20.

ε	Lower	Witness width	Status
0.05	1.55879	0.10794	uncertified upper
0.05	4.24475	0.08548	uncertified upper
0.05	7.76989	0.13408	uncertified upper
0.05	12.45297	0.10222	uncertified upper

CAL-LP ERROR VERSUS WORK AND DEGENERACIES

calculation is retained in full. Its finest tolerances required substantially e table and two generated figures display all 12 executions, not only cern a continuous local relaxation with zero global support floor, and d with the new nonzero-floor global recursion as if their errors were

0.01	1/400	12526	0.004638	82.27
0.05	1/100	171	0.018550	0.51
0.05	1/200	309	0.009275	1.03
0.05	1/400	588	0.004638	2.35
0.05	1/100	168	0.018550	0.40
0.05	1/200	306	0.009275	0.84
0.05	1/400	585	0.004638	1.96

ulation uses a fixed represented partition, not an adaptive nondegeneracy
on. For one operating inequality, it enumerates feasible pure actions and
traddling the boundary. When a pure action lies exactly on the boundary
. When two operating values coincide, no division by their difference is
ir face is infeasible or its linear cost minimum is attained at an included
onal coincidences do not create an unreported termination assumption for
The old logs did not instrument an occurrence count for every degeneracy
f construction. We do not infer such a count from successful termination.
ain subject to their explicitly conditional theorem.

DEPENDENT CHECKS, PUBLICATION CHECKS, AND PRESERVATION

trees are checked using SymPy rationals without importing either solver,
or the LP package. The verifier reconstructs the economic stage contract,
man equalities, operating feasibility, box coverage, contractions, all ac-
multiplier residuals, and the minimum over terminal leaves. Nine distinct
class, probabilities, terminal value, initial law, revision cost, lower endpoint,
t coverage are rejected. These mutation tests supplement rather than prove
correctness.

-state lottery objects and their 42 connected local lower objects receive
, cost, support, and vertex checks. The separate randomized and determin-
nts retain their provenance. Most final lower endpoints are either the newly
or the newly checked exact support integral. A retained R34 randomized
te those at a particular row; such an inherited component is not converted
pendent proof merely by taking a maximum. The publication audit records
s. Likewise deterministic strict separation uses the retained R38 determinis-
own independent checks remain part of the archive. The independent label
local bounds does not expand automatically to all historical components.
ndependent checks cover the whole $(T, N) = (8, 64)$ and $(32, 64)$ objects,
and operating feasibility. They use rational corner geometry and scalar
teger recursions, rather than vectorized construction arithmetic. Other
ept the new R40 $(32, 128)$ replay, have directed construction and structural
independent status. These labels are distinct from hash identity, repeated
ation-integrity checks.

es a witness-driven feasibility repair and a lattice error budget that
g witnesses rather than an exact operating optimum. Its repair slack
ere d bounds the upper witness’s Bellman residual. Approximation
d approximation of the policy remain separately visible. The new
proposition preserves one policy across every model in its uncertainty
ation charges are derived for all 126 primary initial-law configurations,

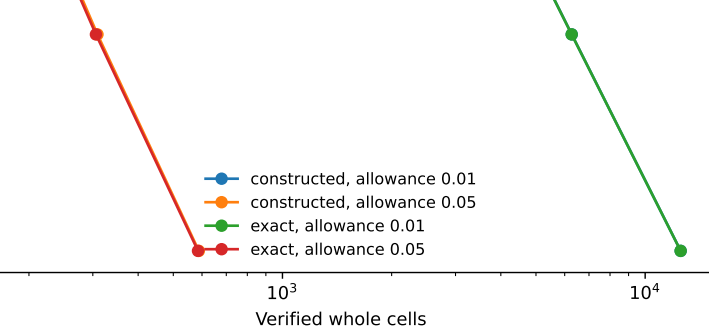
r constructor does not compute or accept the exact optimum. In 48
parate exact operating oracle constructs TEST witnesses; it is not input
stage and terminal rewards, two or three actions, two or three states,
are covered. SymPy matrix evaluation checks every operating and cost
constructor. All 48 deliberately nonpositive witness-slack cases are
t declared infeasible. These tests supplement the analytic theorem.

-affine audit checks 200 discontinuous pairs, including separately
with 5756 independent value comparisons and 400 malformed-partition
20 local programs are checked by an independently assembled SymPy
59 contain coincident operating values and 64 have a pure action
boundary. These are measured degeneracy counts for the new audit, not
counts for the historical adaptive solver.

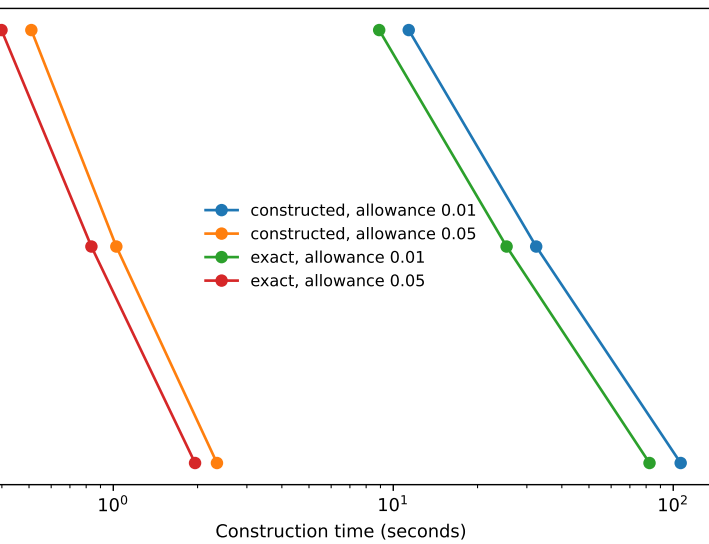
e proof trees, three declared full primary lottery/local objects, and the
) = (32, 128) object using the frozen independent verifiers. The latter
lication to a finer mesh. The full inherited 42-primary-object verifier
act hash linkage; the three new primary replays are not represented as
e of these tests is a statistical false-acceptance guarantee.

10.1. *Preserved scientific content*

cle and supplement contain the complete deterministic proof, fixed-
d frontier, two-period randomized example, adaptive local-LP theorem,
ysis, nonlinear feasibility construction, and killed-operator/stopped-
remain historical documents with their original claims and limitations;
upplement specify the current conclusions. The complete older 349-
dF, prior reviews, failed runs, and source hashes remain unchanged on
scientific content is deleted. Preservation is not presented as additional
ed economic optimum has been solved.



local interval width against whole cells. Every inherited local execution is displayed; the
er bounds solve a different recursive lower problem.



local interval width against construction seconds. The steep work increase is retained rather
te success count.

beyond piecewise-affine closure, while 66 sensitivity
is intervention exposure. Separate rational verification,
nal costs, and nonzero residual intervals accompany the
on connects the method to stopped economic control.

ming, Policy revision, Verified computation, Constrained

1. INTRODUCTION

expensive to operate and expensive to replace. A new rule
but its implementation cost depends on which decisions
rule subsequently visits. Maintenance, organizational
ems share this distinction. A rule that minimizes today's
e the discounted cost of future changes. The economic
ontrol problem, not a pointwise classification problem.

that problem. Its operating requirement is uniform over
ation objective is an expectation under a specified initial
own transition law. These are different scopes: a uniform
n initial-distribution optimum into a cost optimum from
tion in the theory and in every numerical declaration of

stic outer-class construction using two-sided operating
on must be compatible with the performance target even
continuation. Replacing that unavailable continuation by a
urrent value by a lower witness, preserves the necessity
n cost over the resulting outer class gives a valid lower
isible deployment gives an upper bound. Equality solves
ve, not merely an inner approximation. Exact operating
ence specialization, and its full cost is charged.

cy-class-specific constrained reference calculation. A
finite-shock, finite-horizon problems from a fixed restart
tory. Its construction clips operating feasibility at each
ode. A two-period version of the maintenance economy
arkov solution for the uniform initial distribution. The

four cases close. Of the 24 integrated equalities, 23 go functions. The remaining 18 deterministic cases. The largest interval is below 0.249 in normalized space is relaxed to obtain these statements. mechanism and the representation. Across 66 designed is strictly less costly than pointwise revision in 22 in 44. A separate bilinear two-state economy uses through 32, and meshes through 256 per dimension. an information-matched classical grid scalarization model numerical method, not empirical calibration or

n to existing methods

ward induction, convexification, and weak duality are 98). We do not rename them as new principles. Our on of operating witnesses, policy-class-correct outer , and a numerical implementation that closes positive nputation provides a direct constrained reference for er than comparing only with unconstrained operating

dies approximation and sampling errors under spec- Szepesvári, 2008, Scherrer, 2013). Safe policy im- e to uncertainty about a transition model (Laroche, Here the model is known, approximation proposes a tic verification layer checks the resulting rule on the posals are retained on identical terms. We establish riority of an installed neural rule merely because a

c and approximation structure (Judd, 1998, Brumm xation methods provide a different route to bounds Smith, 2014). Both approaches must charge bound nce calculation exploits exact affine structure; the structure with directed box arithmetic. Verified com- well as arithmetic error (Rump, 2010). Accordingly, fied by its actual code boundary, rather than inferred ath.

negative revision weight w_t , our principal specification is

a tolerance $\varepsilon \geq 0$, write

$$\mathcal{J}_D(\varepsilon) = \{\pi : V_t(x) - J_t^\pi(x) \leq \varepsilon \text{ for every } (t, x)\}. \quad (3)$$

Markov policies. A randomized Markov kernel $p_t(\cdot | \cdot)$ in (2); replacing the feasible class gives K_R . An action a_t is drawn from $p_t(\cdot | x_t)$ for the action actually drawn, so its conditional expected

Let $h_t = (x_0, a_0, x_1, \dots, a_{t-1}, x_t)$ be an admissible history of length t . Let $J_t^\pi(h_t)$ be the continuation reward before the date- t action draw. The history-conditioned

$$\text{at every admissible decision history,} \quad (4)$$

at off-path histories. We do not impose the same action draw but before receiving its payoff. Such a mixture of the mixtures studied below. Deterministic history-dependent policies are kept separate. A pointwise integral can be integrated against any initial distribution; an integral over the state cannot simply be integrated to claim a Markov policy.

The installed rule, $J_t^\pi \geq J_t^{\pi^0}$, is checked separately. It is an experiment every selected deployment satisfies both the lower bound and the problem with the

incentives and the unchanged primary cohort

on. Actions 0, 1, 2 mean defer, maintain, and replace. The

$$(c_0, c_1, c_2) = (0, 3/20, 9/20), \quad g(x) = x/2. \quad (5)$$

334 incumbents, not refitted substitutes, are used. partition knots are part of the new proof objects. A not interchangeable with this exact compiled policy

ESSES AND FEASIBLE DEPLOYMENT

$$L_T = U_T = g. \quad (7)$$

and checked, not supplied as a costless oracle. In the positive affine maps, addition of affine rewards, and fine structure. The exact representation stores rational val, and separate values at every knot.

g: Let F be continuous, convex, rational piecewise rational mesh lies above F . Rounding the starting a common grid $2^{-q}\mathbb{Z}$ produces a convex function G chord interpolant. If F is M -Lipschitz and mesh width 2^{-q} . The actual maximum defect is determined exactly

pes; upward rounding on a common lattice preserves 2^{-q} , so integration over a total length of one adds less ts every chord above F , and the endpoint Lipschitz . The difference of the two represented functions is ma occur at endpoint limits and stored points. *Q.E.D.*

U_{t+1} and an encoding satisfying $0 \leq U_t - F_t \leq d_t$. $= U_t - b_t$. Backward comparison proves (7). In the es as $M_t \leq 1 + \beta(17/20)M_{t+1}$ before the explicit precision choices exist for every positive prescribed actual defects rather than relying only on a priori

generator uses exact V only in the reference study. Its search parameters, not estimated economic prices. Ante. The selected full policy is reevaluated before use, and generator itself.

UBOUNDS AND OPTIMALITY CERTIFICATES

set and its cost recursion by

$$T_t^a U_{t+1}(x) \leq \varepsilon\}, \quad (9)$$

$$U_t(x) = \min_{a \in \mathcal{N}_t(L, U; x)} \{k_t(x, a) + \beta P_t^a B_{t+1}^{L, U}(x)\}. \quad (10)$$

ating-optimal action belongs to it. Unlike (8), this is an deterministic action must pass the test, but passing the test does not remain feasible.

outer bound: *Under (7), every feasible deterministic policy $\hat{\pi}$ satisfies $J_t^{\hat{\pi}}(x) \leq U_t(x)$ at every restart. The same inequality holds for every policy satisfying (4), with $x = x_t$ and its history-conditioned cost $\hat{\pi}$ has*

$$J_0^{L, U} d\nu = \int C_0^{\hat{\pi}} d\nu, \quad (11)$$

over the stated deterministic history-dependent class. $J_t^{L, U} = C_t^{\hat{\pi}}$ additionally establishes cost optimality from $J_t^{\hat{\pi}}$ does not establish that stronger assertion.

decision history with current action a , operating compar-

$$J_t(x) - T_t^a V_{t+1}(x) \leq V_t(x) - J_t^{\pi}(h_t) \leq \varepsilon.$$

continuation as an upper bound on the policy's actual cost. Hence $a \in \mathcal{N}_t$. Assuming the cost inequality at the

$$k_t(x, a) + \beta P_t^a B_{t+1}^{L, U}(x) \leq C_t^{\pi}(h_t).$$

if $L \uparrow V$ and $U \downarrow V$, then the sets $\mathcal{N}(L, U)$ of admissible actions increase to $B^{V,V}$ pointwise. For a finite horizon, excluding actions from the threshold ensures that the approximation errors are smaller than that margin. Neither the exact optimum nor a feasible upper policy closes the gap.

increases with n . An action whose exact gap exceeds ε is excluded. The exact gap is at most ε is always included. A finite set of actions is the exact necessary set. By backward induction, the gap decreases; monotone convergence passes through each action. The uniform-margin assertion follows from $\|L^n - L\|_\infty + \beta \|U^n - V\|_\infty$. The exact necessary test set, which explains the final qualification. *Q.E.D.*

$L^{L,U}$ and $B^{V,V}$. The former demonstrates that safe actions are included in the exact necessary set. The latter is a fully charged set of actions. Their difference is measured, not suppressed. When an action is not feasible, it remains a valid lower-bound witness but is excluded from the exact necessary set.

DIRECT CONSTRAINED REFERENCES

to randomized policies. A mixture can satisfy the constraint with a small probability to an action that fails the deterministic constraint. The support/restart bounds for the randomized reference calculation.

Optimal action-frontier recursion

be the achievable pairs of conditional operating value and cost. The set of actions that fail the deterministic constraint is suppressed. Put $\mathcal{K}_T(x) = \{(g(x), 0)\}$.

frontier: *For a finite horizon, finite actions, and finite shocks, the history-conditioned class. Rational primitives and rational state space frontier. Primal convex combinations and supporting hyperplane and its minimum cost. The recursion need not compute different histories reaching the same physical state may*

assertion is immediate. Conditional on a current action a , use a continuation pair in the corresponding successor set $S(a)$ exactly the action-specific set in braces. Mixing current action and half-space enforces current feasibility; successor sets enforce history restriction. Conversely, choose a convex set C and use the selected feasible successor continuation pair to construct every pair in the recursion. Finite Minkowski sum and half-space intersections preserve rational polytopes. Vertices and primal combinations attain the retained frontier and certify entire edges. The construction remembers which action was chosen and imposes no common Markov continuation across actions. *Q.E.D.*

with horizon. Memoization of identical (t, x) frontier pairs does not impose equality of the selected continuation pair across shocks, additional validated set approximation would reduce dimensional complexity rate from a finite tree. Restart horizon-four problems: the 14 primary configurations have objectives are Dirac initial distributions, not the primary randomized history-conditioned value is strictly below the restart. Each full frontier has a separately implemented experiment is a direct constrained comparator for its exact value and a numerical quadrature of K_R in the primary study.

Randomized Markov continuum example

direct comparison under the *same uniform initial objective* maintenance primitives and install defer at both dates. At the operating value and has zero implementation cost. Thus it

the integral of $(1+x)/(ax+b)$ equals $(r-l)/a + \frac{1}{a} \ln \frac{r-l}{a}$, with the evident constant-denominator formula. These yield costs approximately 1.3720075152810114. These widths are below 10^{-20} . At $x=0$, $\Delta = 2323/4000$. Since $1/100$ is $2283/2323$. This counterexample rules out the claim. It supplements rather than replaces the unchanged

UNRESTRICTED CONTINUATION AND ADAPTIVE LOCAL CERTIFICATES

Let $W_T^\lambda = \lambda g$ and

$$k_t(\cdot, a) + \beta P_t^a W_{t+1}^\lambda\}. \quad (14)$$

A stationary policy satisfies $W_t^\lambda \geq \lambda J_t^\pi - C_t^\pi$, hence

$$C_t(x) - \varepsilon - W_t^\lambda(x)] \leq C_t^\pi(h_t). \quad (15)$$

Since its supremum exhausts global duality.

after every successor decision history. Set $q_a = L_t(x) - \varepsilon$. Feasibility implies $\sum_a p_a q_a \geq b$.

$$\sum_a p_a z_a = \sup_{\mu \geq 0} \min_a \{z_a + \mu(b - q_a)\}. \quad (16)$$

Let us continue unrestricted continuation. The local multiplier μ in (16). They are not interchangeable.

Lemma: If F_t and R_{t+1} are valid cost lower bounds for $R_t = \max\{F_t, \Phi_t(R_{t+1})\}$ is also valid. For three cases: (i) attained either by a feasible pure action or by a mixture of two actions, (ii) attained at zero or at a nonnegative intersection of two lines, (iii) handled explicitly.

denominator affine. A fixed dual price gives $D_\mu(x) =$ affine lower bound. We first split at all input knots, eliminating changes. A cell is accepted only if the primal is feasible throughout it. Multiplication by the nonnegative denominator is a quadratic sign test. A quadratic's maximum on a closed interval is at a stationary point. Zero denominators at cell endpoints are isolated knots are solved separately by the exact point LP.

Approximation: Suppose the preceding cell tests cover the isolated point value of the dual lower function downward by 2^{1-q} on $[0, 1]$. Let $e = \rho/2 + 2^{1-q}$ and assume $2^{1-q} \leq$ lower encoding, and the same exact floor F_t , produces

$$\mathcal{R}_t - \widehat{R}_t \leq e \sum_{s=0}^{T-t-1} \beta^s. \quad (18)$$

feasible action margin on each compact cell, refinement degeneracies, finite termination is asserted only when the tests admit a finite cover. A refinement cap causes an *ate.*

slope and intercept by less than 2^{-q} decreases an affine values have the same direction. The cell test bounds the sum before rounding. Changing a continuation cost by at most βd , so minimization over the unchanged feasible set is nonexpansive. The maximum with the same floor is nonexpansive. A strict feasibility margin, feasible primal vertices and moduli on a finite subdivision; the midpoint primal/dual widths shrink. Compactness supplies a finite cover. At this need not follow from strict interior feasibility alone, termination and explicit stopping test are required. *Q.E.D.*

on encountered rational serialization growth. That unsuccessful experiment uses $q = 40$ and $\rho = 1/100, 1/200, 1/400$, enclosures. It solves the two horizon-four neural31001 enclosures and exact V , for 12 complete certificates. The floor

ality.

AND COMPUTATIONAL ACCOUNTING

utable sources and exact or outward-directed proof
ions are not used as observations. The primary model,
erating tolerances are unchanged. The computation
s, execution records, arithmetic libraries, and the

stic optima and residual intervals

and independently feasible deployments agree in 24
ectly positive costs. Ten positive-cost equalities occur
e horizon-eight neural31003 case with $\varepsilon = 1/20$ has
cost-to-go functions. The other 23 exact cases have
n is tested, not inferred from rounded table entries.
e feasible, and their full cost functions match those
supplies a certificate-level explanation for all 14
relaxation itself has a feasible minimizing selector.
ur maintenance specification has this property. At
ates the operating target in 19 cases, by as much as
retain a positive initial cost interval; the remaining
described. Accumulated operating loss, not failure of
on to treating those outer selectors as deployments.
-state operating constraint and preserves the installed
d all 18 residual deterministic intervals are printed in
result ledger. The lower and upper fractions, policy
ash, policy class, objective, and exactness scope are
displays do not determine equality.

lled gap diagnostics

or identical installed rules: constructed L with eight
t prices, and exact V with fifteen prices. The eight
efinement adds $1/2, 2, 8, 32, 128, 512, 2048$. This re-
s across 21 horizon-incumbent pairs; both operating

ce lower	Adaptive lower	Adaptive upper
.076186	0.088078	0.106627
.076186	0.088107	0.097382
.076186	0.088121	0.092759
.020289	0.025620	0.044169
.020289	0.025649	0.034924
.020289	0.025665	0.030302

support floor, 40-bit directed coefficients. The endpoints enclose the trained optimum.

ctions. Pointwise monotonicity of both refinements is

0 support floors, with maximum integrated gain below improves 28 floors, with maximum gain approximately under, exact V raises 30 safe outer bounds, with maximum o enrichment lowers 30 inherited deployment costs, by as measurements show why exact-value availability is not ual deterministic gaps on this cohort.

policy classes. Global support floors are valid for ran- the necessary-action floor is deterministic. The local LP y the two declared horizon-four neural31001 cases. For exact deterministic interval, necessary-selector violation, upper-search improvement are reported as separate coordi- al gap of the continuous global randomized problem has

ws a certified interval around the continuous local relax- he eight-price local recursion is computed on identical The theorem explains the narrowing without assigning it

sensitivities and endogenous exposure

ght, three natural installed rules, two separately declared ne-at-a-time primitive specifications. The baseline con- maintains below 0.6. The preventive rule replaces below

counting to 0.90 or 0.99, the revision weight to 1 or maintenance cost to 0.10 or 0.20, or defer persistence changes in each scenario. Both dynamic and pointwise; their costs are evaluated under their own transitions y. All 66 configurations pass their operating checks. on savings and 44 exact zero effects; the maximum complete scenario table retains the zero effects. The entention exposure across the declared economic rules, e calibration. Computational penalties used elsewhere sion weights.

near two-state economy

ctions defer, repair the first component, repair the components. Rewards are $(x + y)/2 - c_a$, with costs $y)/4$, and revision weight $1 + (x + y)/2$. Discounting $(0.6, 0.4)$. Each coordinate transition has the form $\{$ and z is that coordinate. The action-specific tuples

inate y -coordinate

0, 3)	(70, 3, 0, 2)
0, 2)	(70, 3, 0, 2)
0, 3)	(15, 4, 70, 2)
0, 2)	(10, 2, 80, 2).

and preserve it, but bilinearity destroys the one-ary solver.

, 128, 256 cells per coordinate, and horizons are backward floors/ceilings are performed with signed h full transition image, enumerates every intersected absolute-value bound below 2^{62} . The grid policy is closed transition enclosures cover ties conservatively. d policies, appear in the supplement.

eparate economy. It does not replace 1/100 or 1/20 generators use penalties zero and four. A classical

Median revision (s)	Max. lower pieces	Max. bits
0.0973	11	143
0.3478	90	186
2.3529	430	242

horizon. Revision time includes both exact-value and safe-witness outer independent verification, support diagnostics, and supplementary studies are generated from this publication execution.

constructs a scalarization portfolio at prices 4, 16, 64, 256; its dual and receives the same directed whole-box evaluation. No work is charged. There are 26 certified restart-generator and classical portfolios among 20. A fresh R38 audit of the policy fails in all 16 pilot mesh-horizon configurations. The pilot is not a reconstruction of the unsupported R35/R36 thems' candidate counts differ, so their success counts are reports each matched mesh, horizon, tolerance, runtime, count, and failure. These results demonstrate executed uniform dominance over the classical comparator.

presentation, and verification

only across configurations with the same horizon. Frozen inputs; no neural training is described as newly executed. Extensions, feasible candidate searches, support refinements, action, nonlinear grids, and their verification are charged comparing warm certificate work with cold neural training. None of the Fraction/PW construction, envelope, components partitions using actual primitive transition pullbacks, logs, open-cell endpoint limits, isolated values, Bellman feasibility, terminal conditions, and exact initial integrals. declared corruption categories are rejected, including utilities, action labels, parser failures, objective scope, point identically checks the 70 fixed-restart frontier objects using flexible convex combinations, and supporting inequalities.

$$(19)$$

$$0.2p_s x_s dZ_s^x, \quad d\langle Z^u, Z^x \rangle_s = -0.25 ds. \quad (20)$$

$2] \times [-0.5, 0.8]$. Running utility is $c^{1-u}/(1-u) - (2)^2 + 0.1 \log x$, and exit settlement is $G - 8(1-s)$, the target at $k = 2$ remains $\sup_{t,u,x} (V_2 - J_2^\pi) \leq 0.01$. Other than a claim that maintenance certificates solve and a a control held over the step. Define the killed

$$Z_s, a) ds + e^{-r\tau} g(t + \tau, Z_\tau) \mathbf{1}\{\tau \leq \Delta\} \quad (21)$$

$$\mathbf{1}\{\tau > \Delta\} \Big].$$

is mass at most $e^{-r\Delta}$. Bellman comparison, witness of therefore extend verbatim to this subprobability using β and exit settlements included in the one-step as the backward witness budget additively, and thus counted sum of ξ_t to the operating certificate. killed-transition and action-coverage enclosures. A lower bound for the original continuous-control revision. Uniform action interpolation, time discretization, approximation errors must all be covered. Markov-chain an additional analysis (Kushner and Dupuis, 2001); it ment.

remains intact. At restart $(0, 61/50, 5/4)$ with $k = 2$, $5]$, the archived derivative enclosures are positive on $= 1/5$ as the unique maximizer in that scalar class. ced in the supplementary preservation material. The 823298 is not converted into a 0.01 certificate. The ly which parts of the certificate method transfer and main necessary for the unchanged diffusion objective.

the proofs, source, unsuccessful computations, and actual
these claims. The original stopped-control objective and
ved with an explicit map to their present location.

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class in the article uses a single measurable action at each
 part uses a probability kernel at that date and state. The
 the current action draw. Neither class is replaced by an
 maint. Implementation cost is averaged under the specified
 start operating inequalities have been established.

remember earlier actions, shocks, and previously drawn
 at a decision history before the next action draw. To
 policy includes continuation kernels at every admissible
 ve probability under the initially deployed policy. The
 onstrained Markov value $V_t(x_t)$ because the underlying
 ity is conditional at every such history.

proof uses the action actually selected at that history. Its
 regardless of the particular history. A randomized policy
 er current actions; individual actions in its support need
 -period example is an explicit instance. Conditioning on
 different feasible class and would remove the mixture
 ed here.

parate continuation choices for different histories that end
 te/date frontier sets can be cached because the primitive
 ent points on the cached frontier at different histories is
 zation is not a proof that a randomized Markov policy

ks and finite horizon, the reachable decision tree is finite.
 extended to unvisited decision histories by an operating-
 has finite implementation cost and zero operating regret.
 ctive. This justifies the fixed-restart reference under the
 iming that independently chosen references at different
 continuation.

PRESENTATION AND INDEPENDENT CHECKING

tuple (z_i, m_i, b_i, v_i) , equal to $m_i x + b_i$ on each open
 terminal and operating witnesses are continuous in the

ations, all-restart regret, and exact initial integrals. It
 se equality flags.
 e primitive discount, transition, shock probability,
 terminal conditions, missing dates, knot order, mal-
 onal actions, isolated values, affine coefficients, false
 and provenance/hash failures. The complete category
 independent verification ledger. The independence
 titution implementation. Both programs intentionally
 this is not formal proof-assistant verification or an
 rors are impossible.
 ashes canonical JSON for 187 scientific objects: 42
 12 local LP objects, 21 global support bundles, and
 nment equality check is distinct from independent
 er groups are covered by the independent SymPy
 ese 187 canonical scientific objects.

CIENT ACTIONS, AND OUTER RESTRICTIONS

e $0 \leq U_t - T_t U_{t+1} \leq d_t$. Induction gives $V_t \leq U_t$. If

$$_1 - T_t V_{t+1} \leq d_t + \beta b_{t+1} = b_t.$$

tual realized defects can produce a tighter witness
 e R38 safe outer calculation uses the realized defect

$\eta_t + \beta e_{t+1}$ and an action satisfies $U_t - T_t^a U_{t+1} \leq \eta_t$.
 $J_{t+1} - e_{t+1}$, then

$$_1 \leq \eta_t + \beta e_{t+1} = e_t.$$

ient action sets therefore gives an operating-feasible
 s is ordinary dynamic programming, but it solves
 y feasible deterministic action is included in $\{a :$
 er this latter class is a lower bound on the original

subject to $p \geq 0$, $\sum p = 1$, and $p \cdot q \geq b$. Every vertex
 of the polyhedron is the convex hull of two actions i, j with $q_i < b < q_j$ and weights

$$\frac{-b}{-q_i}, \quad p_j = \frac{b - q_i}{q_j - q_i}.$$

handles the endpoint. When $q_i = q_j$, the pair does not
 define a unit vector or another crossing pair suffices. Ties are
 broken by the very tied optimal value remains valid.

For nonnegative values $(z_i - z_j)/(q_i - q_j)$ for unequal q_i, q_j .
 primal-dual equality at every sampled basis-selection point
 exact point checks select a certificate; they do not verify a

affine. Additional splits ensure that action feasibility and
 interior. The proposed two-action primal has nonnegative
 the cell; these conditions are checked at endpoint limits.
 For $D > 0$. For a fixed $\mu \geq 0$, write $d_a = z_a + \mu(b - q_a)$.
 $-D_\mu \leq \delta$ is equivalent to

$$(\delta) \leq 0 \quad \text{for every action } a.$$

maximum is calculated using both endpoints and, for a
 prior. A zero denominator at an endpoint is treated by the
 separate exact point LP at the endpoint. A denominator that
 is zero.

can change active action within an accepted cell. All such
 changes verify that each stored affine line is the corresponding
 not only at its midpoint. The union of accepted cells must
 be convex. The exact knot LP value is stored independently of the

cept, and point value is rounded downward to a denom-
 inator. Rounding changes the value by at most 2^{-39} in
 precision is checked again before propagation. For arbitrary
 precision enough bits to keep this rounding term below half the
 specified tolerance for the three displayed tolerances, not a

specific Minkowski sums of efficient successor vertices, acts it with the required operating half-space. Pruning the pair has no smaller operating value and no larger supporting lower edge. Because feasibility is a lower cheaper pair dominates for all preceding lower-budget

proof stores either one candidate or a convex com- Its cost-support line $c = \mu j + \eta$ is checked against slopes are nonnegative. Adjacent vertices determine every candidate proves the entire edge, including line. At the first feasible frontier point, either the slope certifies an unconstrained cost minimum. The also checked against the independent operating value. candidate action/child combinations from the primitive s. It compares canonical candidate counts/hashes and report the constructor's hull or optimization functions. These tests certify the serialized recursive frontier, n just its reported initial number.

ized Markov calculation uses a different analytic egrand is rational with an affine denominator. For a age reduction writes $r = 2^m s$ with $1 \leq s < 2$, using $(s + 1)$,

$$, \quad 0 \leq R_n \leq \frac{2y^{2n+1}}{(2n+1)(1-y^2)}.$$

terms give the recorded rational integral enclosures. rmine which endpoint is used. This analytic calcula- pendently verified frontier objects.

ARITHMETIC AND CERTIFICATE SCOPE

$+ 1)/N]$ for range bounding. The represented policy er outside boundary included in the final box. All eriment are nonnegative. Therefore the lower/lower image of the entire box, including the bilinear term.

absolute value is checked below 2^{62} before integer division. In, multipliers are at most 64 and bounded model values signed 64-bit capacity. This finite experiment’s overflow γ -horizon machine-arithmetic theorem.

rule can change inside a box. Cost bounds therefore *le installed actions on the full closed box*. A candidate never it might equal the installed action; it has zero upper possible installed action. Otherwise the upper revision midpoint installed-action comparison from becoming an

uses the installed action at each box midpoint and then it is a proposal, not exactly the continuous installed rule the classical solver uses midpoint rewards and bilinear π policy. Both receive the same whole-box operating and the resulting serialized box policies, not to an unrecorded

s.
 er operating action enclosures bound the unconstrained necessary-action recursion uses the lower operating witness, lower implementation costs. The randomized local-grid over *all* actions. The code checks both nonnegativity and this randomized bound. For a policy with $U_t - J_t^{\text{lower}} \leq$ feasibility follows. Its implementation upper enclosure above; a failed operating certificate leaves only a bound

not prove that the candidate policy is truly infeasible. “te pass” rather than treating every failure as a mathematical pilot, fallback boxes, coarse-mesh failures, and retained. Witness construction is charged once for each reevaluation is charged to its method. This is a nonlinear timed advantage at arbitrary dimension.

PRIMARY CONFIGURATIONS

0.000000	0.000000	0.000000	I/P
0.000000	0.000000	0.000000	I/P
0.000000	0.000000	0.000000	I/P
0.000000	0.000000	0.000000	I/P
0.810725	10.812347	0.001622	–
0.750714	10.770159	0.019445	–
0.468642	0.479631	0.010989	–
0.124513	0.227986	0.103473	–
0.336417	0.347406	0.010989	–
0.044313	0.114248	0.069935	–
0.865305	0.871739	0.006434	–
0.345015	0.345015	0.000000	I
9.490961	9.490961	0.000000	I/P
9.248564	9.250944	0.002380	–
0.000000	0.000000	0.000000	I/P
0.000000	0.000000	0.000000	I/P
0.000000	0.000000	0.000000	I/P
0.000000	0.000000	0.000000	I/P
5.612820	15.614178	0.001359	–
5.471299	15.537400	0.066101	–
3.493934	3.508053	0.014119	–
2.402580	2.580313	0.177732	–
2.536828	2.546365	0.009537	–
1.764791	1.840578	0.075787	–
1.227595	1.234256	0.006661	–
0.140483	0.388955	0.248472	–
3.699252	13.702695	0.003443	–
3.242542	13.302762	0.060220	–
0.000000	0.000000	0.000000	I/P
0.000000	0.000000	0.000000	I/P
0.000000	0.000000	0.000000	I/P
0.000000	0.000000	0.000000	I/P

quality; P additionally denotes all-restart function
ashes are in results/primary_exact_values.json; no
readable exact table contains both cost endpoints as
point, the exact gap, both equality flags, the installed-
function hash, and full proof-object hash. In each

dition	0.05	8.721989	8.721989	0.000000
dition	0.20	10.434669	10.434669	0.000000
entive	0.20	10.640825	10.630435	0.010390
endar	0.20	8.653418	8.653418	0.000000
dition	0.05	8.933154	8.933154	0.000000
entive	0.05	9.131155	9.131155	0.000000
endar	0.05	7.458884	7.458884	0.000000
dition	0.20	8.906508	8.906508	0.000000
entive	0.20	9.075142	9.074068	0.001074
endar	0.20	7.387071	7.387071	0.000000
dition	0.05	11.907352	11.907352	0.000000
entive	0.05	12.107270	12.102384	0.004886
endar	0.05	9.894955	9.894955	0.000000
dition	0.20	11.866806	11.866806	0.000000
entive	0.20	12.085757	12.068172	0.017585
endar	0.20	9.842486	9.830059	0.012428
dition	0.05	5.733254	5.733254	0.000000
entive	0.05	5.898295	5.893513	0.004782
endar	0.05	4.826153	4.826153	0.000000
dition	0.20	5.733254	5.733254	0.000000
entive	0.20	5.896158	5.880418	0.015740
endar	0.20	4.803912	4.803912	0.000000
dition	0.05	15.208582	15.208582	0.000000
entive	0.05	15.463492	15.462576	0.000916
endar	0.05	12.617824	12.617824	0.000000
dition	0.20	15.136084	15.136084	0.000000
entive	0.20	15.385492	15.379338	0.006154
endar	0.20	12.502923	12.502923	0.000000
dition	0.05	10.897170	10.897170	0.000000
entive	0.05	11.004643	11.003208	0.001435
endar	0.05	8.787898	8.787898	0.000000
dition	0.20	10.870687	10.870687	0.000000
entive	0.20	10.973545	10.964602	0.008943
endar	0.20	8.703869	8.703869	0.000000
dition	0.05	10.044666	10.044666	0.000000
entive	0.05	10.357144	10.354904	0.002240
endar	0.05	8.656080	8.656080	0.000000
dition	0.20	9.998652	9.998652	0.000000
entive	0.20	10.308105	10.296267	0.011838
endar	0.20	8.602967	8.602967	0.000000

	0.20	8.257042	8.170051	0.087011
n	0.05	10.469741	10.469741	0.000000
e	0.05	10.679481	10.678541	0.000940
r	0.05	8.721618	8.721618	0.000000
n	0.20	10.434168	10.434168	0.000000
e	0.20	10.640244	10.630954	0.009290
r	0.20	8.653691	8.653691	0.000000
n	0.05	8.309909	8.309909	0.000000
e	0.05	9.281022	9.281022	0.000000
r	0.05	6.696913	6.696913	0.000000
n	0.20	7.808322	7.792568	0.015754
e	0.20	8.734932	8.692720	0.042212
r	0.20	6.142199	6.131385	0.010814

e-class operating check for both policies. All values
 costs; exact rational values, changed primitives, and
 proof objects.

RT CONSTRAINED REFERENCES

restart history-conditioned references

andomized	Det. lower	Nodes	Vertices
4.149907	4.365233	253	314
4.447063	4.658161	257	318
4.743143	4.951089	253	314
5.032875	5.244017	256	316
5.422816	5.615933	212	264
3.287636	4.365233	253	314
3.600854	4.658161	257	318
3.909072	4.951089	253	314
4.188360	5.244017	256	316
4.647604	5.615933	212	264
0.000000	0.000000	253	253
0.000000	0.000000	257	258
0.000000	0.000000	253	254

0	0.000000	0.000000	212	212
0	0.000000	0.000000	253	253
4	0.000000	0.000000	257	257
2	0.000000	0.000000	253	253
4	0.335335	1.750000	256	258
1	0.000000	0.000000	212	212
0	0.000000	0.000000	253	253
4	0.000000	0.000000	257	258
2	0.000000	0.000000	253	254
4	1.467067	1.750000	256	261
1	0.000000	0.000000	212	213
0	0.000000	0.000000	253	253
4	0.000000	0.000000	257	258
2	0.000000	0.000000	253	254
4	0.335335	1.750000	256	261
1	0.000000	0.000000	212	213
0	4.149907	4.365233	253	312
4	4.447063	4.658161	257	316
2	3.774393	3.954007	253	303
4	4.036576	4.234366	256	302
1	0.000000	0.000000	212	237
0	3.287636	4.365233	253	312
4	3.600854	4.658161	257	316
2	3.051087	3.954007	253	303
4	3.252671	4.234366	256	302
1	0.000000	0.000000	212	239
0	0.000000	0.000000	253	253
4	0.000000	0.000000	257	257
2	0.000000	0.000000	253	253
4	0.000000	0.000000	256	256
1	0.000000	0.000000	212	212
0	0.000000	0.000000	253	253
4	0.000000	0.000000	257	257
2	0.000000	0.000000	253	253
4	0.000000	0.000000	256	256
1	0.000000	0.000000	212	212
0	0.000000	0.000000	253	253
4	0.000000	0.000000	257	257
2	0.000000	0.000000	253	253
0	0.000000	0.000000	253	253
4	0.000000	0.000000	257	257
2	0.000000	0.000000	253	253

R CANDIDATES AND FAILURES

candidates, including failed certificates

at bound	Global LB	Policy UB	Seconds
0.4977	0.0000	2.6751	0.011
0.4977	0.0000	2.4974	0.010
1.1074	0.0000	1.6213	0.007
0.4151	0.0000	3.0150	0.024
0.5157	0.0000	2.1002	0.012
0.4996	0.0000	2.1219	0.012
0.9009	0.0000	0.6582	0.012
0.3095	0.0000	2.6452	0.038
0.5000	0.0000	1.7191	0.030
0.4996	0.0000	1.9953	0.031
0.8199	0.0000	0.3489	0.028
0.2182	0.0000	2.6183	0.083
0.5000	0.0000	1.3502	0.103
0.5000	0.0000	1.5348	0.105
0.7748	0.0000	0.1618	0.100
0.1651	0.0000	2.5997	0.279
0.5000	0.0000	0.9991	0.369
0.5000	0.0000	1.1250	0.377
0.1539	0.0000	2.5915	1.049
0.7217	0.0000	7.8440	0.014
0.7207	0.0000	7.5944	0.014
1.9218	0.0000	3.5345	0.014
1.8675	0.0000	3.2901	0.048
0.5306	0.0000	5.3481	0.024
0.4999	0.0000	5.1263	0.024
1.4938	0.0000	1.0518	0.023
0.4683	0.0000	5.2667	0.075
0.5000	0.0000	4.5263	0.060
0.4998	0.0000	3.9348	0.060
1.3586	0.0000	0.5982	0.057

5.0440	0.0000	5.9728	0.095
0.6609	0.0000	13.1739	0.047
0.6250	0.0000	12.3093	0.046
2.3354	0.0000	1.6657	0.045
2.3030	0.0000	1.8973	0.148
0.5000	0.0000	8.7753	0.124
0.5000	0.0000	8.7555	0.119
2.0828	0.0000	1.1280	0.107
0.4225	0.0000	9.0306	0.348
0.5000	0.0000	7.9683	0.400
0.5000	0.0000	7.6643	0.393
1.9345	0.0000	0.2580	0.372
0.2597	0.0000	8.9885	1.118
0.5000	0.0000	7.3118	1.436
0.5000	0.0000	7.4019	1.430
0.1931	0.0000	8.9669	4.103
1.6264	0.0000	17.9085	0.055
1.6261	0.0000	17.8356	0.056
4.3523	0.0000	8.9351	0.054
4.2975	0.0000	8.8716	0.188
0.8575	0.0000	24.2374	0.091
0.8417	0.0000	23.5439	0.094
3.2834	0.0000	2.3702	0.087
3.2552	0.0000	2.6072	0.293
0.5056	0.0000	16.2448	0.242
0.5013	0.0000	14.0445	0.238
2.8959	0.0000	1.7564	0.211
0.4440	0.0000	16.7883	0.684
0.5000	0.0000	12.5711	0.797
0.5000	0.0000	12.5225	0.832
2.6814	0.0000	0.3477	0.705
0.3209	0.0000	13.1873	2.276
0.5000	0.0000	11.8492	2.867
0.5000	0.0000	12.1034	2.876
0.2305	0.0000	13.1578	8.256

is the classical four-price portfolio; P is the midpoint-
 B is the deterministic outer-class lower bound. Policy UB

0186	0.088107	0.097382	6272
0186	0.088121	0.092759	12526
0289	0.025619	0.044169	171
0289	0.025649	0.034924	309
0289	0.025665	0.030302	588
0289	0.025620	0.044169	168
0289	0.025649	0.034924	306
0289	0.025665	0.030302	585

ct feasibility and quadratic gap tests; stored isolated
 e floor is zero in all rows.

R PROPAGATION AND ACTION COVERAGE

- Q_t^av , where Q_t^a is a positive subprobability kernel
 and the estimate $\|\mathsf{T}_t^av - \mathsf{T}_t^aw\|_\infty \leq \alpha_t \|v - w\|_\infty$
 imation obeys $\left\|\widetilde{\mathsf{T}}_t^av - \mathsf{T}_t^av\right\|_\infty \leq \xi_t$ for every action
 If upper encoding adds defect d_t , then the backward

$$+ \xi_t + \alpha_t b_{t+1}.$$

king the corresponding lower envelope supplies safe
 ent proves the deterministic necessary-action lower
 h compatible stopping and nonnegative costs.
 A discrete subset gives a lower operating value but
 s an overly narrow implementation policy class. To
 rol target, one must enclose all omitted actions, for
 d interval covers. State interpolation, killed-transition
 discretization error must remain separately identifiable
 enance calculation validates a distinct finite-action
 onstants.

R STOPPED-CONTROL OPTIMUM

$c = 3/4$, $p = 0$, and constant $\theta \in [-1/5, 1/5]$, the
 ictly increasing in θ . The derivative lower bounds

$= 50/49$, hence $x(t) \geq 25/49 > 1/2$. It is decreasing boundary is reached. Let y denote preference. Direct ment gives

$$+ 8 - .04\theta(y - 2) - .00005 + .002 - \frac{.075}{x(t)} \\ 2)^2 - .004 \log x(t) + .32(1 - t), \tag{2}$$

$$2). \tag{3}$$

$/5, 9/5]$ and $L = \log(4/3) < 1/3$. The utility is $-e^{Lz}/z$, because $Lz < 1$. Its lower bound is $-5e^{1/15} \geq -75/14$. $\log x(t) \leq \log(5/4) < 1/4$. Dropping nonnegative terms

$$\frac{1}{1000} - \frac{4}{625} - \frac{1}{25} - \frac{75}{14} = \frac{342357}{140000} > 2. \tag{4}$$

$^2 \geq 10/81$. Thus

$$- \frac{4}{3125} - \frac{1}{125} = \frac{28901}{253125} > 0. \tag{5}$$

$4/5$ to larger y . The extension remains bounded, nonde- meter derivative retains the bounds in (3). Conservative 2 hold for this extension.

g lower bound for the derivative

$6/5$, where $u_0 = 61/50$ and $\sigma = 1/20$. If drift increases, date and lower-boundary survival events are nested. For θ , paths surviving under both drifts have a nonnegative paths contribute at least two. Taking a positive difference

$$\mathbf{1}\{t < \tau_a\} \geq 2\partial_\theta \Pr_\theta(t < \tau_a). \tag{6}$$

be added, not included in this frozen-integrand compari- $\theta \in [-1/5, 0]$ and at least $-54/125$ for $\theta \in [0, 1/5]$, after

1 inequalities

$$2) > 1/50, \quad \Phi(-14/5) > 1/500, \quad (8)$$

$$5, \quad \Phi(6/5) > 22/25.$$

drift half interval.

s at $\theta = 0$ and $\theta = 1/5$ are at least $-4/5$ and $6/5$.

ft half interval. Since $e^{-.04t} \geq 24/25$, the two lower

d occupation expression are

$$- \frac{4}{125} = .066304, \quad (9)$$

$$- \frac{54}{125} = .379008. \quad (10)$$

ument. The intervals cover every admissible constant

lary error and rational checks

scarded. Under all admissible drifts, a path hitting it

st $(b - u_0 - 1/5)/\sigma$. Reflection and a Gaussian tail

$$\frac{-1/5)^2}{5^2} \Big] = 2e^{-380.88} < 2^{-379}. \quad (11)$$

on that event, the drift likelihood score has second

ounds the derivative discrepancy between the actual

extended lower-killed one by

$$02^{-189} + 22^{-379} < 1/1000. \quad (12)$$

half-line bounds gives precisely $8163/125000$ and

nuous function on the compact drift interval has the

ality or satisfy the original all-domain current-state target.

N MAP AND HISTORICAL EVIDENCE

review is removed. The branch starts at the latest R36
le, supplement, response, and computational report from
ISTORY_R38.pdf, followed by the earlier recursive
consumption–portfolio specification, its full-domain target,
the 47-coordinate derivative/residual bridge discussion,
ns, inventory comparisons, and operating-mode studies
ages.

main unchanged in the repository. Their unmaterialized
verified observations. R38 supplies a newly executed
ce. In particular, the fresh compiled-rule nonlinear pilot
nalty run. The complete prior unfavorable neural and
converted into a new neural acceleration claim.

not additional experiments for the R38 cohort. Its former
read at their original dates. The current main article and
ectives have new certificates and which original targets
anization shortens the main narrative without deleting
prints the references cited in its current theorem and
inherited bibliography is retained in `history/refe`
d R34 source, and in the historical PDF annex; uncited
removed from the scientific archive.